EH-2080B/EH-2080T/EH-2080C/

EH-2080B Pro/EH-2080T Pro/EH-2080C Pro

Urine Formed Element Analyzer

OPERATOR'S MANUAL

Description

Thank you for purchasing EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro Urine Formed Element Analyzer.

Please carefully read and understand this operator's manual prior to the use of the product, so as to correctly use this product.

Please properly keep and use this operator's manual after reading them, so that this manual can be looked up as required at any time.

Product name: Urine Formed Element Analyzer

Model: EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro

Performance, construction and composition of the product: this medical electrical system is consisted of four parts, i.e., medical electrical machine, non-medical electrical computer (including software), display and printer.

Applicable scope: this analysis is applicable to the detection of the visible component in urine in clinical lab.

Contraindication :There is no contraindication about our product present.

Manufacturer: Suzhou Hyssen Electronics Co., LTD

Address: Unit 213-216, Building A3, Biobay, 218 Xinghu Street, Suzhou Industrial Park 215123, P. R. China
Unit 501,Building 2, 218 Sangtian Street, Suzhou Industrial Park 215123, P. R. China.

Revision number of the Operator's manual: 1.0

Issued date of the Operator's manual: 2017-08

Service life :5 years

Date of manufacture : On the label

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(2)All the parts to be replaced during maintenance, and the associated accessories as well as consumables are the materials that are originally equipped (or assembled) or approved by the manufacturer.

(3)The relevant electrical equipment conforms to national standards and the operator's manual.

(4)The operation of the product is in accordance with the operator's manual.

(5) The reagent system to be used shall be supplied or approved by the manufacturer, within the period of validity and not polluted.

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1. When necessary, the manufacturer will, in accordance with user's requirements, provide circuit diagram, list of element & instrument, notes on drawings, details for correction or the information helpful and necessary for the equipment parts that are approved to be maintained by the manufacturer and the user's qualified technical personnel.

2. We promise to provide one year of warranty and life-long maintenance in the event of quality issue regarding the manufacturer's product starting from the procurement date (the installation date shall govern).

3. In the event of quality issue and/or absence of accessory during unpacking inspection, do not use the product immediately; instead, call our after-sale hot-line (0512-87182916) immediately.

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b) Instrument is damaged due to improper use or maintenance of user.

After-sale service organization

Suzhou Hyssen Electronics Co., LTD
Unit 501,Building 2, 218 Sangtian Street, Suzhou Industrial Park 215123, P. R. China.

Service hot-line: +86 512 87182916

Fax: +86 512 87182901

⚠ Warning

- This system may only be used by the professional inspection personnel, doctors or laboratory assistants who have been trained by the manufacturer or the manufacturer's agents.
- It is important for the hospital or organization that employs this equipment to carry out a reasonable service/maintenance plan. Neglect of this may result in machine breakdown or injury of human health.
- Ensure to use this system under the condition specified in this manual; otherwise, the system would not work normally, and the analysis result would be unreliable; which will damage the system components and caused personal injury.

NOTE

- The readers of the operator's manual shall be the following professional personnel of inspection section:

1. Those who perform daily operating task;
2. Those who perform maintenance and troubleshooting;
3. Those who study the operation of the system.

Table of Contents

Chapter 1 General Description for the Manual **1-1**
1.1 Introduction 1-1
1.2 Who Should Read This Manual 1-1
1.3 How to Find Information 1-1
1.4 Convention Used in this Manual 1-2
1.5 Symbols 1-3
1.6 Safety information 1-5

Chapter 2 Overview for the System **2-1**
2.1 Introduction 2-1
2.2 Intended Use 2-1
2.3 Parameters 2-1
2.4 Product Description 2-2
2.4.1 Overview 2-2
2.4.2 Modules and Components 2-3
2.5 User Interface 2-6
2.5.1 Title bar 2-6
2.5.2 Shortcut icon area 2-6
2.5.3 Menu bar and submenu bar 2-7
2.5.4 Status bar 2-7
2.5.5 Analyzer control area 2-8
2.6 Reagents and Controls 2-9
2.6.1 Reagent 2-9
2.6.2 Controls 2-10

Chapter 3 Operating Principle **3-1**
3.1 Introduction 3-1
3.2 Analysis Principle of the machine vision technology 3-1
3.2.1 Processing Flow 3-1
3.2.2 Identification Flow of Digital Images 3-1

Chapter 4 Installation **4-1**
4.1 Introduction 4-1
4.2 Installation Requirements 4-1
4.2.1 Power Requirement 4-1
4.2.2 Fuse Requirements 4-2
4.2.3 Space Requirement 4-2
4.2.4 Environmental requirements 4-2
4.2.5 Transportation and Installation 4-3
4.3 Connecting the analysis system 4-3

4.3.1 Connection of tube 4-3
4.3.2Connection of external devices 4-4

Chapter 5 Settings 5-1
5.1Introduction 5-1
5.2Maintenance 5-1
5.2.1 Maintenance 5-2
5.2.2 Reagent Information 5-3
5.2.3 Restore Factory 5-6
5.3 System setting 5-6
5.3.1 Basic setting 5-7
5.3.2 Report 5-7
5.3.3 Print template 5-9
5.3.4 LIS 5-9
5.3.5 Other 5-10
5.3.6 Hospital Information 5-18
5.4 Parameters Setting 5-18
5.4.1 Optional parameters 5-19
5.4.2 Selected parameters 5-19
5.4.3Reference group 5-20
5.4.4 Range of reference value 5-21
5.4.5 Range of children age 5-21
5.4.6 Units 5-21
5.4.7 Threshold 5-22
5.5 Setting of Chemistry 5-23
5.6 Others 5-27
5.6.1 Data Dictionary 5-27
5.6.2 User management 5-27

Chapter 6 Routine Operation 6-1
6.1 Introduction 6-1
6.2 Preparation prior to operation 6-2
6.3 Startup and user log-in 6-3
6.4 Daily Quality control 6-4
6.5 Preparation of sample 6-4
6.6 Sample analysis 6-6
6.6.1 Enter worklist information 6-6
6.6.2 Matching Chemistry information 6-8
6.6.3 Sample inspection procedure 6-9
6.6.4 Special functions 6-11
6.7 Overview for worklist 6-12
6.7.1 Introduction of the screen 6-13
6.7.2 Operation functions 6-13
6.8 Shutdown 6-14

6.8.1 Exit the terminal software 6-15
6.8.2 Shutdown the external computer 6-15
6.8.3 Power off the analyzer 6-15

Chapter 7 Result Output 7-1
7.1 Introduction 7-1
7.2 Review/Report 7-1
7.2.1 Introduction to review interface 7-1
7.2.2 Button Functions 7-2
7.2.3 Report 7-3
7.2.4 Button functions 7-9
7.3 Review interface 7-10
7.3.1 Introduction to review interface 7-10
7.3.2 Button functions 7-11
7.4 Statistics 7-19
7.4.1 Workload Statistics 7-19
7.4.2 Export 7-20

Chapter 8 Customizing Print Template 8-1
8.1 Introduction 8-1
8.2 Entering the customization screen 8-1
8.2.1Setting paper property 8-1
8.2.2 Editing the template 8-3
8.3 Edit new print template 8-8
8.4 Edit of existing print template 8-8
8.5 Managing the Template 8-9
8.5.1 Template loading 8-9
8.5.2Template preview 8-9

Chapter 9 Quality Control 9-1
9.1 Introduction 9-1
9.2 Quality control 9-1
9.2.1 Setting 9-1
9.2.2 Quality control 9-4
9.2.3 Review 9-5

Chapter 10 Services 10-1
10.1 Introduction 10-1
10.2 Reagent management 10-2
10.3 Maintenance 10-3
10.3.1 When and Why 10-3
10.3.2 Startup liquid line initialization 10-3
10.3.3 Clean the sample probe 10-3
10.3.4 Normal curing 10-3

10.3.5 Forward flush 10-3
10.3.6 Backward flush 10-4
10.4 Version Information 10-4
10.5 Manually cleaning 10-4
10.5.1 Cleaning of the surface of counting cell 10-4
10.5.2Cleaning the enclosure of the analyzer 10-4
10.5.3 Cleaning the lens of microscope 10-4
10.6 Replacement 10-5
10.6.1 When to Replace and Tools Needed 10-5
10.6.2 Replacing the Reagent A and B 10-5
10.6.3 Replacing the waste container 10-5
10.7 Transportation and storage 10-5
10.8 Long-time not use dispose 10-6

Chapter 11 Troubleshooting 11-1
11.1 General 11-1
11.2 Error Information and Handling 11-1

Appendix A Specification A-1
A.1 Classification A-1
A.2 Associated reagents of the product A-1
A.3 Parameters A-1
A.4 Sampling features A-1
Sample volumes required for each analysis A-1
A.5 Parameter range A-1
A.6 Carryover A-2
A.7 Input/output devices A-2
A.8 Display resolution A-2
A.9 Ports of the analyzer A-2
A.10 Power supply A-2
A.11 Operating environment A-2
A.12 Electromagnetic compatibility A-2
A.13 Dimension and weight A-3
A.14 Inventory of attachment A-3
A.15 Performance of products A-3

Chapter 1 General Description for the Manual

1.1 Introduction

This chapter explains how to use your EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro operator's manual. The operator's manual will be supplied with the equipment, in which the purpose, functions and operation for EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro system are described in details. Read this manual carefully before operating your analyzer and operate your analyzer strictly as instructed in this manual, so as to ensure that EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro can be correctly used, and its optimal performance can be exerted, and the operator's safety can be guaranteed.

1.2 Who Should Read This Manual

This manual contains information written for clinical laboratory professionals to:

- Learn about the hardware and software of EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro;
- Customize the system setting;
- Perform daily operating task;
- Perform system maintenance and troubleshooting.

1.3 How to Find Information

This manual comprises 11 chapters and 1 appendix. Refer to the table below to find the information you need.

If you want to…	Refer to…
Understand the operator's manual of EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro.	**Chapter 1 General description**
Understand the purpose, measurement parameters, reagent, quality control, instrument composition, software operation and so on.	**Chapter 2 Overview for the system**
Understand the measurement principle in the application of EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro	**Chapter 3 Operating principle**
Learn about the installation requirements and methods of	**Chapter 4 Installation**

EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro.	
Learn about the parameter settings and printing parameter settings of the EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro	**Chapter 5 Settings**
Learn about the process of sample collection and analysis	**Chapter 6 Routine operation**
Carry out the routine operation of EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro	**Chapter 6 Routine operation**
Review sample results	**Chapter 7 Result output**
Learn about how to customizing the print template of the EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro.	**Chapter 8 Customizing print template**
Learn about how to use the quality control programs of the EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro.	**Chapter 9 Quality control**
Learn about the maintenance and testing methods of EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro	**Chapter 10 Services**
Learn about how to solve the problems of the EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro.	**Chapter 11 Troubleshooting**
Learn about the technical specifications of the EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro.	**Appendix A Specifications**

1.4 Convention Used in this Manual

This manual uses certain typographical conventions to clarify meaning in the text:

Form	Meaning
‘XX’	Bold letters included in ‘ ’ indicate text you can find on the screen of the EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro

XX	Italic letters are emphases
[XX]	All capital letters enclosed in [ ] indicate a key name on keyboard

All illustrations in this manual are provided as examples only. They may not necessarily reflect your EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro setup or data displayed.

1.5 Symbols

You will find the following symbols in this manual:

Symbol	Meaning
	Read the statement below the symbol. The statement is alerting you to a potentially biohazardous condition.
Warning	Read the statement below the symbol. The statement is alerting you to an operating hazard that can cause personnel injury.
Caution	Read the statement below the symbol. The statement is alerting you to a possibility of analyzer damage or unreliable analysis results.
NOTE Note	Read the statement below the symbol. The statement is alerting you to information that requires your attention.

You may find the following symbols of the analyzer system:

Symbol	Meaning
	CAUTION, CONSULT ACCOMPANYING DOCUMENTS.
	BIOLOGICAL RISK
	HIGH VOLTAGE

	EXERCISE CAUTION WHEN WORKING AROUND TO AVIOD PRICKING
	PROTECTIVE EARTH (GROUND)
	ALTERNATING CURRENT
	FOR IN VITRO DIAGNOSTIC USE
	BATCH CODE
	USE BY
	SERIAL NUMBER
	DATE OF MANUFACTURE
	MANUFACTURER
	TEMPERATURE LIMITATION
	CONSULT INSTRUCTION FOR USE
	HANDLING PROMPT

	THE FOLLOWING DEFINITION OF THE WEEE LABEL APPLIES TO EU MEMBER STATES ONLY: THE USE OF THIS SYMBOL INDICATES THAT THIS PRODUCT SHOULD NOT BE TREATED AS HOUSEHOLD WASTE. BY ENSURING THAT THIS PRODUCT IS DISPOSED OF CORRECTLY,YOU WILL HELP PREVENT BRINGING POTENTIAL NEGATIVE CONSEQUENCES TO THE ENVIRONMENT AND HUMAN HEALTH. FOR MORE DETAILED INFORMATION WITH REGARD TO RETURNING AND RECYCLING THIS PRODUCT, PLEASE CONSULT THE DISTRIBUTOR FROM WHOM YOU PURCHASED THE PRODUCT.
CE	THE DEVICE IS FULLY CONFORMANCE WITH THE COUNCIL DIRECTIVE CONCERNING IN VITRO DIAGNOSTIC MEDICAL DEVICES 98/79/EC.
EC REP	AUTHORISED REPRESENTATIVE IN THE EUROPEAN COMMUNITY

1.6 Safety information

The following symbols are used to prompt danger or the information to be particularly paid with attention in this Manual:

- The tested object of this product is urine. Since the pollution degree or infectivity of urine cannot be determined, it is recommended that operators wear breathing masks and gloves during operation, and should be completely disinfected after operation.
- In case of any leakage of the machine, the leaked liquid may be biohazardous.

Warning

- Please check the firmness of all the doors/covers/plates before running the analyzer, and ensure that they will not be opened or loosened during the operation of the instrument.
- Make sure all the safety measurements are adopted. Disabling any safety

device or sensor is prohibited.

- Please take action to any alarm and problem indication immediately.
- Do not touch the moving parts.
- Contact the manufacturer or manufacturer -authorized distributors in time if any damaged part is found.
- Be careful when opening/closing and removing/installing the doors, covers and boards of the analyzer.
- Discard the analyzer according to government regulations.

⚠ Caution

- Please use the analyzer strictly as instructed by this manual.
- Make sure only software authorized by manufacturer is installed on the computer.
- Please install the original edition software to prevent the computer from being infected by virus.
- Please adopt proper measurements to prevent the reagents from being polluted.
- It is recommended that you install anti-virus software on the computer and scan for viruses periodically.

Chapter 2 Overview for the System

2.1 Introduction

EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro Urine Formed Element Analyzer is an in vitro analysis instrument for quantitatively testing the visible component in urine in clinical laboratories.

2.2 Intended Use

EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro Urine Formed Element Analyzer is a quantitative instrument for In Vitro Diagnostic Use in clinical laboratories; it provides visible component measurement (such as red blood cell, white blood cell, crystal, cast, epithelial cell, etc) in urine in clinical laboratories.

2.3 Parameters

NOTE

- The purpose of this analysis system is to identify the normal patient, with all normal system-generated parameters, and to flag or identify patient results that require additional studies.

The testing parameters for urine include 9 auto identification parameters, 1 research parameter and 30 artificially-identified and self-defined parameters.

The user can add or delete a maximum of 30 reportable items (but not all these items are auto identification items) according to the situation of the hospital.

Table 2-1 Parameters to be automatically identified and reported

Parameter	Abbreviation	Unit
Red blood cell	RBC	cell/μL
White blood cell	WBC	cell/μL
WBC clump	WBCC	cell/μL
Epithelial cell	EC	cell/μL
Small round cell	SRC	cell/μL
Hyaline cast	HYA Cast	cell/μL
Unclassified Cast	UNCC	cell/μL
Crystal	CRYS	cell/μL
Yeast	YST	cell/μL
Mucous Strands	MUC	Qualitative parameter
Bacteria	BACT	

Table 2-2 Research parameters

Parameter	Abbreviation	Unit
Poikilocyte	POI	%

2.4 Product Description

2.4.1 Overview

The EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro Urine Formed Element Analyzer includes four parts, i.e., medical electrical analyzer, non-medical electrical computer (including software), display and printer as shown in the following figure:

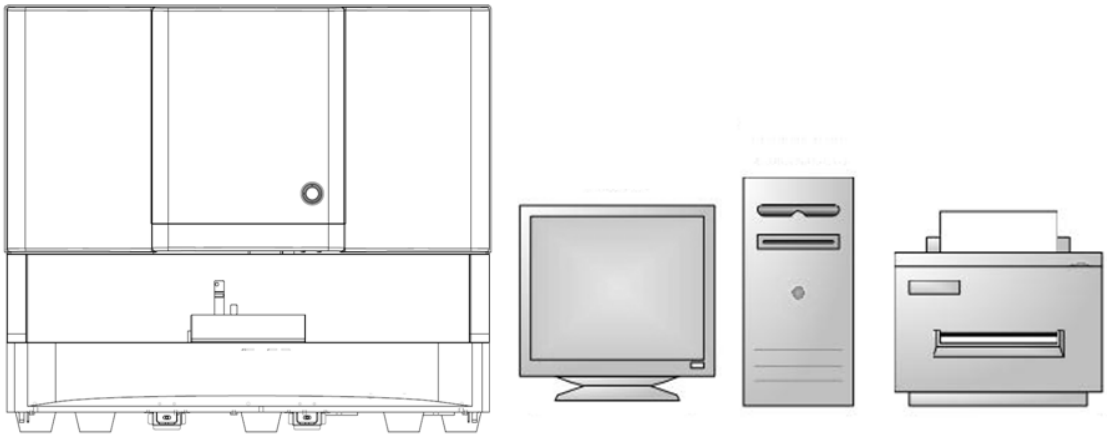


Figure. 2-1 EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro Urine Formed Element Analyzer

2.4.2 Modules and Components

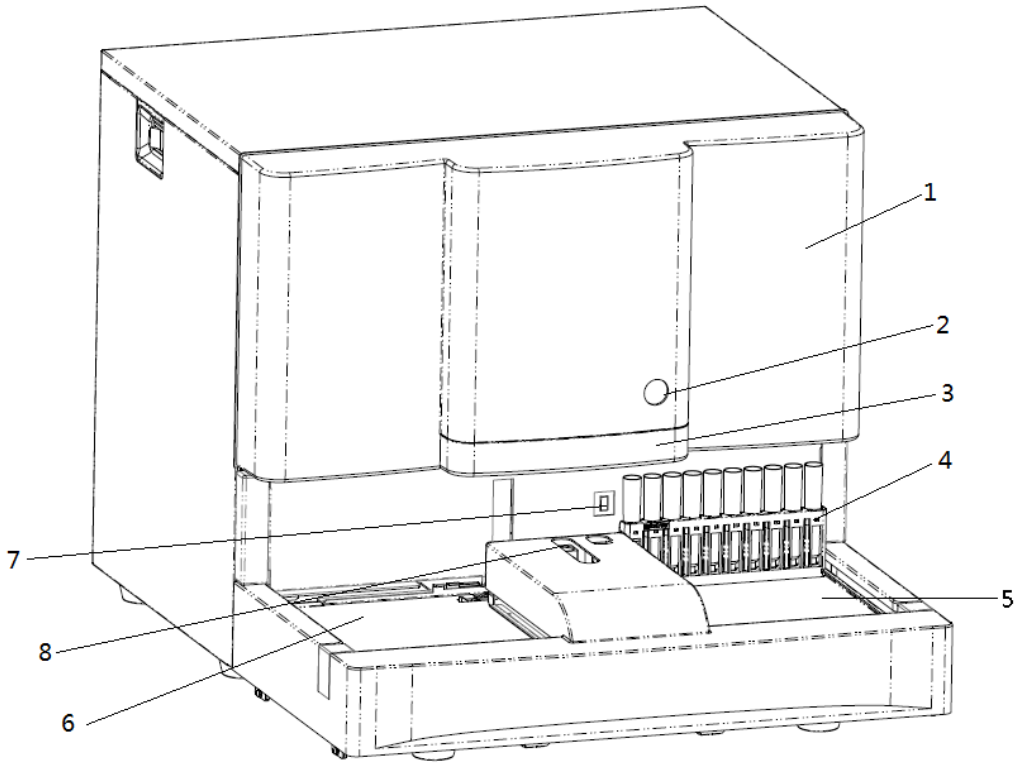


Figure. 2-2 Front of the analyzer

1 --- front movable cover plate

2 --- start button

3 --- front casing indicator-lamp

4 --- tube rack

5 --- loading area

6 --- unloading area

7 --- Test tube detection area

8 --- emergency treatment position

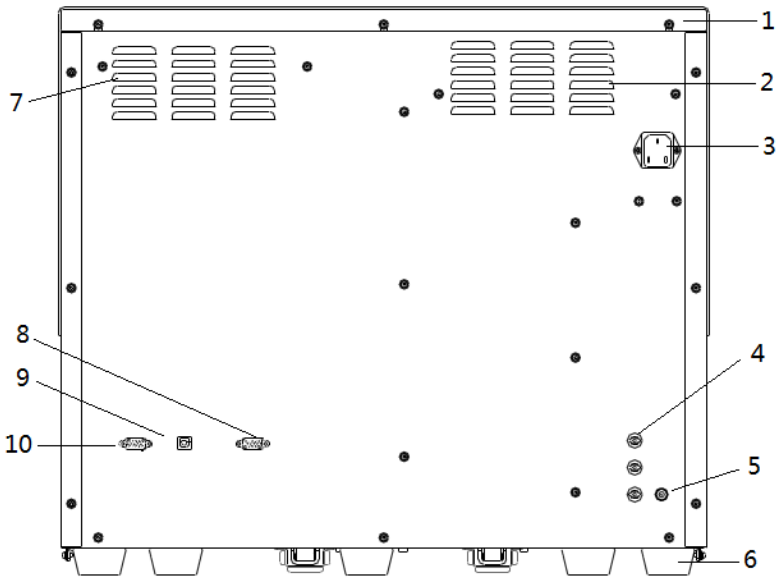


Figure. 2-3 Back of the analyzer

1 --- top plate

2 --- louver

3 --- power plug

4 --- liquid line connector

5 --- reagent alarm port

6 --- foot

7 --- louver

8 --- Cascade port

9 --- USB port

10 --- Serial port

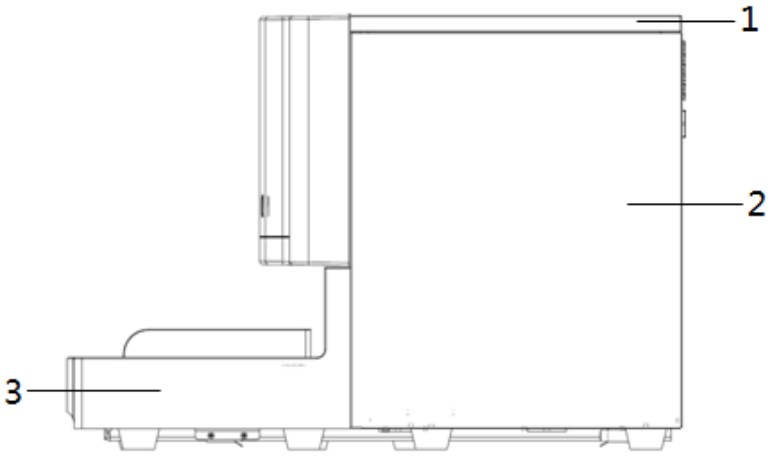


Figure. 2-4 Right side of the analyzer

1 --- top plate

2 --- right door

3 --- automatic sampler

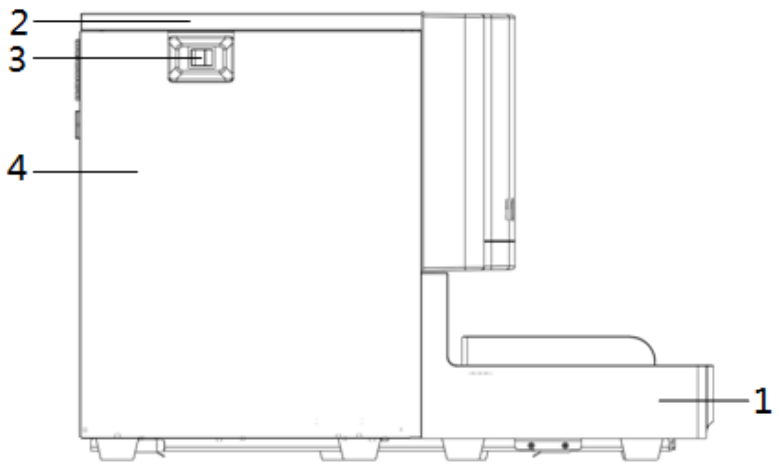


Figure. 2-5 Left side of the analyzer

1 --- automatic sampler

2 --- top plate

3 --- power switch

4 --- left door

2.5 User Interface

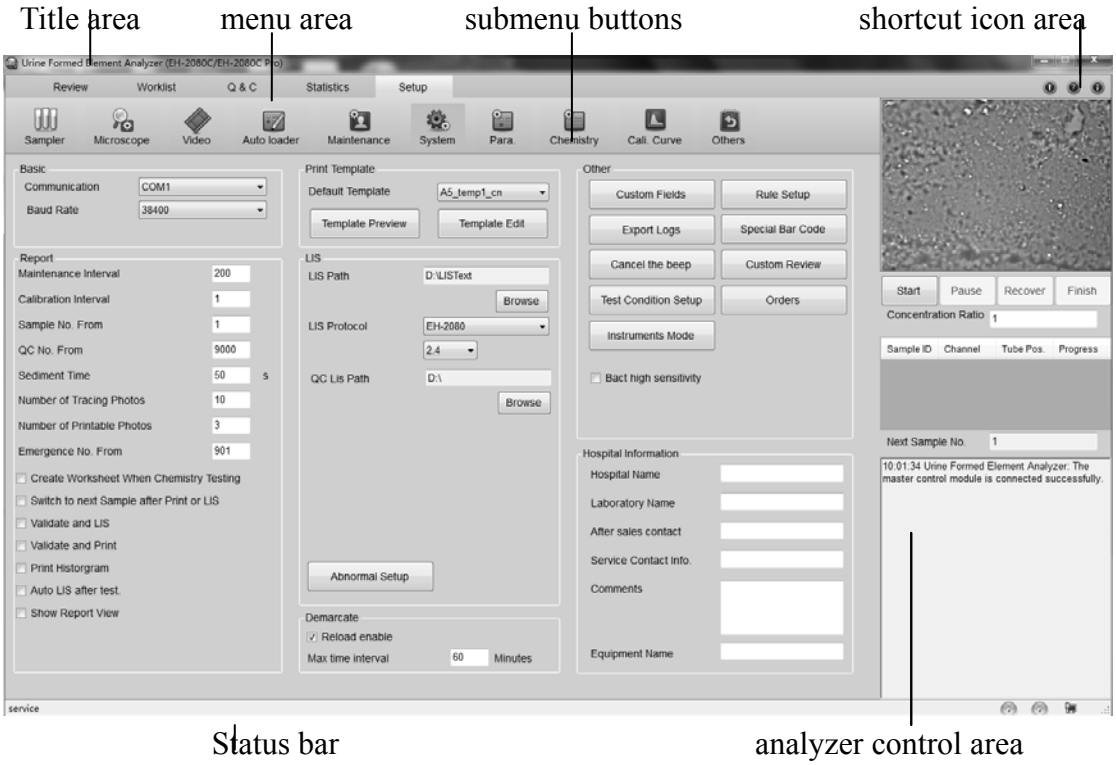


Figure. 2-6 Main interface

2.5.1 Title bar

Show the title of the analyzer and the current screen.

2.5.2 Shortcut icon area

Icon	Function
	Changeover user
	Display the operator’s manual of software.
	Click to check basic information of the software

2.5.3 Menu bar and submenu bar

Hierarchy of the menu

Menu Area	Submenu Area		Menu Area	Submenu Area
Review	Query		Quality Control	Q&C
	Validate			Review
	Cancel Validate			Setting
	Print		Statistics	Export
	Print Preview		Setting	Sampler
	LIS Sample			Microscope
	Select All			Video
	Delete			Maintenance
	Histogram			System
	Real View			Parameters
	Low-power Lens Unclassified			Chemistry
	High-power Lens Unclassified			Calibration Curve
Work list	Create			Others
	Modify			
	Save			
	Delete			
	Skip Test			

2.5.4 Status bar

This bar is used to:

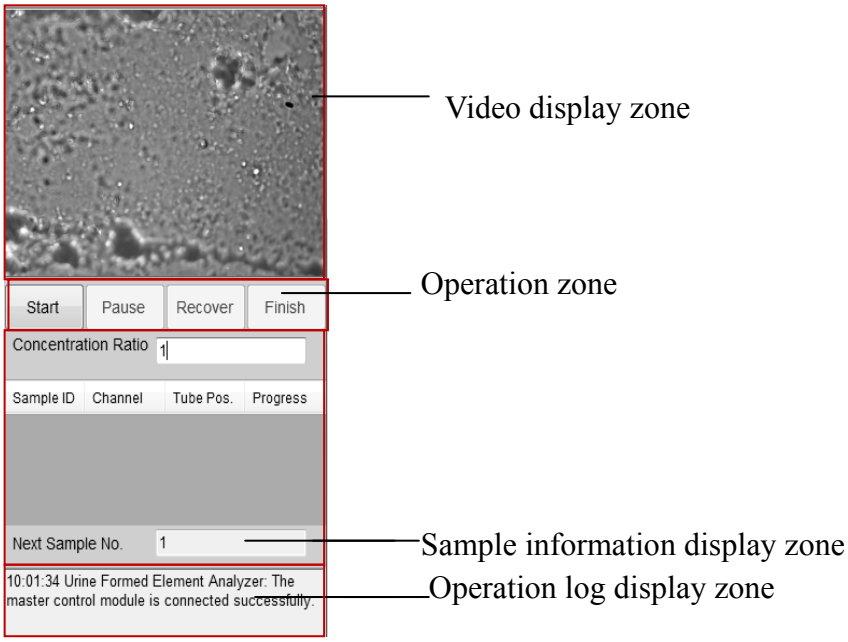
Display the current log-in user name;

Display reagent status;

Display the connection status between computer and analyzer.

2.5.5 Analyzer control area

Analyzer control area is mainly divided into the following parts:



Name	Function
Video display zone	Update and display the current video under microscope in a real-time manner.
Operation zone	Perform start/pause/recover/finish for sample to be inspected. Select sample style and test model. After select the concentrated model, you should input the sample volumes of you required in the box of sample volume.
Sample information display zone	Display the number, inspection status, channel, inspection progress of the sample being inspected; Display next sample No. and tube position.
Operation log display zone	Display the operation log of analyzer, including alarm prompt, etc.

2.6 Reagents and Controls

Reagent, quality control substance and analysis instrument totally form a system, and they must be used as a whole, so as to ensure the performance of the system. As the analyzer, reagents and controls are components of a system, performance of the system depends on the combined integrity of all components. You should only use the specified reagents (see Appendix A for Specification), which are formulated specifically for the fluidic system of your analyzer in order to provide optimal system performance. Do not use the analyzer with reagents from multiple suppliers. In such use, the analyzer may not meet the performance specified in this manual and may provide unreliable results. All references related to reagents in this manual refer to the reagents specifically formulated for this analyzer.

Each reagent package must be examined before use. Product integrity may be compromised in packages that have been damaged. Inspect the package for signs of leakage or moisture. If there is evidence of leakage or improper handling, do not use this reagent.

NOTE

- Store and use the reagents as instructed by instructions for use of the reagents.
- Ensure to use reagent within the period of validity indicated in the instructions of reagent.
- Before using the reagent, keep it still for a while before use.

2.6.1 Reagent

■ **Reagent A**

Reagent A is a specified reagent in the EH-20 Series Urine Formed Element Analyzer, and is used to clean the tube and counting cell of the analyzer after testing visible component in the urine and startup/shut down maintenance. This wash solution does not participate in the sample analysis process.

■ **Reagent B**

Reagent B is a specified reagent in the EH-20 Series Urine Formed Element Analyzer, and is used to clean the inner and outer walls of the sample probe, sampling tube and counting cell of the analyzer after testing visible component in urine sample and startup/shut down for maintenance. This wash solution does not participate in the sample analysis process.

■ **Cleanser**

Cleanser is specified reagent in the EH-20 Series Urine Formed Element Analyzer, and is used for the startup/shutdown maintenance, and for the maintenance after certain amount of sample is tested. The solution is a kind of Reagent that contains

sodium hypochlorite, and can be used to clean the measuring channels and tube so as to remove the pollutant such as albumen.

2.6.2 Controls

The controls are substance used in clinic medical inspection and have a known content and substrate property identical to that of actual sample. The control is used to monitor and control the operation condition of analyzer, to assess the precision of the inspection result of the analyzer so as to guarantee the reliability of analysis result.

Refer to the operator's manual for the use and storage methods of the controls.

All references related to controls in this manual refer to the controls specifically formulated for this analyzer by the manufacturer. You should buy those controls from the manufacturer or authorized distributors.

Chapter 3 Operating Principle

3.1 Introduction

EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro accomplishes intelligent analysis and quantitatively counting and testing of the shape of the visible component such as red blood cell in urine using the technology of the machine vision imaging and automatic coordinate tracing and identification.

3.2 Analysis Principle of the machine vision technology

Machine vision uses machine, video-sensing equipment and computer to replace the hands, eyes and brain of inspection doctor for measurement and judgment. When used in the analysis realm of the visible component in urine, machine vision usually converts various kinds of visible component in urine into digital and image signal via the vision-sensing equipment such as CCD and image acquisition card, etc, and then, the signals are transferred to the graph processing and analysis system of computer; and then, analysis software extracts useful objects of the images based on the information such as image element distribution and grey scale variation; and the analysis software calculates the characteristic sets of various divisional objects by modeling the characteristic space of visible component, and identifies visible component, and carries out classification and counting and automatically generates inspection report by simulating the artificial neural network of human brain.

3.2.1 Processing Flow

First of all, inject sample into the testing and counting cell by automatic sampler; the control system carries out automatic focusing and acquires pictures; the computer identifies, analyzes and positions the targets of the pictures of low power lens; the object lens change over to high power lens for high-power tracing and photographing; and then, the computer carries out the identification, classification and counting of the pictures of high power lens again.

3.2.2 Identification Flow of Digital Images

Usually, this identification flow includes the preprocessing of images to remove noise interference; and then, the targets of images are divided and characteristic calculation is conducted for the divided area; finally, the characteristic result is sent to neural network for classification and identification.

Chapter 4 Installation

4.1 Introduction

Warning

- Installation by personnel not authorized or trained by the manufacturer may cause personal injury or damage your analyzer. Do not install your analyzer without the presence of authorized personnel.
- The installation, test, update, and modification of the software associated with the analyzer shall only be performed by personnel authorized by the manufacturer.

The analyzer is checked and packed with care before it is shipped from the factory. Inspect the carton carefully after arrival. If you see any sign of damage is found, contact manufacturer or your local distributor immediately.

4.2 Installation Requirements

Warning

- If the analysis system is handled or installed by unauthorized or untrained personnel, damage may be caused. Do not unpack or install the analysis system without the presence of authorized personnel.

4.2.1 Power Requirement

Warning

- Make sure the analysis system is properly grounded and must not be used in flammable and explosive atmosphere.
- Before turning on the analyzer, make sure the input voltage meets the requirements.

Caution

- Using pin board may bring electrical interference and the analysis results may be unreliable. Please place the analyzer near the electrical outlet to avoid using pin board.
- Please use the original electrical wire shipped with the analyzer. Using other electrical wire may damage the analyzer or cause unreliable analysis results.

If this voltage requirement is not met, special AC regulated power supply shall be used.

Name	Voltage	Input power	Frequency
Analyzer	(100V-240V～)±10%	300VA	(50Hz/60Hz)±1Hz

4.2.2 Fuse Requirements

NOTE

- Only install fuses of specified specification on the analyzer. The manufacturer will not take responsibility if the instrument is damaged due to the use of the fuse of other models.

Fuse size: Φ5 T3A L250V

Fuse model: 5TT3-R

Fuse Requirement: fuse current 3A
Voltage shall not exceed 250V

4.2.3 Space Requirement

Since the exhaust fan is at the back of the analyzer, at least 15cm of net distance from wall shall be reserved during installation for heat emission.

The operating table shall exceed 70cm in width and 160cm in length, and shall be strong enough to bear the weight 80kg or above.

4.2.4 Environmental requirements

■ Indoor use;

■ Altitude does not exceed 2000m;

■ Temperature is within the range 5℃- 40℃;

■ Relative humidity ≤80%;

■ Rated pollution class: 2;

■ Table top must be stable and level, avoiding vibration and mechanical shock;

■ The analyzer shall be kept away from the equipment having strong electric field and strong magnetic field, as well as high-voltage equipment;

■ Display screen shall not be exposed to the direct sunshine; there shall be no strong light inside room so as for image observation;

■ Keep ventilation, damp-proof and dust-proof.

Do not use this analyzer in close proximity to sources of strong electromagnetic radiation (e.g. unshielded intentional RF sources), as these may interfere with the proper operation. It is advisable to evaluate the electromagnetic environment prior to operation of this analyzer.

4.2.5 Transportation and Installation

Warning

- Installation by personnel not authorized or trained by the manufacturer may cause personal injury or damage your analyzer. Do not install your analyzer without the presence of -authorized personnel.

Caution

- Installation of unrelated software on the computer (e.g. play games, log on the internet, etc.) may cause system damage and/or data error. Therefore, please make sure the computer is used for the analysis system only.

This analyzer must be handled and installed by the personnel authorized by the manufacturer. Do not handle or install the analyzer without contacting the after-sale service department or regional agent of the manufacturer.

4.3 Connecting the analysis system

4.3.1 Connection of tube

Warning

- Operator is in duty bound to obey the relevant regulations of the local region and nation regarding the release and treatment of invalid reagent, waste liquid, waste samples and consumables.
- The reagents are irritating to eyes, skin and diaphragm. Wear proper personal protective equipment (e.g. gloves, lab coat, etc.) and follow safe laboratory procedures when handling them in the laboratory.
- If the reagents accidentally spill on your skin, wash them off with plenty of water and if necessary, go see a doctor; if the reagents accidentally spill into your eyes, wash them off with plenty of water and immediately go see a doctor.

Caution

- There are two ways to discharge the waste: discharge to a waste container, or to directly to a waste disposal channel.
- If the waste is directly discharged to a waste proposal channel, make sure the outlet of the waste piping is placed lower than the waste outlet of the analyzer.
- The positions of the reagent containers of solution A and solution B shall not be higher than the reagent inlet of the analyzer.

The connection mode of the tube line of the analyzer is shown in the following figure.

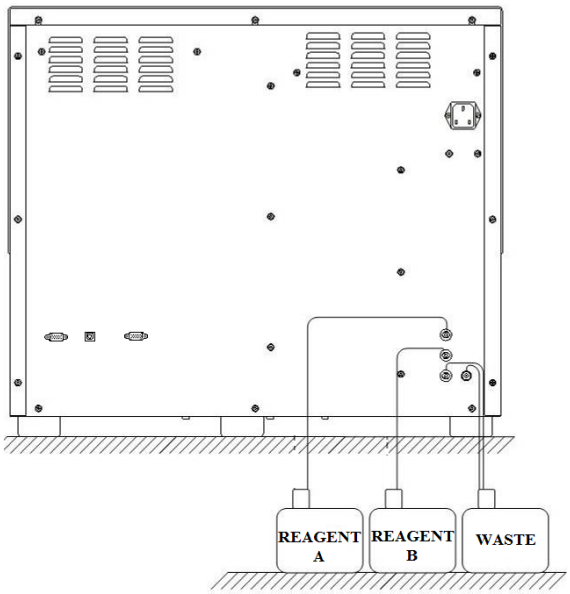


Figure. 4-1 Connecting the reagents

4.3.2Connection of external devices

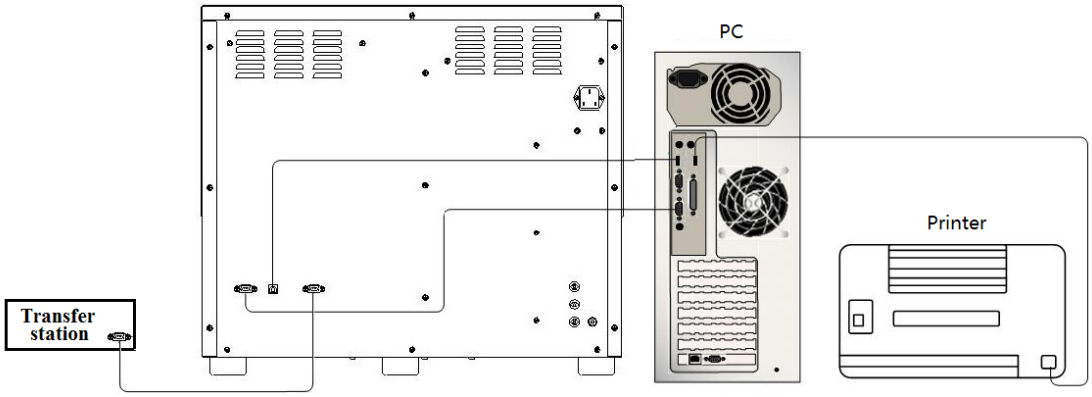


Figure. 4-2 Connection of external devices

Chapter 5 Settings

5.1Introduction

This analysis system has been initialized in the factory, and interface seen by user is the default value of the system after the initial startup. To meet different requirements in actual applications, some of the parameters of the analysis system may be set.

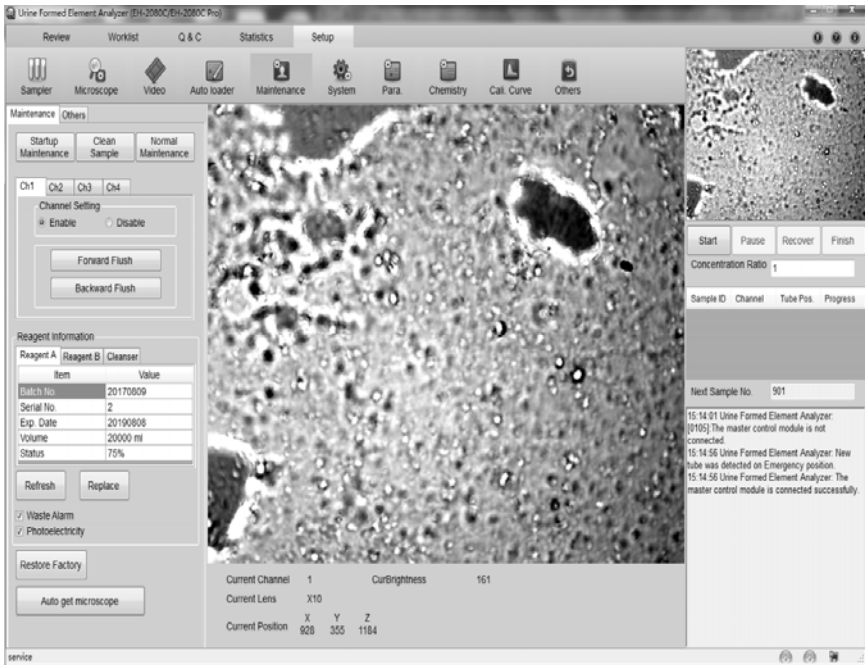
For the security of the settings and data, two access levels are provided to the operator of the analyzer. The administrator access level provides the operator with access to more functions or settings, some of which can be configured to be accessible to operators.

⚠ Warning

- If parameters are not set correctly, the operation of the analyzer may be damaged and analysis accuracy may be affected. It is recommended that parameters of the analysis system be set by the personnel authorized or trained by the manufacturer; the manufacturer will not take any responsibility for the damage to the analysis system due to the parameters set by the personnel not authorized or trained by the manufacturer.
- Administrator or his authorized common user has verification authority.

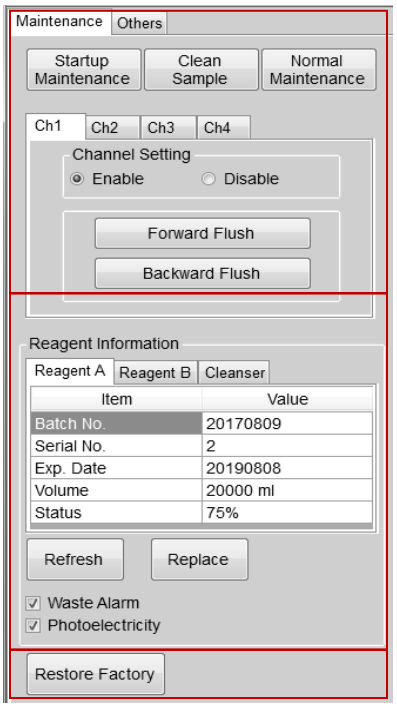
5.2Maintenance

Click ‘ Setup’＞ ‘ Maintenance’ to get into ‘ Maintenance’ screen, as shown in the following figure:



User can carry out the following operations on the ‘Maintenance’ screen:

- **Maintenance**
- **Reagent information**
- **Restore Factory**



5.2.1 Maintenance

The following operations are performed in maintenance stage:

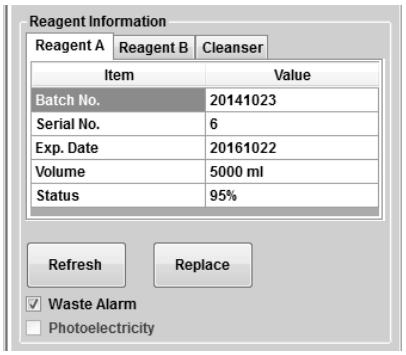
Function key	Description

Startup Maintenance	Execute startup maintenance
Clean Sample	Clean sample probe
Normal Maintenance	Choose channel for normal maintenance
Channel Setting	Choose channel for enable/disable
Forward Flush	Choose channel for forward flush
Backward Flush	Choose channel for backward flush

5.2.2 Reagent Information

After the system is started up for initialization, the system will automatically update the information from the interior computer to epistatic computer, and the reagent information is updated accordingly.

Reagent information screen



The following contents of RAGENT A, REAGENT B are displayed on the 'Reagent information' screen:

Display name	Description
Batch No.	Production date of reagent
Serial No.	The serial number that is used for reagent on that day
Exp. Date	Period of validity of reagent
Volume	The total volume of reagent carton
Status	Percentage of remaining amount reagent in total reagent

NOTE

- Reagent A and Reagent B are set by the system by default; waste is set by the user manually.

- After being used up, reagent A and reagent B need to be timely replaced; otherwise the instrument cannot continue.

Waste alarm

When 'Waste Alarm' is selected, the analysis system will judge if waste is full according to the float status on the waste container cap assembly of the waste; when waste is full, the following window will pop up to remind the user to dump the waste.

In this case, the user may click on 'OK' or ' 'to close this window to c replace the waste container (see section 10.6.4 for detailed operation).

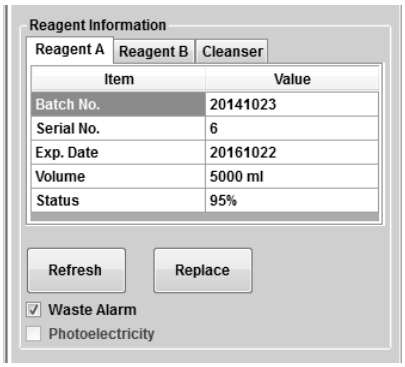
Photo-Electric

Check "Photo-Electric"; reagent alarm will be set to photo-electric alarm mode.

NOTE

- If 'Waste Alarm' is not selected, the analysis system will not make judgment according to the float status of the assembly of the waste; the user needs to regularly check waste container or to adopt direct drain method.
- If the user fails to replace the waste in time after closing this window, the window will pop up repeatedly prior to the inspection of each sample till the user replaces new waste container.
- After being used up, reagent A and reagent B need to be timely replaced; otherwise the instrument cannot continue.

Button Functions

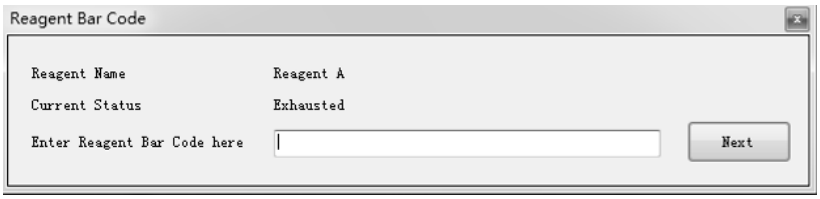


■ Refresh

When 'Refresh' is clicked on, the software will automatically update reagent information.

■ Replace

Click 'Replace', and the following dialog box will pop up:

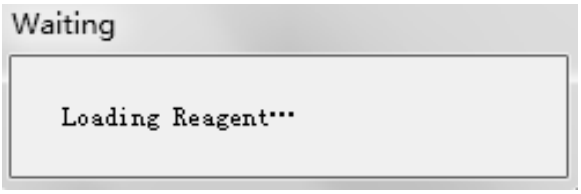


Enter the reagent bar code in the dialog box manually, click on 'Next' button or press Enter key to confirm, after the bar code verification is passed, the software will generate the type, production date, serial number, remaining amount and status of the reagent automatically.

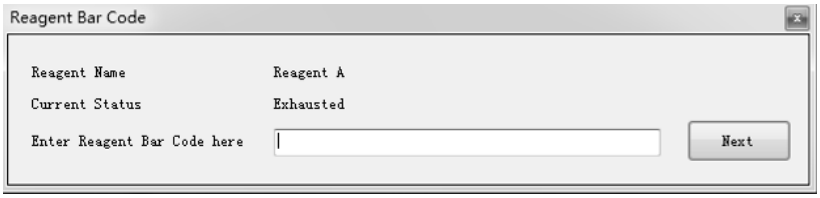
Reagent alarm

1. During check of instrument, the software will pop up the following dialog box:

2. Replace reagent (refer to 10.6.2 for detailed operations); click button "Replaced Already"; the software will automatically load reagent and display the following dialog box:



3. After loading of reagent, the following dialog box will pop up:



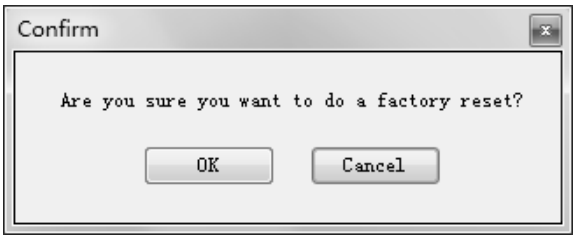
Please insert reagent IC card to replace reagent bar code. For particular method of operation, refer to the function "Replace" in 5.2.2.

NOTE

- The digit of the reagent bar code of is 22; if the digit is incorrect, the software will clear it automatically.

5.2.3 Restore Factory

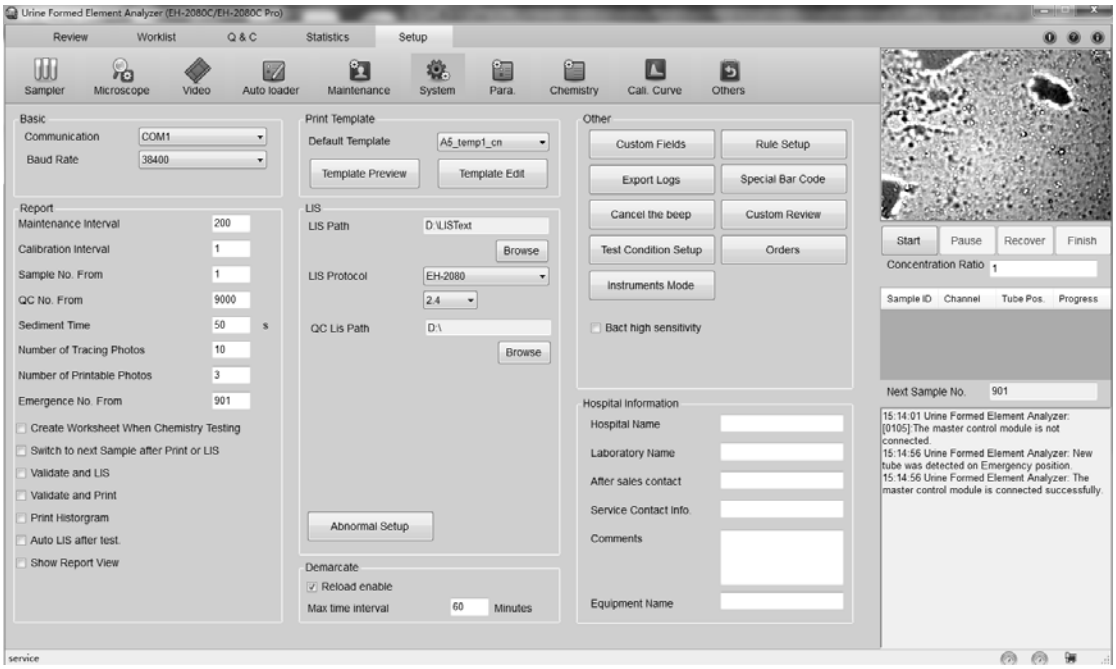
Click 'Restore Factory', the following dialog box will pop up.



Click 'OK', all the set parameters modified by the user will be restored to default values.

5.3 System setting

Click 'Setting'＞ 'system' to get to 'system' screen shown as follows:



The user can complete the following settings in ‘system’ screen:

- **Basic**
- **Report**
- **Print template**
- **LIS**
- **Other**
- **Hospital information**

5.3.1 Basic setting

Communication

Click the pull-down list to choose the desired communication ports from the following options:

- Not used
- Com 1
- Com X

NOTE

- COM X is an actual port which is automatically generated according to the position of the serial ports of the computer installed.

5.3.2 Report

Setting item	Description
Maintenance Interval	Maintenance is performed every several

	samples
Calibration Interval	Calibration is performed every several samples
Sample No. From	Starting number of sample
QC No. From	Starting number of QC sample
Sediment Time	The time of sedimentation
Number of Tracking Photos	The number of tracking photos
Maximum amount of Printable photos	The maximum amount of printable photos
Emergency No. From	Starting number of emergency sample
Linked Sample No. From Manual Input	The setting of this item is effective only under the cascading mode with 1905 instrument; if this item is ticked, the start sample number of each batch of samples shall be inputted by user manually when linked test is started.
Report interface, Switch to next Sample after Print or LIS	This item is only effective when “Show report View” item is enabled, if this item is ticked, software will automatically shift to next sample after sample executes printing /LIS operation.
Validate and LIS	Tick off this item, the software uploads LIS of this sample automatically after the sample is verified
Validate and Print	Tick off this item, the software prints out the report sheet of this sample automatically after the sample is verified
Print Histogram	Tick this item, software will print histogram in printing sample report
Automatically upload LIS after test	Tick this item, software will automatically upload LIS of this sample after sample is tested
Show Report View	Tick this item, software will provide report and review interface display; if this item is not ticked, software only provides review interface display

NOTE

- The parameter range of maintenance interval is [0,1000];
- The parameter range of the starting number of sample is [1,10000];

5.3.3 Print template

Default template

Click the pull-down list of “Default template” and select default print template from the list options.

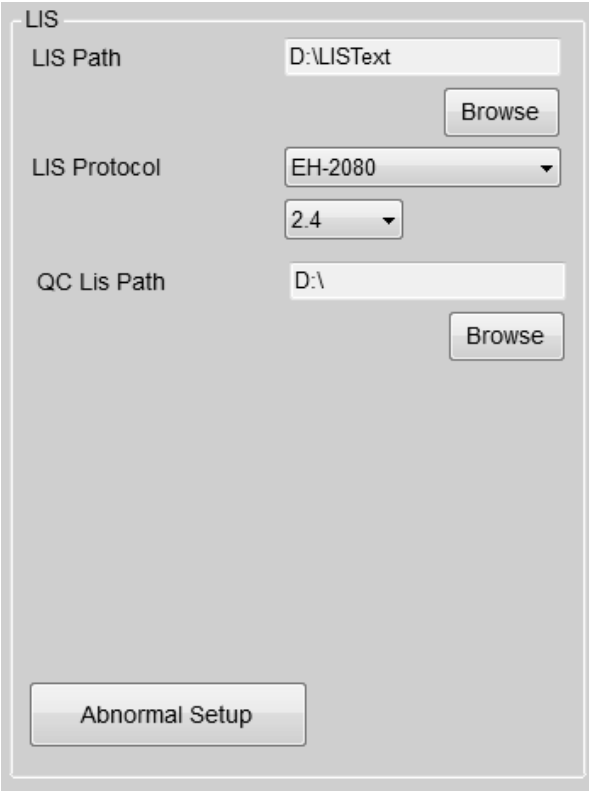
Template preview

Click “Template preview” button to preview the currently-selected default template.

Template editing

For detailed operations, please refer to Chapter 8.2.2.

5.3.4 LIS



LIS system export path

Click “Browse” and select the export path of data that needs to be exported to LIS.

NOTE

- LIS export path shall adopt English as much as possible.
- After LIS export path is set, generally, don’t modify this path, otherwise, it may cause LIS communication failure.

Work order path

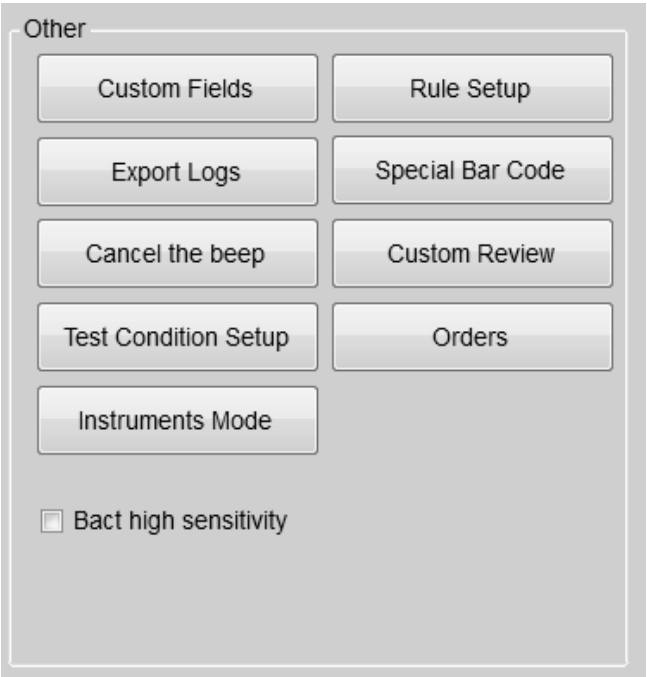
Click “Browse” to select the query path of work order data which needs to be downloaded from LIS system.

“Enable LIS work order”: tick this item, the software will read LIS work order under the specified path and confirm the working mode that should be executed by the sample;

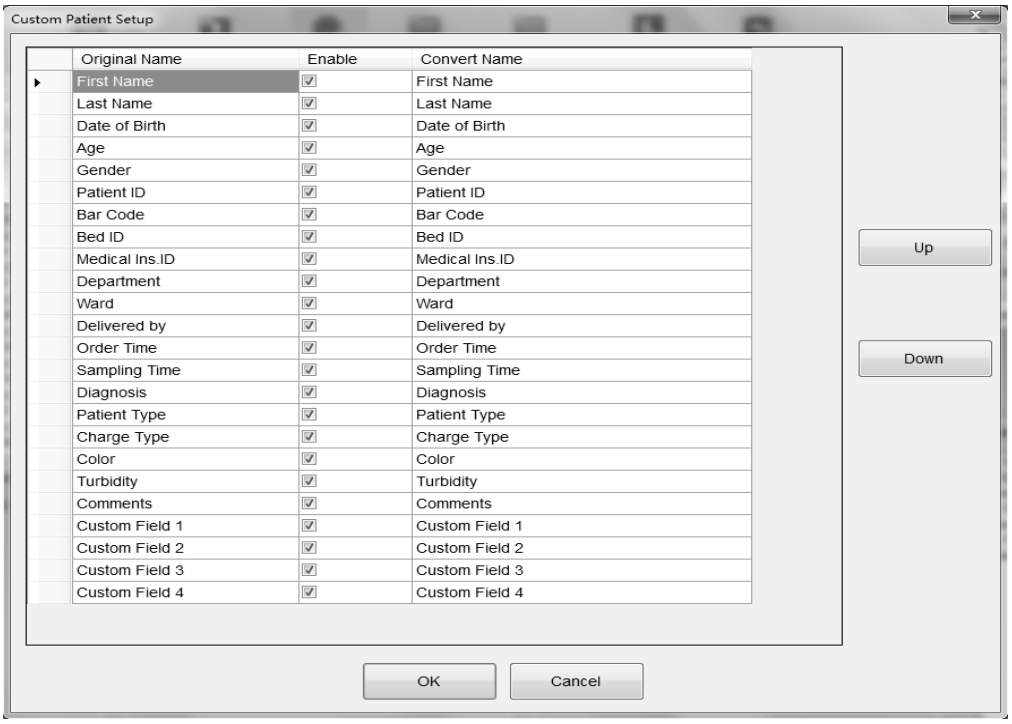
NOTE

- Work order path shall adopt English as much as possible.
- After the work order path is set, generally, don’t modify this path, otherwise, it may cause LIS communication failure.
- Work order path is effective only when “LIS worksheet enable” is ticked.

5.3.5 Other



Custom patient information field



1. Enabled column: Users can check and set the information field for all patients, if the field has been ticked off, the field will be automatically updated to the patient information area in the "work list" interface and "review" interface.
2. Convert Name: Users can input text on all the contents of any field interface, after the completion of the input, the content will be automatically updated to the patient information area in the "work list" interface and "review" interface.
3. Move up and down: Users can adjust the order of the patient information field, after the adjustment is completed, the “work list” interface and the “review” interface will be displayed in this revised order.

For example, set the " patient's symptom" field

Check the “custom field 1”, enter the “patient's symptom” in the corresponding box ".

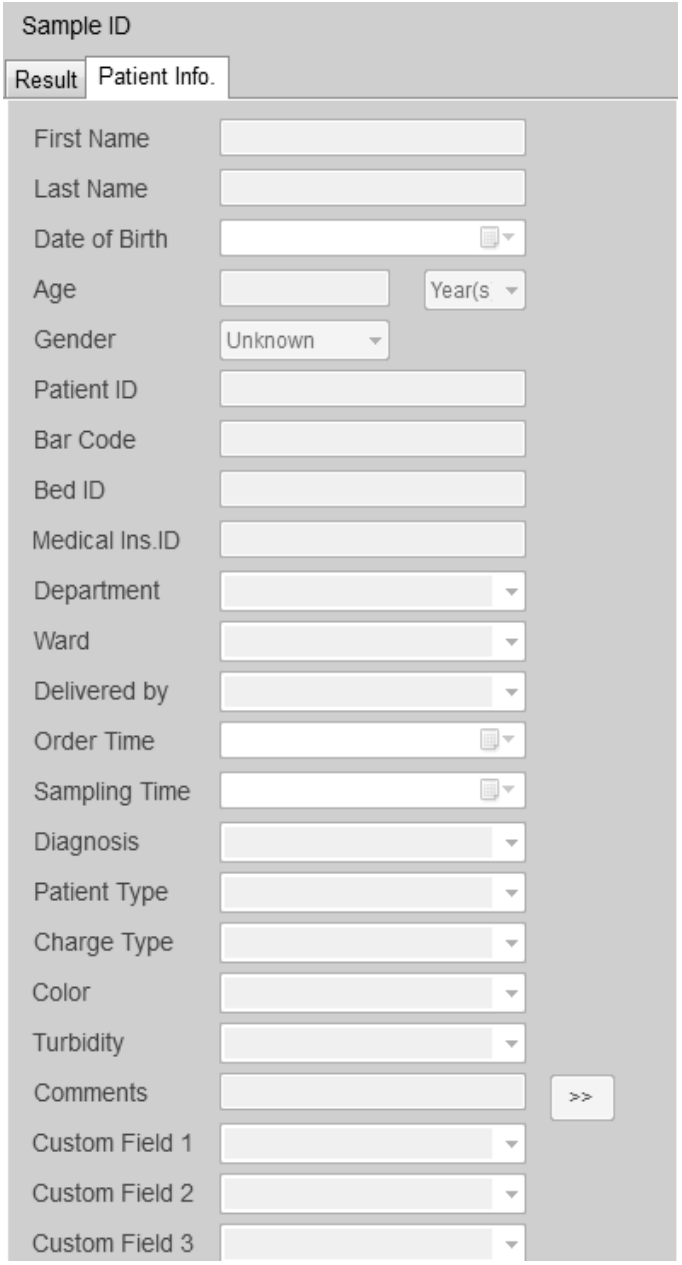
Custom Field 1	☑	patient's symptom
Custom Field 2	☐	Custom Field 2
Custom Field 3	☐	Custom Field 3
Custom Field 4	☐	Custom Field 4

After the completion of the input, the field of the patient information area will be automatically updated in the "work list" and "report" interface.

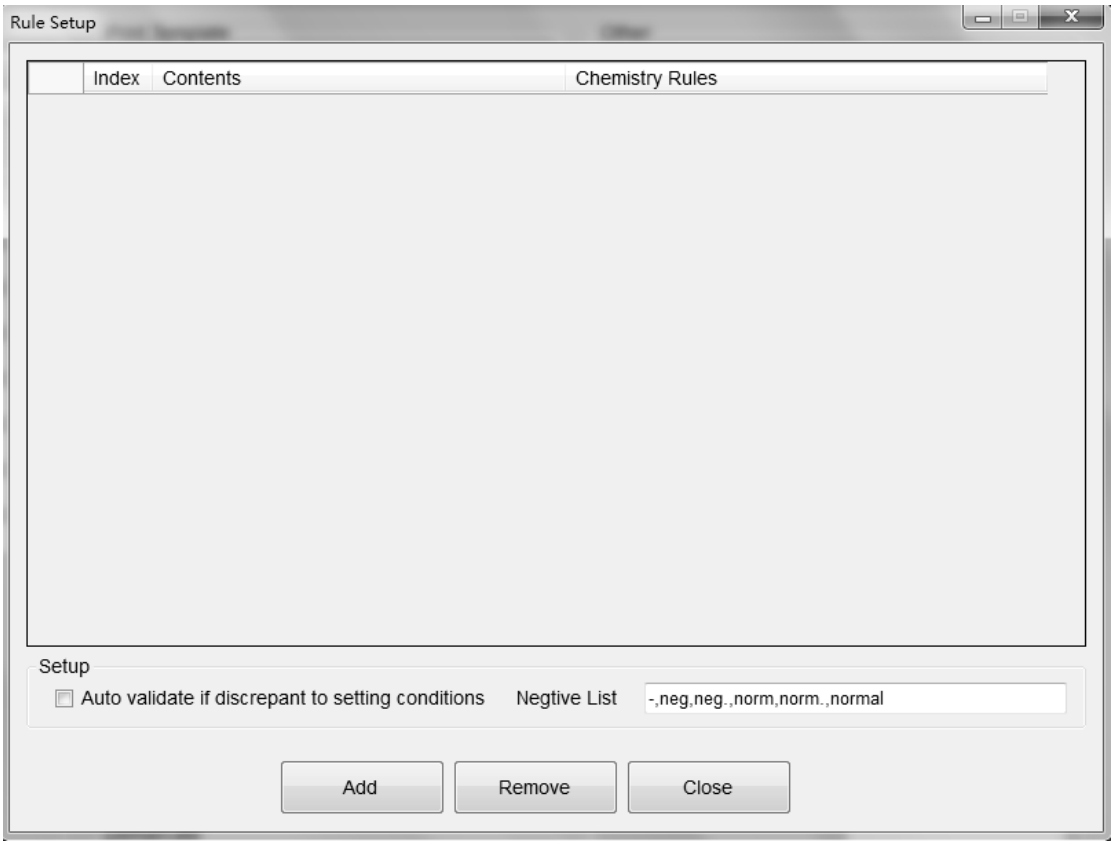
Patient information area in the “work list” interface

Sample ID	1		Emergency
First Name		Last Name	
Date of Birth		Age	Year(
Gender	Unknown	Patient ID	
Bar Code		Bed ID	
Medical Ins.ID		Department	
Ward		Delivered by	
Order Time		Sampling Time	
Diagnosis		Patient Type	
Charge Type		Color	
Turbidity		Comments	
Custom Field 1		Custom Field 2	
Custom Field 3		Custom Field 4	

Patient information area in the "review" interface



Rule setup



1. Setting

When “Auto validate if discrepant to setting conditions” is ticked, those samples which have executed urinary sediment and dry chemistry tests and fail to meet the retest conditions after the tests will execute automatic check operation.

Negative result table: It’s used to set character display content identifying negative result.

2. Add：It’ s used to increase new retest rules.

3. Remove：It’ s used to remove the existing retest rules.

Export Logs

Click “Export log” button, the software will automatically export the log file that is currently operated by the instrument.

Special Bar Code

Click “Special bar code” button, the software will provide setting which can judge whether to execute, terminate or skip the test according to the bar code.

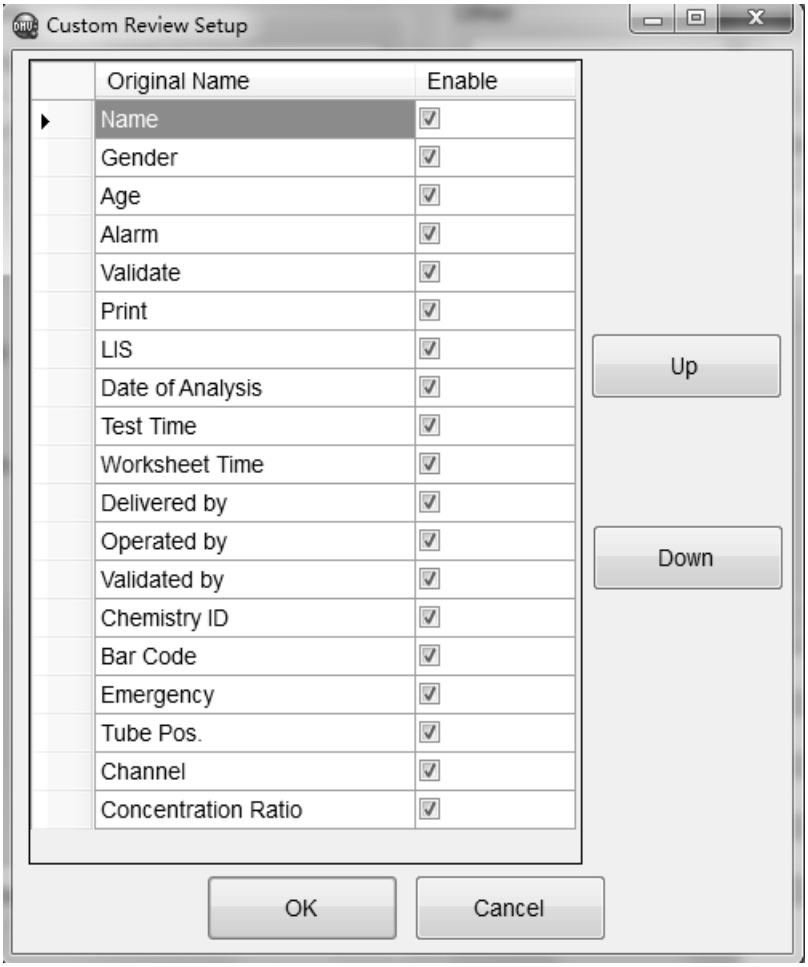
NOTE

- The setting of special bar code will come into effect only after “Start test tube bar code scanning” is ticked.

Cancel the beep

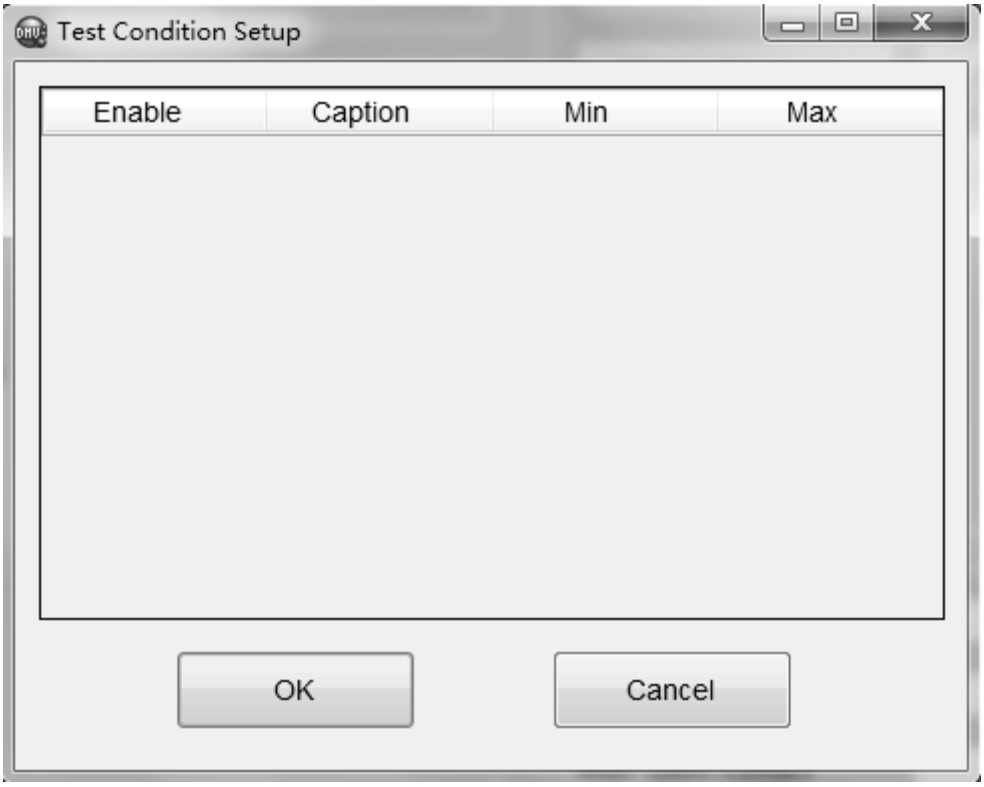
Click “Cancel alarm sound” button, the alarm sound of host will be eliminated.

Custom Review



1. Enable column: user can tick and set the fields in sample list header in “Review” interface; when this field is ticked, this field will be automatically updated to sample list header area in “Review” interface.
2. Move up and down: user can adjust the order of fields in sample list header in review interface; after the adjustment, the sample list header of “Review” interface will be displayed in this order.

Test condition setup



1. Test formed elements for all samples: After this option is ticked, all the samples will be tested.
2. Test formed elements for following conditions: the premise of setting conditions: first, link with dry chemistry; second, set up dry chemistry resolution protocol, tick the received data and the data options after resolution; third, send the corresponding dry chemistry result data of dry chemistry protocol; fourth, import the data, see dry chemistry setting function for details. If there are still no setting conditions in this interface, please restart the software to check.

If any of conditions enabled in this list is met, urinary sediment test will be conducted; if all the conditions are not met, the test will be skipped.

Cascading

If this option is ticked, cascading function will be opened.

Link with dry chemistry software

If this option is ticked, the cascading function with 1905 software will be opened.

Skip all the test tubes upon test

If this option is ticked, test will not be conducted for all the samples and the samples will be unloaded to unloading platform directly.

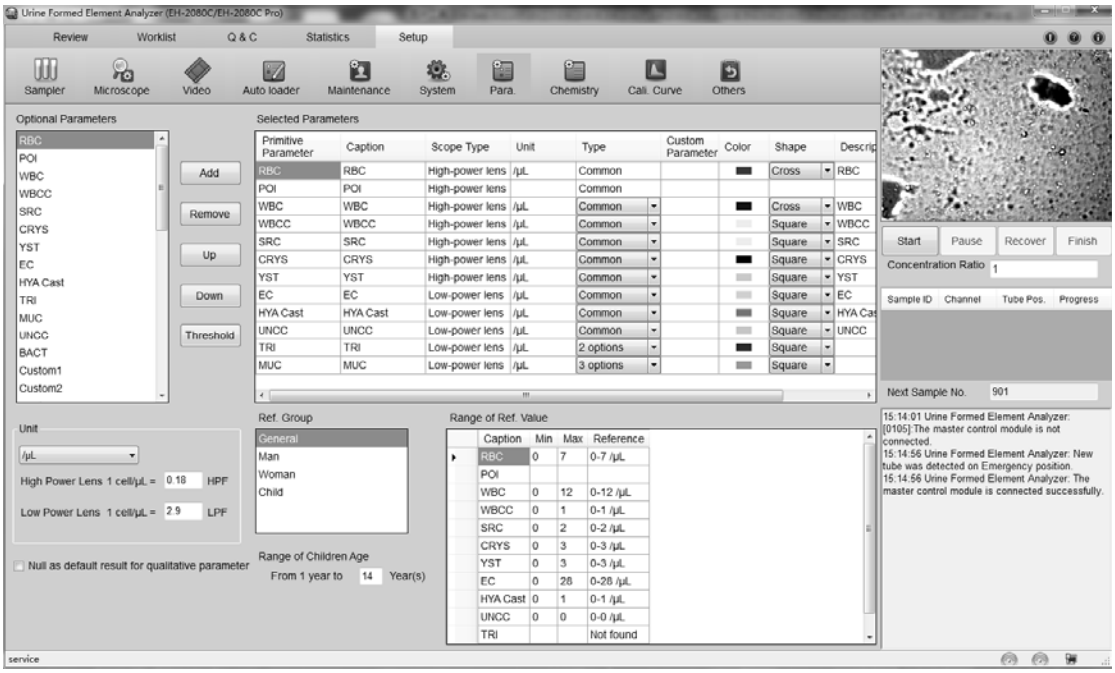
5.3.6 Hospital Information

The user can click o edit boxes to enter relevant lab information as required.

Information name	**Description**
Hospital Name	Name of the hospital where the analyzer is installed
Laboratory Name	Name of the lab where the analyzer is installed
After sales contact	Contact information of the supervisor who is in charge of the use of the instrument
Service Contact Info.	Service Contact Information
Comments	Comments information

5.4 Parameters Setting

Click on ‘Setting’＞ ‘Parameter’ to go to ‘Parameter’ screen shown as follows:



The user can complete the following setting on the ‘Parameter’ screen:

- **Optional parameters**
- **Selected parameters**
- **Reference group**
- **Range of reference value**
- **Range of children age**
- **Uints**
- **Threshold**

5.4.1 Optional parameters

Optional parameters include auto test parameters and customized test parameters of this analyzer.

The user can choose the test parameters from the following ‘optional parameters’:

Red blood cell, poikilocyte, white blood cell, WBC clump, small round cell, crystal, yeast, epithelial cell, hyaline cast, trichomonad, mucin, non-hyaline cast, bact and customize（1, 2, 3……30）.

5.4.2 Selected parameters

Definition

‘Selected parameters’ refer to the test parameters that can be chosen by user from the ‘optional parameters’ screen.

Change of parameters

User can change the name, applicable microscope, units and type of the customized test parameters on the 'Selected parameters' screen.

User can change the color, mark, shape and description of the auto test parameters of the analyzer on the 'Selected parameters' screen.

User may change the name of the selected parameter on the 'selected parameter' screen; after name change in the 'selected parameter' screen is completed, the name corresponding to it in the 'reference value range' screen is updated simultaneously.

Button functions

When choosing certain parameter in the 'optional parameters', click 'Add', this parameter will be added to the 'Selected parameters' screen and become 'Selected parameter'.

When choosing certain parameter on the 'Selected parameters', click 'Remove', this parameter will be deleted, and the list will be automatically refreshed.

The parameters in the 'Selected parameters' can be 'Top', 'Up', 'Down' or 'Bottom'.

NOTE

- The sequence of the parameters on the 'Selected parameters' screen is consistent with the sequence of the parameters on the 'report' screen.

5.4.3Reference group

Definition

The reference groups are set according to different sexes and ages, including four reference groups, i.e., general, children, man and women.

The children group refers to the reference group in which the age is smaller than setting;

The man group refers to the reference group in which the age is bigger than setting but the sex is male;

The women group refers to the reference group in which the age is bigger than setting but the sex is female;

The general group refers to the reference group in which the age is bigger than setting but sex or age (including 0 years old) has not been set.

Change of parameters

Parameters in the reference groups are consistent with the parameters on the 'Selected parameters' screen; if the user wants to add or remove any parameter, see section 5.4.2 for detailed operation.

User can change the maximum value and minimum value of the test parameters in each reference group, in this case, the range of reference value would be changed accordingly.

Application of parameters

The analysis system assign the reference groups in which patients are located according to patient information; when analyzing the result, the analysis system will automatically call the reference range of reference groups and will finally output the inspection result of samples.

5.4.4 Range of reference value

Definition

The range of reference values refers to the range of the normal values of various test parameters in the reference groups.

Change of parameters

Click and highlight the parameter to be changed from the 'range of reference values' screen; and then, change the maximum value and minimum value of the parameters.

NOTE

- The maximum value of reference values must be bigger than minimum value.

5.4.5 Range of children age

The purpose of this parameter is to distinguish the reference range of the inspection result of different ages; user may set the age range of the children by himself, the system defaulted age is started from 1 year old; the suggested range is '1~14' years old.

5.4.6 Units

In the drop-down box for unit selection, there are three kinds of unit modes,they are /μL, /μL + LPF/HPF and custom unit.

The default conversion unit of system is "/μL", only the result value of "/μl" unit is displayed in review interface.

When unit "/μL + LPF/HPF" is selected, the unit displayed in parameter interface is still /μL, LPF/HPF is the second unit, specifically related to high- and low-power microscopes; it's unified to be LPF for low-power microscopeand, HPF for high-power microscope. The conversion value of high- and low-power microscopes can be inputted through customization; after the setting, the result values of two units "LPF/HPF and "/μL" will be displayed in review interface respectively.

When unit "Custom" is selected, the unit displayed in parameter interface is a custom unit, namely one kind among /μL, /LPF and/HPF, user can customize the

setting; review interface will display the result values according to the units of various parameters.

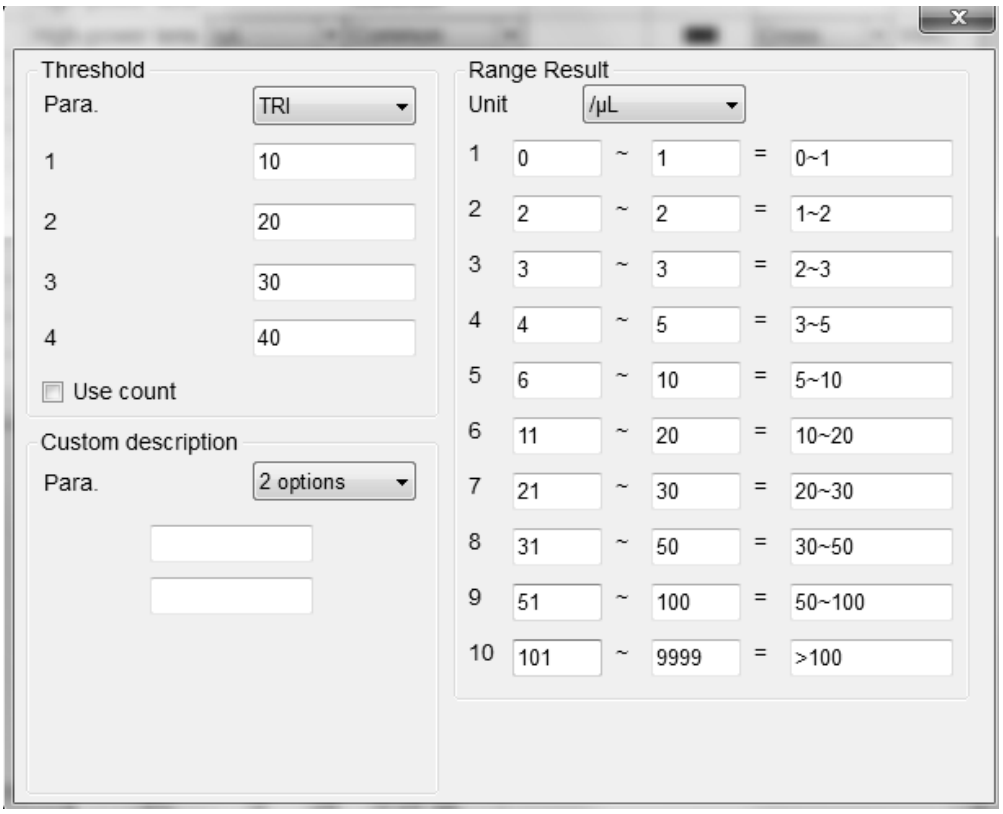
After the unit “Custom” is selected, the "selected parameters" interface will increase “scope result” and “no display of empty result” columns, providing the selection of whether to display scope results and whether to display empty results for various parameters in review interface. If “scope result” is ticked for a parameter, the result of this parameter will be displayed to be an interval result value in review interface, printing and LIS. If “no display of empty result” is ticked for this parameter, if the result value of this parameter is 0 or unseen, this parameter will not be displayed in printing and LIS.

NOTE

- When the parameter type is selected to uncommon, i.e. qualitative type, the unit will be modified to /μL automatically, which is not modifiable.

5.4.7 Threshold

Through clicking “Threshold” button, threshold setting interface can be opened, as shown below:



In threshold interface, the functions provided include：

1. Threshold scope of qualitative parameter can be set, providing selection for whether to divide threshold scope according to quantity.
2. Provide setting for “Custom description” of qualitative parameter.

3. Provide scope result display setting of three units.

5.5 Setting of Chemistry

Click ‘setting’＞ ‘Chemistry’ to go to the ‘Chemistry’ screen shown as follows:

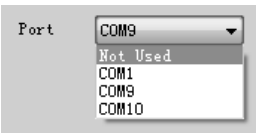
User can complete the following settings on the ‘Chemistry’ screen:

- **Chemistry configuration**
- **Units type**
- **Chemistry Analytic configuration**
- **Data monitoring**
- **Chemistry Match**

Connect Chemistry instrument and the Analysis System properly before setting the Chemistry.

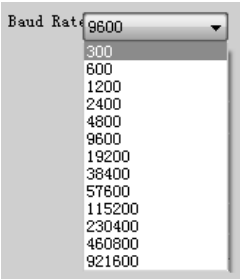
First step: set the configuration of the serial ports of Chemistry Port

Click the pull-down list of ‘Port’, and choose the desired port from the following options as shown in the following figure:



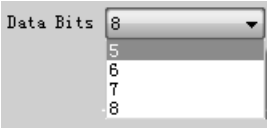
Baud rate

Click the pull-down list of ‘Baud rate’, and choose the desired baud rate from the following options as shown in the following figure:



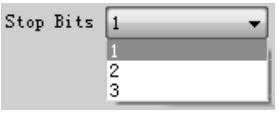
Data bit

Click the pull-down list of ‘Data bit’, and choose the desired data bit from the following options as shown in the following figure:



Stop Bits

Click the pull-down list of ‘Stop Bits’, and choose the desired Stop Bits from the following options as shown in the following figure:

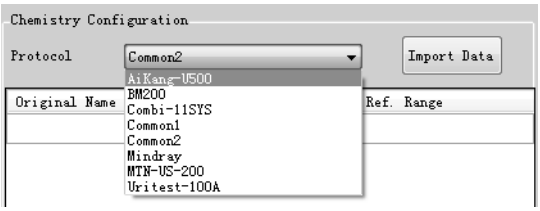


Check digit

Click the pull-down list of ‘Check digit’, and choose the desired check digit from the following options.

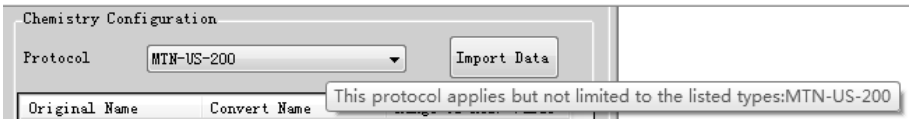
Second step: set the configuration of Chemistry analysis

Click the pull-down list of ‘Protocol’, and choose the desired protocol from the following options as shown in the following figure:



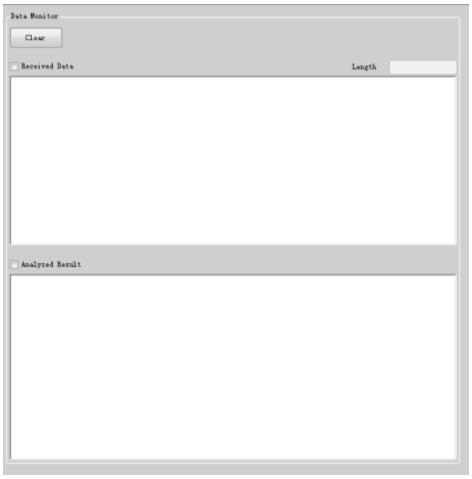
Customer in the pull-down list is the agreement customized by the customer service according to customer’s demand.

After the cursor of the mouse is held at the chosen protocol for 1 second, the analyzer model applicable to the protocol will be displayed, as shown in the following figure:



Third step: data monitoring

Tick off 'Received Data' as shown in the following figure:



After 'Received Data' is ticked off, when data is sent through the Chemistry instrument, the data will be automatically transferred to the first box of the above-mentioned figure, and the data length will be displayed automatically.

If user wants to see the analyzed data, tick off 'Received data' and 'Analyzed Result' at the same time. When data is sent through the Chemistry instrument, the Analyzed Result will be automatically displayed in the second box of the above-mentioned figure.

Fourth step: Export the data

Click 'Export data', the software will automatically export the name of the test item of the latest data on the 'Analyzed Result' screen into the left display area of the test parameters of the Chemistry.

■ Auto configuration

The analyzer automatically configures the initial item, conversion item and reference values of the test parameters of Chemistry in the display area of the test parameters of the Chemistry.

■ Change of parameters

User can change the conversion item and reference range of the test parameters of Chemistry.

The changed Chemistry item will be saved automatically; the latest change content will be called when Chemistry is configured the next time.

If the parameters are not changed, the initial item is consistent with the conversion item.

■ **Addition of parameters**

Add new parameters to the last blank line of the display area of the test parameters of the Chemistry; in this case, a blank line will be added under the new parameter while new parameter is being generated.

■ **Deletion of parameters**

Choose the parameter to be deleted, and Click 'Remove' or press [delete] key on the keyboard to delete.

■ **Up**

Choose the parameter to be upped, Click 'Up' button. The item will be upped a line.

■ **Down**

Choose the parameter to be downed, Click 'Down' button. The item will be downed a line.

NOTE

- If Chemistry data is not displayed in the diagram box due to fault, check if the connections of connecting wires and serial ports are correct; if these situations are excluded, check if the configuration of the serial ports of Chemistry instrument is set correctly.
- The test parameters of Chemistry vary with Chemistry instrument, so when a set of new Chemistry instrument is used, this instrument shall be reset according to the requirements described in section 5.5.
- The conversion item and the range of reference values that have been changed in the display area of test parameters of Chemistry will be used till the next change.

Fifth step: Chemistry match

The user based on their demands to decide whether to check the 'Auto Match'.

If you tick 'Auto Match', the software will auto match the untested sample chemistry; if you not tick 'Auto Match', you need manual match chemistry. (See section 6.6.2 for detailed operation)

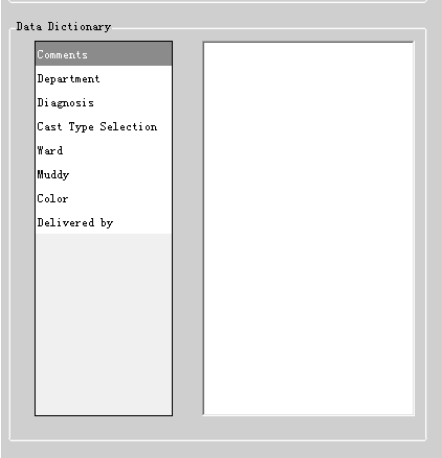
Sixth step: Units type

According to requirements, user can decide to tick one of them: "plus sign system", "international system of units" and "conventional system of units"

5.6 Others

5.6.1 Data Dictionary

The ‘Data Dictionary’ screen is used to set up the entry area of the sample information of ‘work list’ screen and the ‘Report’ screen, the relevant content of the shortcut entry of auxiliary patient information.



Edit ‘Data Dictionary’

1. Click item to be edited in the ‘Data Dictionary’ which will be highlighted.
2. Enter relevant content of the corresponding items in the box on the right of the ‘Data Dictionary’.

Setup the ‘corresponding shortcut code’

1. Click to select the item you want to edit in the ‘Data Dictionary’, this shows the item highlighted.
2. Input ‘corresponding shortcut code’ and ‘shortcut item name’ at the input box which is located at the right of the ‘Data Dictionary’

I.e. setup the ‘corresponding shortcut code’ of Department

Click the ‘Department’ to highlight it, input ‘6|department’in the input box at the right side. Take effect immediately.

On the page of worklist or report, input ‘6’ from the drop down of ‘Department’, then click [enter] key, the input box of department will appear ‘department’ automatically.

Hold down the [shift] and [\] to generate ‘|’.

‘6’ can be any number or letter.

5.6.2 User management

Level of common user

■ Change of password

The current log-in user can change the password of current log-in account.

Click 'Edit' on the 'User management' screen, the following dialog box will pop up:

Tick 'Modify Password' to change the password; revise the pass word and click 'OK' to save the password and close the dialog box.

NOTE

- User name cannot be null.
- User name shall only be letter and number.
- User name shall not be started with number.
- Only the current user can edit his user information.

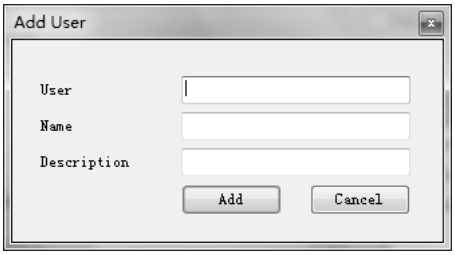
Administrator level

■ Change of password;

Refer to common user

■ Add user

Click 'Add' button on the 'User Management' screen, the following dialog box will pop up:

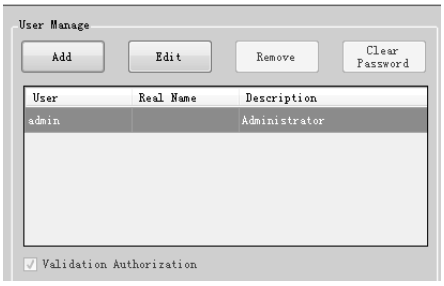


Enter relevant information in the edit box such as 'User', 'Name' and 'Description'.

After the entry, click on 'Add' to save the entry and close the dialog box.

■ Assignment authority

Select user on the 'User management' screen, the selected user column will be highlighted as shown in the following figure:



After a user is chosen, tick off the Validation Authorization; this authority will be effective immediately once ticked off.

■ Removal of user

Click 'Remove' on the 'user management' screen, the following dialog box will pop up:

Click 'OK', the removal succeeds; click 'Cancel', the removal is canceled.

■ Clear Password

When the 'Clear password' on the 'User Management' screen is clicked on, the following dialog box will pop up:

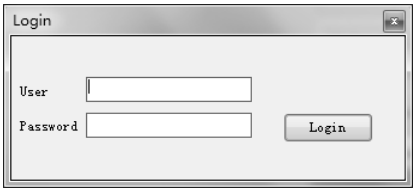
Click 'OK' and the removal succeeds; click 'Cancel' and removal is canceled.

NOTE

● Administrator can remove the password of common user but cannot change the password of common user.

■ Change user

1. Click on icon ‘ ’ on the software screen upper right corner, the prompt box shown below will be popped up



2. Enter name of the user to be changed over and the password in the prompt box, click on ‘Login’.

Chapter 6 Routine Operation

6.1 Introduction

This Chapter describes the entire routine operation process from the startup to the shutdown of the analysis system, and particularly describes the analysis and operation process of samples in different working modes.

See the following Fig. 6-1 for the flow chart of routine operation:

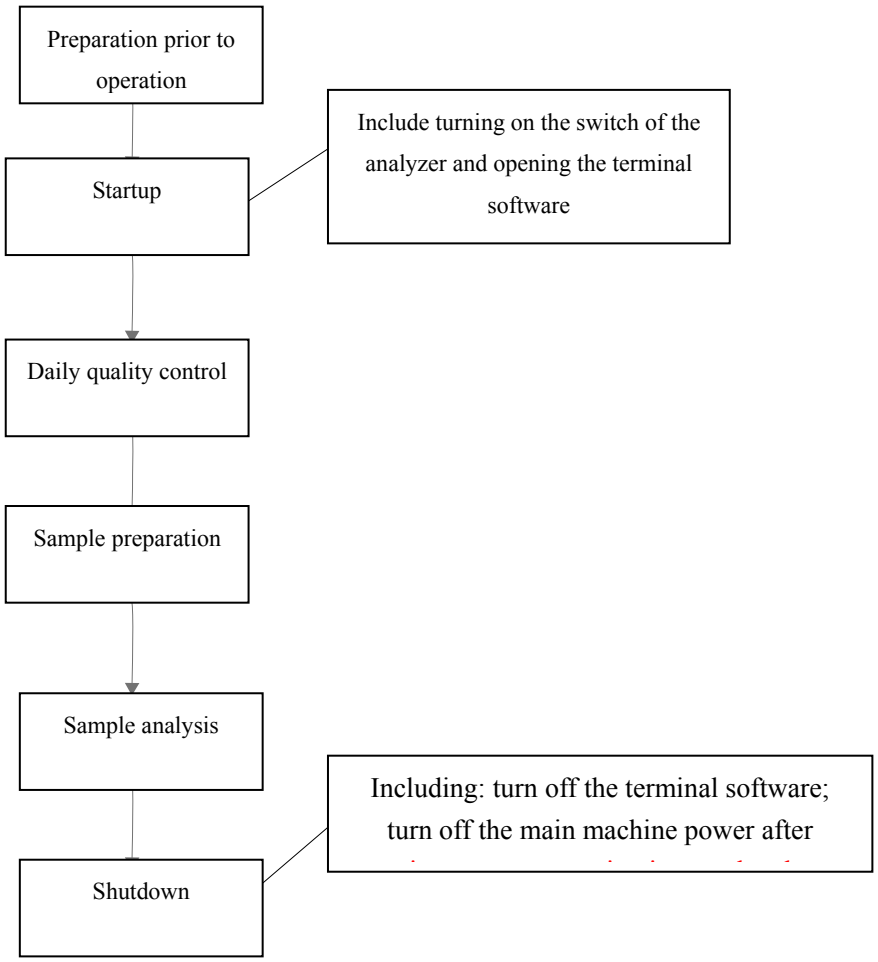


Figure. 6-1

6.2 Preparation prior to operation

Before power on the analyzer, user must carry out check as per the following requirements, so as to ensure that the system is ready for operation.

- The object to be tested with this project is urine. Since the concrete pollution degree or infectivity of urine cannot be determined, it is recommended that operators wear breathing masks and gloves during operation, and should be completely disinfected after operation.
- The substance such as waste liquid and waste sample that contain biocontamination must be treated according to local regulations.

NOTE

- Operator shall use the reagent designated by the manufacturer, and shall store and use reagent strictly according to operator's manual.
- Before this analysis system is used, check and ensure that the reagent is correctly connected.
- Before the reagent is used, it shall be stationary placed for a period of time till being stable.

Check waste container

Operator must provide waste container, and must ensure that the waste container have been emptied prior to daily startup.

Check liquid tube line and power supply

Check and make sure the reagents, waste tubes are properly connected and not bent.

Check and make sure the power cords of the analyzer are properly plugged into the power outlet.

Check printer (optional)

6.3 Startup and user log-in

Start up the machine

1. Place the power switch at the left side of the analyzer in the ON position (I). The power indicator light will be on.

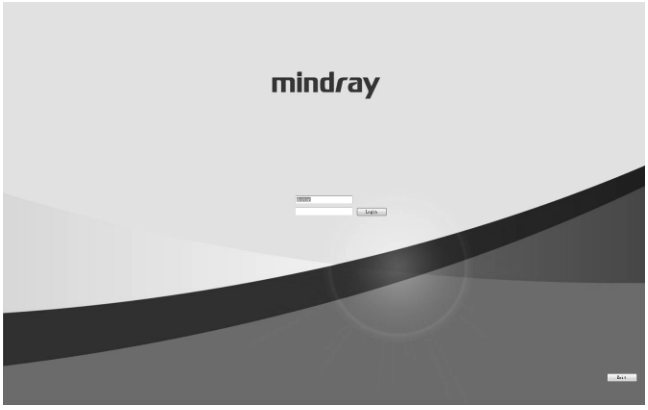
2. The analyzer will automatically perform the self-test procedure.

Start up the external computer and run the analyzer software

1. Power on the PC and display.

2. Get into the operating system after startup is complete; and then, double click the 'EH-2080 Urine Formed Element Analyzer' icon to run the specified software installed.

3. Get into the startup screen after the software is initiated; the screen is as shown in the following figure:



4. Enter correct user name and password on the startup screen.

5. Click 'OK' button or the [Enter] on the keyboard to enter the software interface; if you do not want to log in, click 'Exit'.

6. After entering the software screen, the instrument starts liquid line initialization automatically, the prompt box below is popped up on the screen:

7. After liquid line initialization is completed, the instrument starts microscope initialization automatically, the prompt box below is popped up on the screen:

8. Only after the liquid line initialization and microscope initialization are completed, the user can carry out the inspection operation.

NOTE

- The system judges whether the access of the operator is administrator or common user according to the user name and password used at log-in, and enables different functions in various interfaces according to user access.
- There is no restriction on the launching order of the analyzer software and PC software.
- If you fail to launch the software after tried for many times, please contact the manufacturer customer service department or your local distributor.
- After the startup procedure completes, check if the date/time on the analyzer and the PC are correct.

6.4 Daily Quality control

Before running any samples, run the controls. See Chapter 9 'Quality Control' for the detailed operation method of quality control.

6.5 Preparation of sample

- The object to be tested with this project is urine. Since the detail pollution degree or infectivity of urine cannot be determined, it is recommended that operators wear breathing masks and gloves during operation, and should be completely disinfected after operation.
- The substance such as waste and waste sample that contain biocontamination must be treated according to local regulations.

Caution

- Do not re-use disposable products

1. Prepare test tube

Requirement of test tube: the diameter of test tube is 15mm~16mm and its length shall not exceed 120mm.

2. Pour sample into test tube

■ Demand of sample amount

After sample is fully mixed, pour it into the test tube till the level reaches 2ml or above.

■ Requirement of sample

Use fresh urine sample; if the urine sample has not been tested in an hour, seal the urine sample and store it under refrigeration condition. The sample needs to be restored to room temperature prior to test.

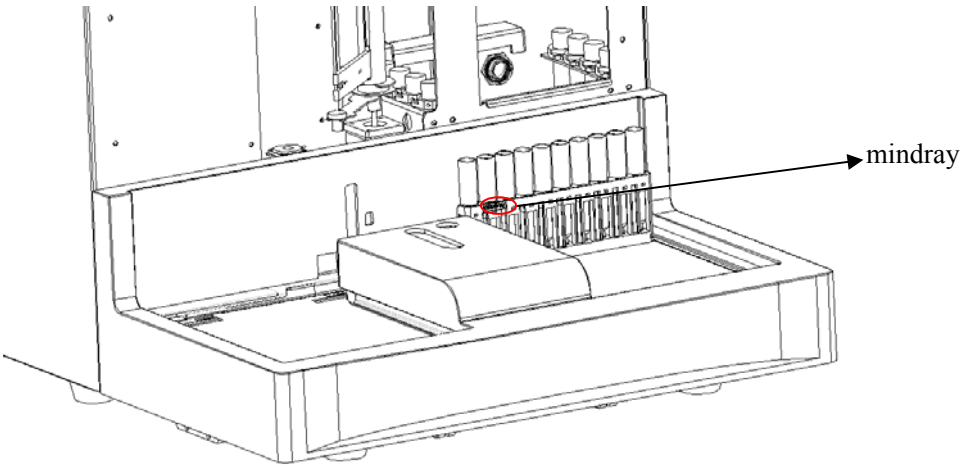
Do not add preservative, disinfectant or detergent into urine sample. Urine sample shall be kept away from the exposure of direct sunshine.

3. Put sample onto test tube rack.

Place the sample tube on the racks vertically, or it will impact the movement of sample probe.

4. Placement of test tube rack

Place the test tube rack in the automatic sampler loading area. User should be able to see mindray. As shown in figure below:



6.6 Sample analysis

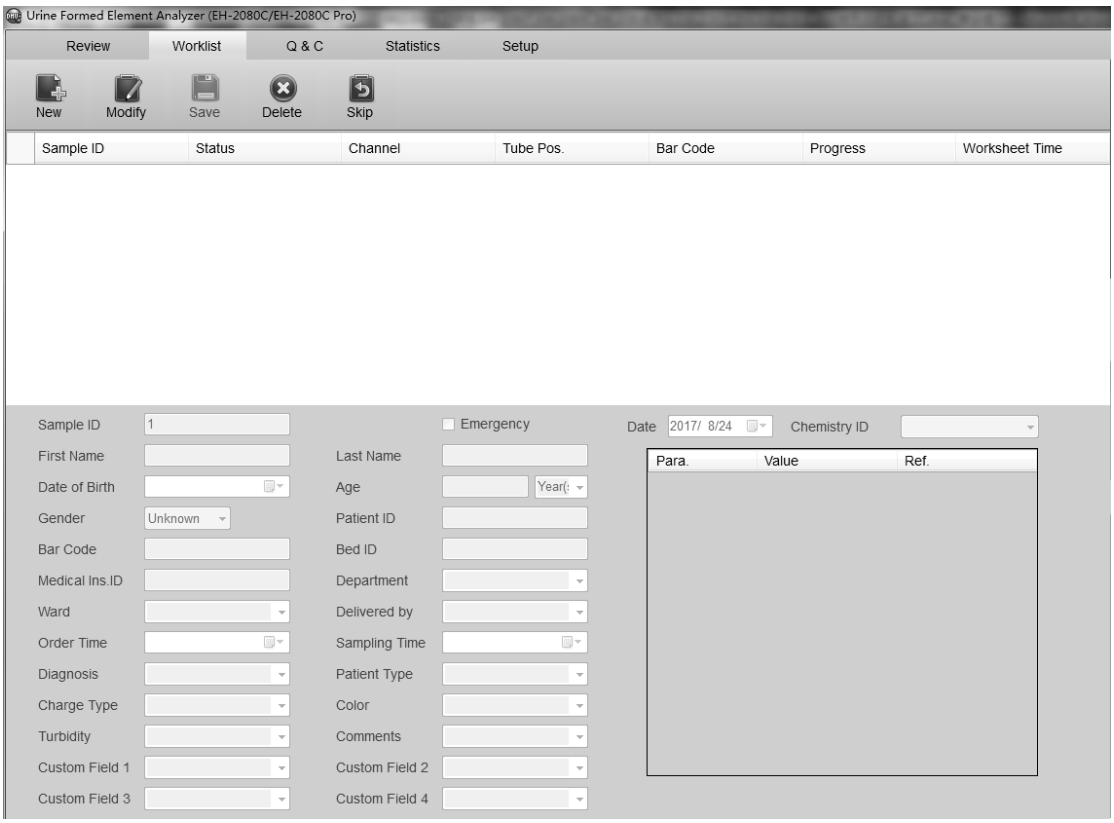
6.6.1 Enter worklist information

NOTE

- If the sample is emergency sample, tick off the ‘Emergency’ box.’

Entry of worklist prior to inspection

Click ‘worklist’ button to go to ‘worklist’ screen.



Click ‘Create’, a new entry will be built in the list area, and then you can enter the complete information in the data entry area below.

■ **Input sample No.**

Input sample ID in the ‘Sample No.’ box respectively.

The default ‘Sample ID’ will increase by 1 when you add a new entry again.

■ **Input first name**

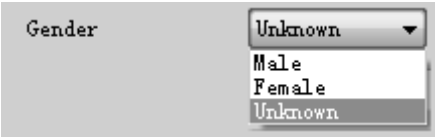
Input patient’s name in the ‘first name’ box.

■ **Input last name**

Input patient’s name in the ‘last name’ box.

■ **Input gender**

Choose the patient's gender from the pull-down list.



■ **Input age**

Input age in the 'age' box; the defaulted unit of age is year old.

■ **Input bar code**

Input bar code number in the 'bar code' box.

■ **Input patient ID**

Input the patient ID in the 'patient ID' box.

■ **Input medical insurance**

Input medical insurance number in the 'medical insurance' box.

■ **Input Bed No.**

Input the Bed No. of patient in the 'Bed No.' box.

■ **Input ward**

Choose the ward name of patient in the 'ward' box or click the pull-down list to choose the ward name of patient.

■ **Input department**

Input the name of order department in the 'department' box or click the pull-down list to choose the name of order department.

■ **Input draw time**

Choose the draw time of sample in the 'draw time' box or click the pull-down list to choose the draw time of sample.

■ **Input deliverer name**

Choose sample submitter in the 'delivered by' box or click the pull-down list to choose sample submitter.

■ **Input diagnosis**

Input diagnosis type in the 'diagnosis' box or click the pull-down list to choose diagnosis type.

■ **Input color**

Input the color of sample in the 'color' box or click the pull-down list to choose the color of sample.

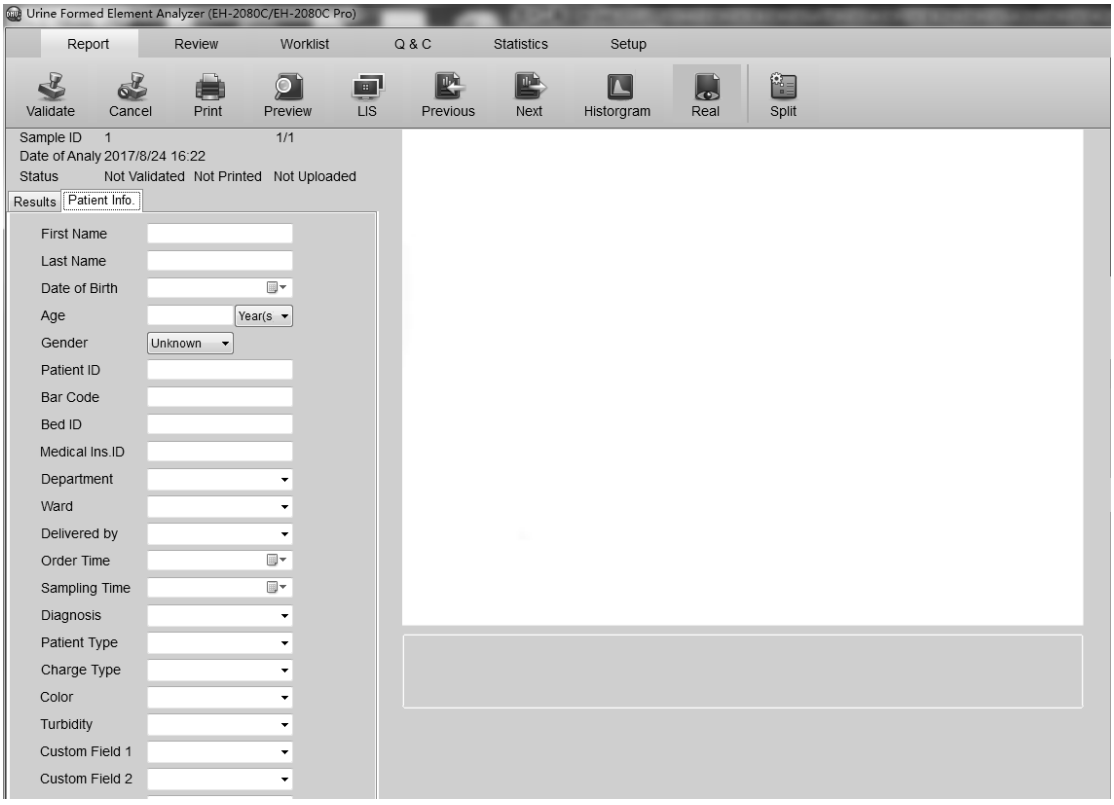
■ **Input Muddy**

Input the muddy of sample in the ‘Muddy’ box or click the pull-down list to choose the muddy of sample.

Entry of working list after inspection

If a sample has not been entered in worklist prior to inspection, the software will automatically generate a copy of worklist for this sample; after this sample is inspected, this working list will be automatically transferred to the report screen. User may reenter this worklist in the report screen.

Click ‘report’＞ ‘patient information’ to go to ‘patient information’ screen as shown in the following figure:



User can enter the name, sex and the like of the applicable patient on the ‘patient information’ screen. See ‘entry of worklist prior to inspection’ for detailed operation.

6.6.2 Matching Chemistry information

Setting-up of Chemistry

See section 5.5 for detailed operation.

Matching Chemistry information

Setting-up of Chemistry shall be completed prior to matching Chemistry information.

There are three methods to match Chemistry information:

■ **Matching Chemistry prior to inspection**

Click ‘Create’ under ‘working list’, and choose the ‘Chemistry ID’ corresponding to the sample from the pull-down box beside the ‘Chemistry ID’. The parameter, result value and reference value of the Chemistry are displayed in the list under the Chemistry ID accordingly.

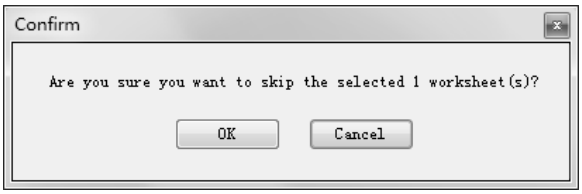
■ **Matching Chemistry after inspection**

Click the pull-down box beside the ‘Chemistry ID’ and choose the Chemistry corresponding to the sample ID on the ‘report’ screen. The parameter, result value and reference value of the Chemistry are displayed in the list under the Chemistry code accordingly.

■ **Only Match Chemistry and do not inspect urinary sediment**

If user conducts the matching of Chemistry other than urinary sediment inspection, operation steps are as follows:

1. Create new worklist (see ‘Entry of worklist prior to inspection’）.
2. Match Chemistry（see point 1 of this section）
3. Choose the worklist that matches with the Chemistry, and click ‘skip’ button; in this case, the following dialog box will pop up:



4. Click ‘OK’, this worklist will skip over inspection and automatically go to review and report screens.

6.6.3 Sample inspection procedure

- The object to be tested with this project is urine. Since the detailed pollution degree or infectivity of urine cannot be determined, it is recommended that operators wear breathing masks and gloves during operation, and be completely disinfected after operation.
- The substance such as waste and waste sample that contain biocontamination must be treated according to local regulations.

⚠ Warning

- The sample probe tip is sharp and may contain biohazardous materials. Exercise caution to avoid contact with the probe when working around it.

⚠ Caution

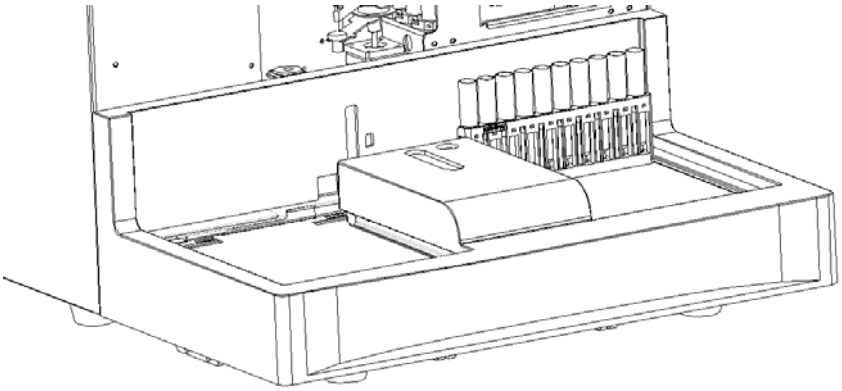
- Do not re-use disposable products.

NOTE

- Sample shall be calibrated prior to inspection; inspection cannot be carried out without calibration.

1. Enter the work list information (For detailed operation, refer to 6.6.1).

2. Place the test tube rack containing samples onto the loading area, as shown in figure below.



For the samples requiring dyeing, just place the dyeing agent in the dyeing position.

3. Select the sample type and inspection mode on the software screen and click on 'start' or press start button on the panel, the instrument starts inspection automatically.

4. The test tube rack will be delivered to unloading area after all the sample on the rack finish analysis.

5. Take away the test tube rack in the unloading area, and dispose the remaining waste liquid as per the local regulations.

6. After adding sample ,the instrument will realize automatic identification ,mixing ,quantitative sampling and counting ,chamber sampling and sedimentation ,finally ,the sample is analyzed and the sample is processed after inspection .

7. After adding sample ,the instrument will automatically control microscope stage , and realize the automatic switching of high-low images , automatic return ,automatic focus ,automatic scanning ,automatically adjust the brightness and automatic position of the microscope these above functions. Finally producing the photos and automatically recognize them ,coming into being sample results.

8. The sample results after analysis are saved to the review, report screen automatically.

NOTE

- The test tube rack cannot cover the light coupler.

6.6.4 Special functions

Analysis Result Saving

The analyzer saves analysis result automatically. When the quantity of sample result reaches the upper limit of storage, the new sample analysis result will automatically cover the previous sample analysis result.

Parameter flags

This system is available with the function of the photograph and storage of image; low power lens can automatically identify epithelial cell, cast and granule and can position through coordinates, and can screen and check negative samples; high power lens can carry out automatic coordinate tracing, and can identify and interpret other visible components; this system can also automatically count and convert the test result (/μL)of visible component, and can compare the test result to reference values; If any test value exceeds the range of reference values, the 'Flag' column on the review screen will be indicated in form of symbol '↑'; the parameters that exceed the range of reference values will be displayed in red font on the 'report' screen.

Emergency treatment

This system is provided with emergency treatment function to provide priority inspection for the samples need to make emergency inspection.

Place the emergency samples on the emergency position, after the samples under inspection are finished the inspection, the instrument will turn to the latest emergency work list and inspect the samples on the emergency position automatically.

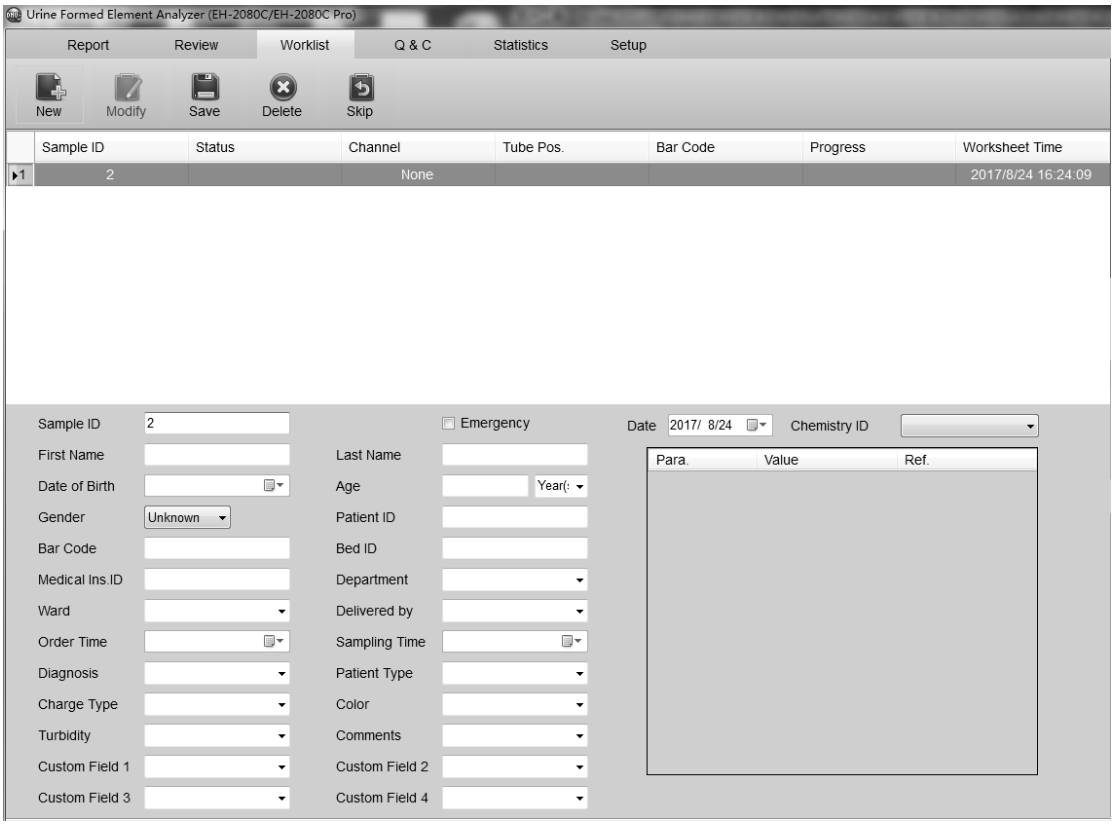
Instrument indicator-lamp and alarm tone control

Instrument status	Indicator-lamp status	Remarks
Instrument is fault, not operating	Red lamp flashing	Alarm tone prompt
Instrument is fault, operating	Green lamp flashing	Instrument operation is normal, and the results are not affected
Instrument reagent is used up, suspended	Yellow lamp flashing	Alarm tone prompt
Dick space is insufficient, suspended	Yellow lamp flashing	Alarm tone prompt

Instrument starts self-check	Green lamp flashing	/
Instrument is at standby status	Green lamp is constantly ON	/
Emergency operation status	Green lamp flashing	Emergency sample inspection
Instrument is free of fault, operating	Green lamp is constantly ON	/

6.7 Overview for worklist

Click ‘Worklist’ button to go to ‘Wordlist’ screen.



6.7.1 Introduction of the screen

The 'worklist' screen is mainly consisted of worklist area and the entry area of sample information.

User can add, modify, save, delete and skip over the working sheet in the working list area; the user can enter sample information and patient information and match Chemistry information in the entry area of sample information.

NOTE

- The worklist at the top of the screen can store a maximum of 4000 items of record.
- The field information in the list except for serial number, 'status' and 'entry time' is originated from the information entry area at the bottom of the screen.
- The information consistent with current records (i.e., records that are identified with arrow in the worklist) is displayed in the information entry area.
- The number of the sample being inspected cannot be changed.

6.7.2 Operation functions

The user can carry out the following operations on the 'Worklist' screen:

Create

Click 'Create', a new entry will be built in the list area. (See section 6.6.1 for detailed operation).

Modify

1. Click the worklist to be changed, allowing it to be highlighted.
2. Click 'modify' button, and change the information of this sample in the information entry area below this record.
3. Click 'Save' after change is complete; if the user executes other operations rather than clicking 'save', the system will pop up a dialog box, reminding the user whether to save the worklist.

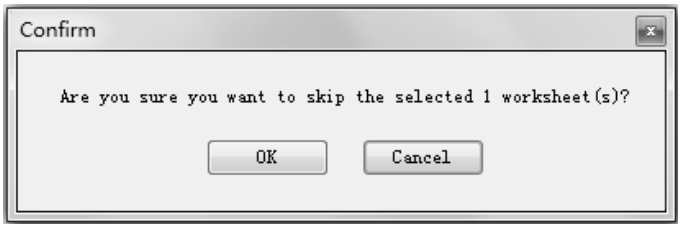
Delete

1. Choose an item or several items of worklist to be deleted.
2. Click 'Delete'.
3. The following prompt box will pop up:

4. Click on 'OK' to delete; meanwhile, the working list is refreshed.

Skip

1. Choose an item or several items of worklist to be skipped over.
2. Click 'Skip'.
3. The following prompt box will pop up:



4. Click 'OK' to execute skipping operation; meanwhile, the worklist is refreshed. As for the worklist that has been skipped over, inspection will not be performed; instead, the worklist will automatically go to the review and report screens.

NOTE

- When several items of worklist are to be deleted and skipped over, depress [Ctrl] or [Shift] to choose multiple items.

6.8 Shutdown

- The object to be tested with this project is urine. Since the detailed pollution degree or infectivity of urine cannot be determined, it is recommended that operators wear breathing masks and gloves during operation, and be completely disinfected after operation.
- The substance such as waste and waste sample that contain biocontamination must be treated according to local regulations.

- The sample probe tip is sharp and may contain biohazardous materials. Exercise caution to avoid contact with the probe when working around it.

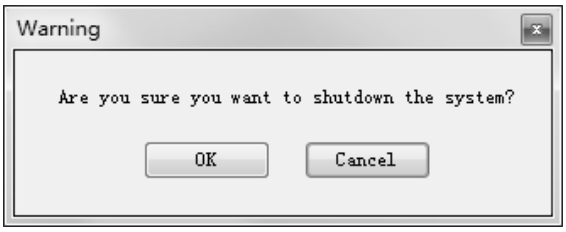
NOTE

- To ensure the stability of the analysis system and the accuracy of analysis result, please shut down the system for half an hour after 8 hours of continuous operation, and then, restart the system.
- Be sure to shut down the analyzer strictly as instructed below.

Shut down the analyzer:

6.8.1 Exit the terminal software

1. Click ‘X’ on top right of the screen of the screen, the following dialog box will pop up:



2. Click ‘OK’ to exit the system.
3. The analyzer carries out shutdown maintenance before exiting the system.

6.8.2 Shutdown the external computer

Click ‘start/shut computer/shutdown’ to shut down the computer; and then, shut down the display and printer in sequence; shut down the analyzer after it completely stops operation, i.e., the shutdown maintenance procedure is complete.

6.8.3 Power off the analyzer

After the operation of the instrument is ended, turn off the power supply of the analyzer.

Chapter 7 Result Output

7.1 Introduction

After every analysis cycle, the analyzer will save the analysis result automatically in the sample data. Up to 10000 analysis results can be stored in the sample data.

User may review the analysis result, edit and process the result report.

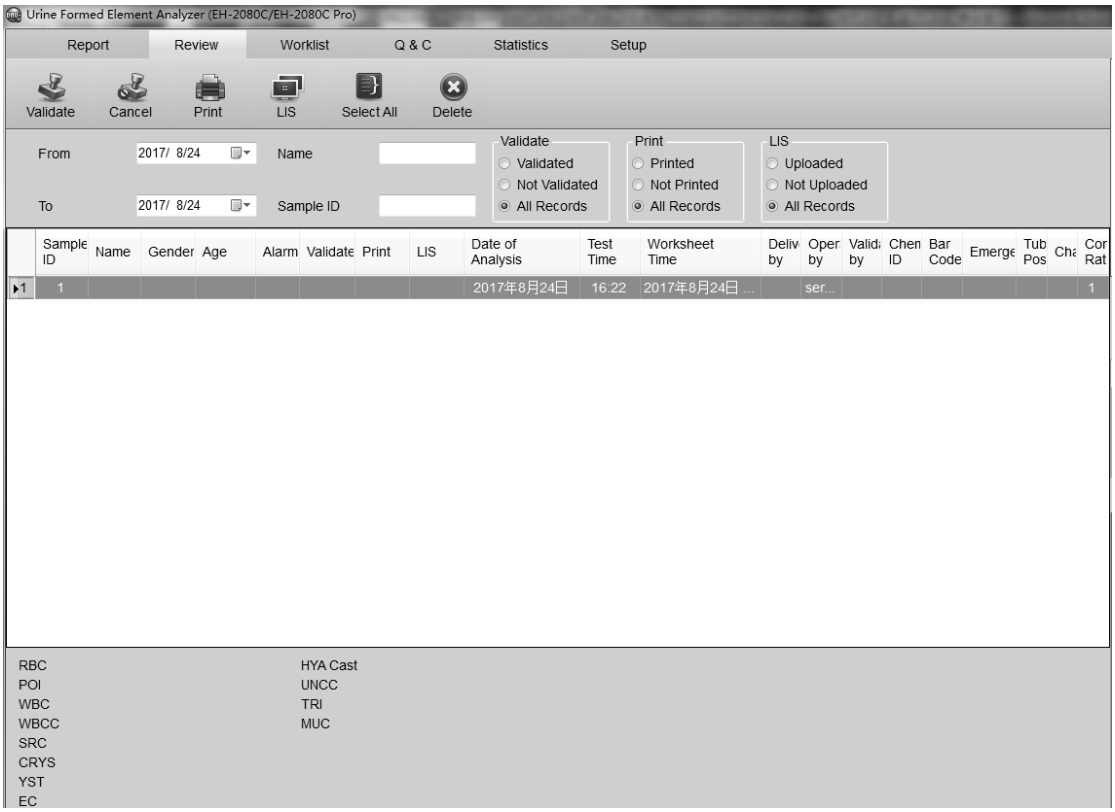
7.2 Review/Report

NOTE

- When “Display report interface” isticked in system setting interface, the software provides review interface and report interface for sample details and contents.

User can view, verify and search the historical results on the review screen.

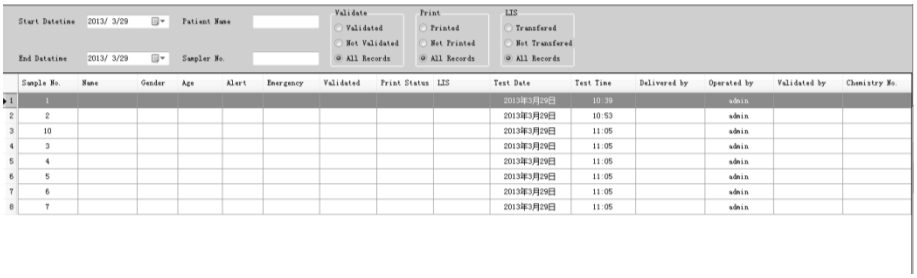
Click ‘Review’ to go to the ‘Review’ screen as shown in the following figure.



7.2.1 Introduction to review interface

List area

The lists of the analyzed samples under designated condition are displayed in the list area, including the basic information of sample, such as sample ID, name, gender and age of patient, flag, emergency treatment sample, verification statue, print statue, LIS, inspection date, inspection time, sample submitter, sample inspector, verification person and Chemistry ID.



■ **Change to report screen**

The existing screen can be changed to the report screen by clicking certain sample information.

Result display area

Display the test result of sample and the test parameters of Chemistry.

7.2.2 Button Functions

Validate

The selected sample record(s) to be validated in the list area will be highlighted.

Click the ‘Validate’ button to validate, and the check boxes of validated sample(s) will be ticked.

Cancel

The selected sample record(s) which are not validated in the list area will be highlighted. Click the ‘Cancel’ button to remove the validation check mark(s).

Print

Choose the sample records to be printed in the list area, and then click ‘Print’.

LIS

Choose the sample information to be transmitted.

Click ‘LIS’ pushbutton to upload sample information to LIS.

Select all

Click ‘Select all’; all the sample records in the list area are selected.

Delete

Choose an item or multiple items of sample record to be deleted in the list area; click ‘Delete’, in this case, the following dialog box will pop up:

Click ‘OK’ to delete the chosen sample and close the dialog box.

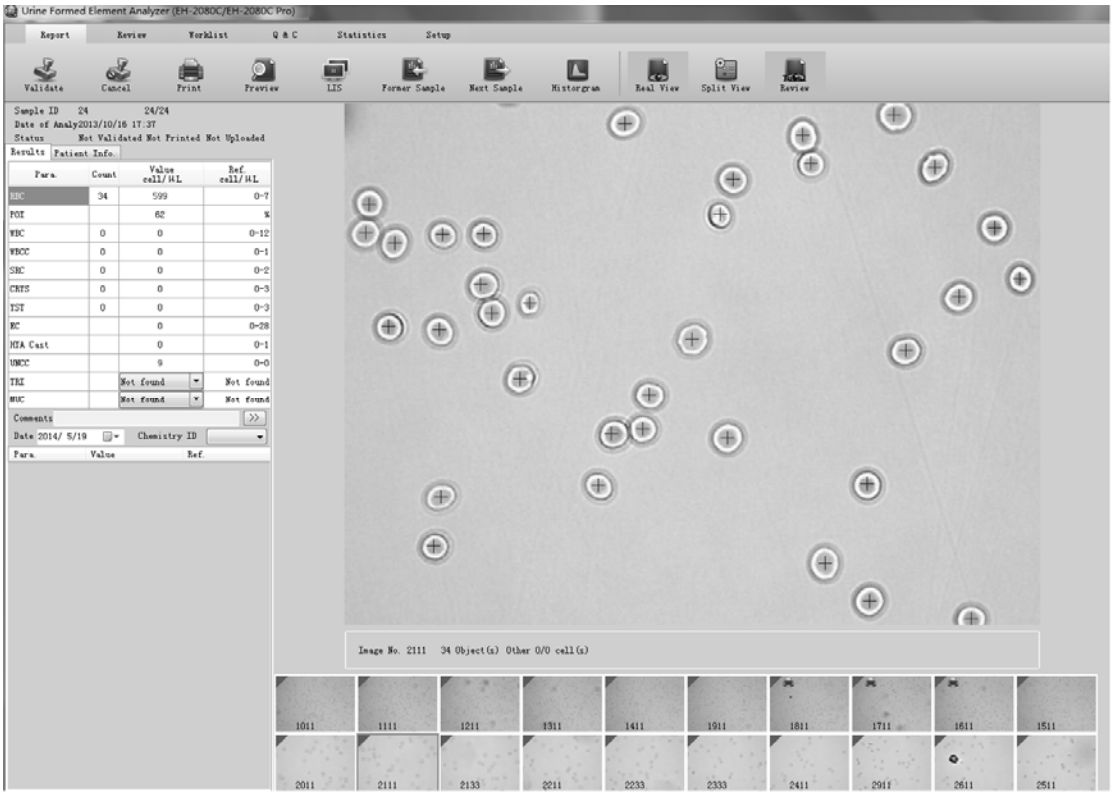
NOTE

- Only administrator or common user whose setting authority has been assigned by administrator can have the authority of validation and canceling validation. See section 5.3.5 for the detailed operation about authority setting.
- Since deletion cannot be canceled, operation shall be carefully performed.

7.2.3 Report

User can edit patient information, view and verify inspection result and print test result.

Click the ‘Report’ button to go to the ‘Report’ screen as shown in the following figure.



The ‘Report’ screen is mainly consisted of information display area and picture display area.

Information display area

This area is used to display the patient information and inspection result corresponding to the current sample; user can edit patient information and inspection result in this area.

Sample ID

Result | Patient Info.

Para.	Value /μL	Ref. /μL	
RBC		0-7	
POI		%	
WBC		0-12	
WBCC		0-1	
SRC		0-2	
CRYS		0-3	
YST		Not found	
EC		0-28	
HYA Cast		0-1	
UNCC		0-0	
TRI		Not found	
MUC		Not found	
BACT		Not found	

The information display area is correlative to the picture display area.

■ **Edit patient information**

User can edit patient information in this area. If the information being edited is the existing sample record, the edited content will be saved in real time; if the information being edited is the added sample record, the added sample information will be saved when switching to other samples.

■ **Edit inspection result**

This software provides three edit mode:

1. Directly modifying the result value would cause the fact that number may be inconsistent with the result.

2. When the parameter number on a single picture is modified, the result value would be recalculated, which is applicable for estimating the particle number on the pictures.

3. Change mark (see the subsequent sections).

NOTE

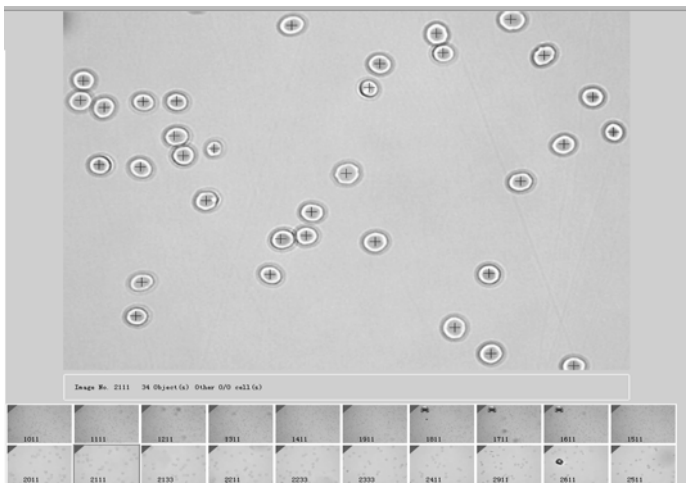
- User needs to select the result values of trichomonad and mucus the 'Inspection result' screen.
- User can rematch Chemistry information on the 'Inspection result' screen (see section 6.6.2 for detailed operation).

Resultant image display area

User may change over at the label page above the resultant image display area to view the real image, division image, cast check image of the sample results.

■ **Real image**

A picture to display the sample inspection result, user may modify the sample picture result in this area.



Amplification area of resultant picture

Picture information bar

Picture area of sample result

M

Changeover of pictures

When clicking certain picture in the picture area of sample result, the picture can be switched to in the amplification area of the result picture; alternatively, changeover of pictures can be accomplished by clicking on mouse when the cursor becomes

'◂•' at the sides of the amplification area of result picture.

Modification of result value

When directly modifying result values in the 'inspection result' screen, the flag number on pictures will not be changed.

Picture information bar

Picture number

Display the current picture number of the amplification area of resultant picture.

Object number

It refers to the total number of the test parameters that are identified with the instrument in the sample result picture area.

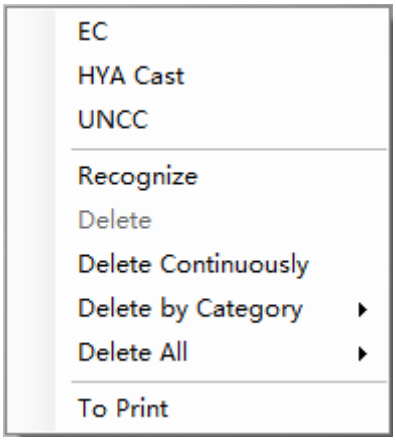
Unclassified

Unclassified number: 0/total 0

It refers to the number of sample in certain picture (high power lens)/total number of all unclassified samples.

Functions of right key

Click the right key on the resultant picture amplification area, menu shown below will be popped up:



EC

1. Click right key and select 'EC'.
2. Click on the unidentified epithelial cell in the resultant picture amplification area.

HYA Cast

1. Click on right key and select 'HYA Cast'.
2. Click on the unidentified hyaline cast in the resultant picture amplification area.

UNCC

1. Click on right key and select 'UNCC'.
2. Click on the unidentified pathological cast in the resultant picture amplification area.

Recognize

Click on right key and select 'Recognize', the software will make Recognize of the current picture automatically.

Delete

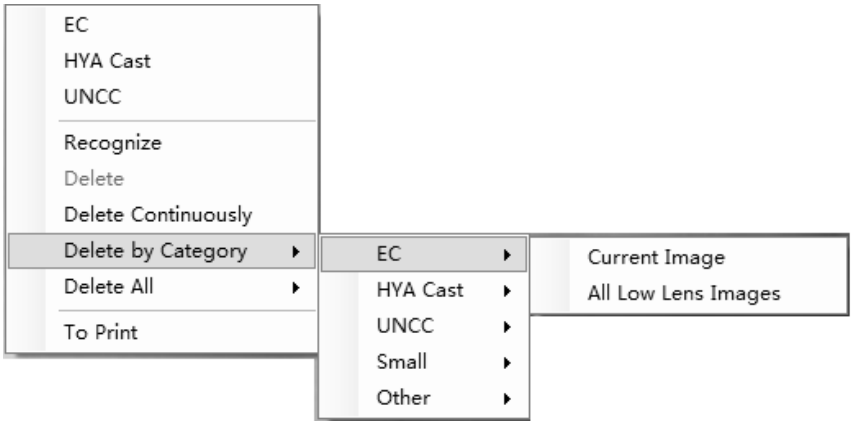
1. Select the visible components to be deleted in the resultant picture amplification area.
2. Click right key and select 'Delete'.

Delete Continuously

1. Click right key and select 'Delete Continuously', '√' will be displayed before continuous deletion.
2. Click on the visible component to be deleted continuously.

Deletion by Category

1. Click right key and select 'Deletion by Category'



2. Select the visible components to be deleted and the deletion area, the selected category visible components within the specified scope are all deleted.

Delete All

1. Click right key and select 'Delete All'

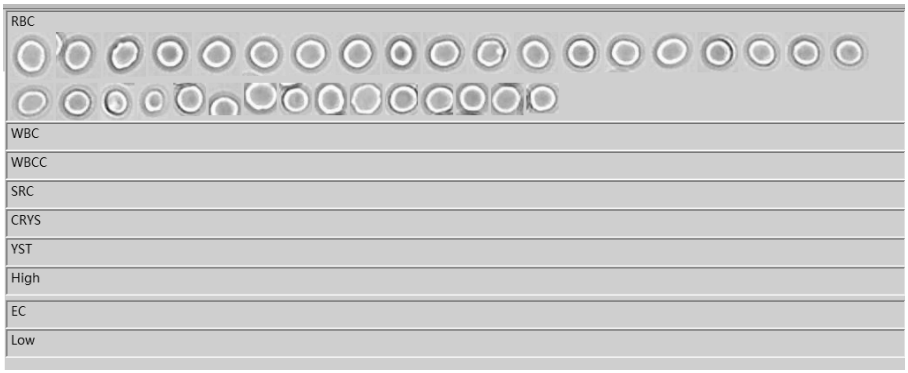
2. Select the deletion area, the test results in this area will all be deleted.

Selected for printing

Click right key and select 'To print', the current picture is set to be printed.

■ **Split View**

Make centralized and classified display for all identified visible components.



Result modification

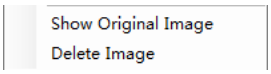
1. Select the division image to be modified and click right key, the list below is popped up:

2. Select the test parameters in correspondence with the division image from the list above.
3. The division image after modification is moved to the corresponding classification area automatically.

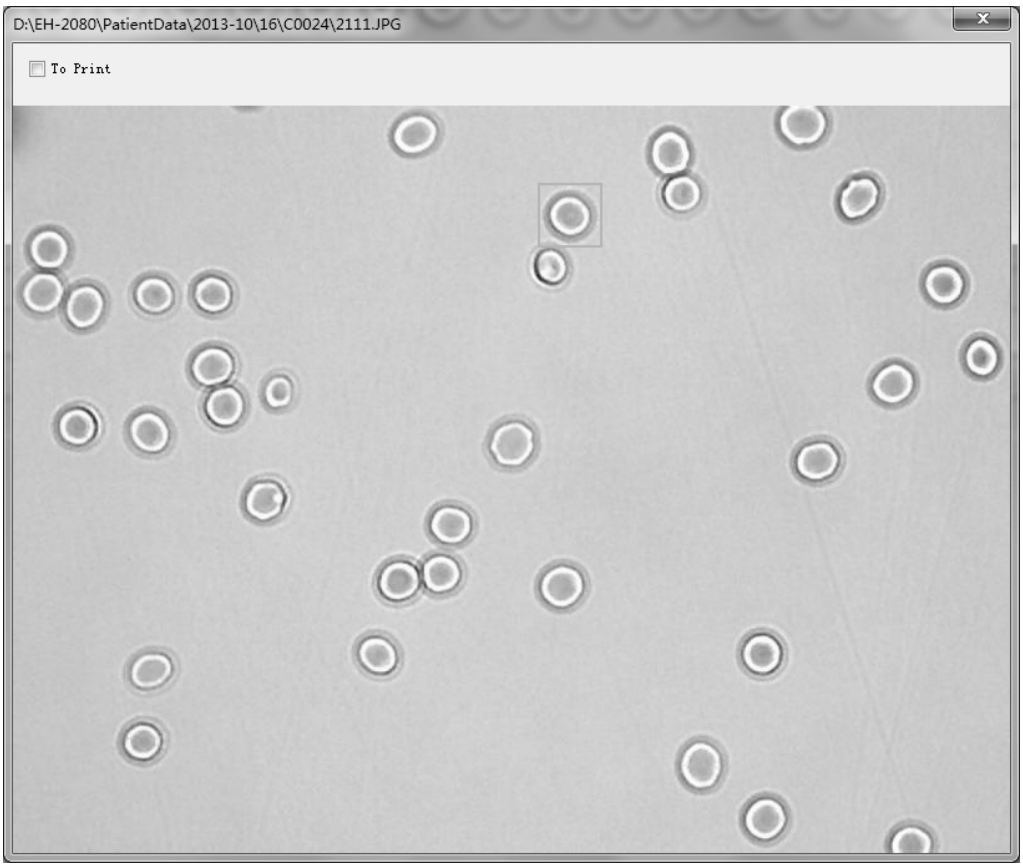
Function of right key

Display original image

1. Select the division image to display the original image and click right key, the list shown below is popped up:

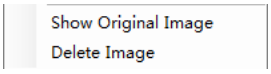


2. Select ‘display original image’, the window shown below is popped up:



Delete the original image

1. Select the division image to be deleted, click right key to pop up the list shown below

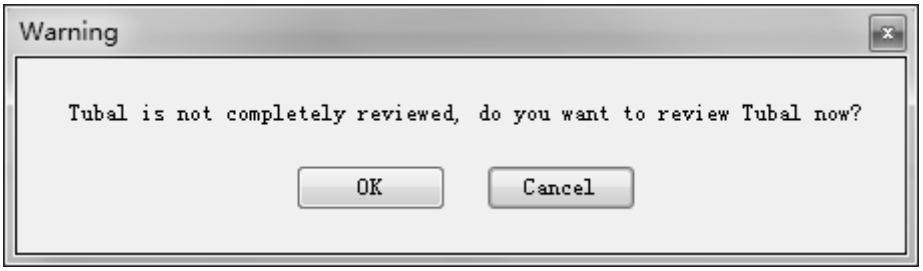


2. Select ‘Delete Image’, the dialog box shown below is popped up

3. Click on ‘OK’ to delete this division image.

■ **Check cast**

1. After division image check is completed, change to cast check screen.
2. On the cast check screen, user makes cast check for 10 pieces of low power lens pictures (for specific check operation, refer to the relevant real image operation).
3. After all 10 pieces of low power lens pictures are checked, change over to the next sample; if sample changeover is made without completing the check, prompt box below will be popped up:



Click on ‘OK’, the screen is changed over to cast check screen automatically, the user continues to complete the unchecked cast pictures; click on ‘Cancel’, change over to the next sample automatically.

NOTE

- The viewed picture is displayed in green at the top left corner of the screen.
- See the following total unclassified number in the high-power lens if the selected pictures are of high-power-lens picture, so as to the low-power-lens picture; similarly, when the font is red, it is indicated that all pictures (whether high-power lens or low-power lens) are questionable.

7.2.4 Button functions

Validate

Click ‘Validate’, applicable sample information will be immediately updated to be validated information.

Cancel

Click ‘Cancel’, applicable sample information will be immediately updated to be unvalidated information.

Print

Click ‘Print’, the content of applicable interfaces will be printed.

Preview

Click 'Preview' to view the print preview of applicable interfaces.

LIS

Click 'LIS' and upload the report information of applicable samples.

Former sample

Click 'former sample', the report information of the previous sample will be displayed on the screen.

Next sample

Click 'Next sample', the report information of the next sample will be displayed on the screen.

NOTE

- User can modify patient information, inspection result and Chemistry inspection result on the report screen, and the modification result can be saved in real time.
- User can match Chemistry and modify result on the report screen.
- Only administrator or common users whose setting authority has been assigned by administrator have the authority of verification and canceling verification.

7.3 Review interface

NOTE

- When "Display report interface" fails to be ticked in system setting interface, the software only provides the following one review interface for displaying sample details and contents.

7.3.1 Introduction to review interface

List Area

List area is used to display the list of analyzed samples with specified conditions, which includes the basic information of samples, such as sample number, name, gender, age, , alarm, audit status, printing status, LIS, test date, test time, work order creation time, submitter, proofer, reviewer, dry chemistry number, test tube position, bar code, sample type, test type and emergency treatment.

	Sample ID	Name	Gender	Age	Alarm	Validate	Print	LIS	Date of Analysis	Test Time	Worksheet Time
1	1								2016年7月19日	9:13	2016年7月19日 9:10
2	2				↑				2016年7月19日	9:13	2016年7月19日 9:11
3	3				↑				2016年7月19日	9:15	2016年7月19日 9:13
4	4								2016年7月19日	9:16	2016年7月19日 9:14
5	5				↑				2016年7月19日	9:38	2016年7月19日 9:36
6	7								2016年7月19日	10:13	2016年7月19日 10:13
7	6				↑				2016年7月19日	10:14	2016年7月19日 10:12
8	8				↑				2016年7月19日	10:15	2016年7月19日 10:13
9	9				↑				2016年7月19日	10:16	2016年7月19日 10:14
10	10				↑				2016年7月19日	10:22	2016年7月19日 10:19
11	12								2016年7月19日	10:22	2016年7月19日 10:22
12	13							√	2016年7月19日	10:22	2016年7月19日 10:22
13	11				↑		√	√	2016年7月19日	10:22	2016年7月19日 10:20
14	1								2016年7月28日	17:32	2016年7月28日 17:30
15	2								2016年7月28日	17:33	2016年7月28日 17:30
16	3								2016年7月28日	17:35	2016年7月28日 17:32

Result Display Area

It displays the test results of samples and dry chemistry detection parameters.

Sample ID 6 7/25

Result | Patient Info.

Para.	Value	Unit	Ref.	
RBC	0	/μL	0-7	
POI	0		%	
WBC	20	/μL	0-12	Details
WBCC	0	/μL	0-1	
SRC	0	/μL	0-2	
CRYS	0	/μL	0-3	
YST	0	/μL	0-3	
EC	111	/μL	0-28	Details
HYA Cast	0	/μL	0-1	
UNCC	0	/μL	0-0	
TRI	Not found		Not found	
MUC	Not found		Not found	

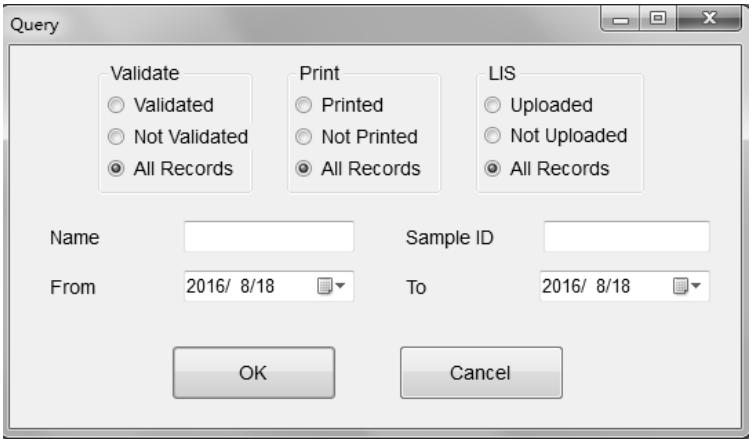
Date 2016/ 7/19 Chemistry ID 06

Para.	Value	Ref.
LEU	+	
URO	norm.	
MCA	0.01g/L	
PRO*	3+	
BIL	-	
GLU	-	
VitC*	2+	
S.G.	1.000	
KET*	2+	
NIT*	+	
CRE	8.8mmol/L	
pH	8.5	
BLD*	1+	
CA*	10.0mmol/L	
Color	Light Yellow	
Turb.	Slight Turbid	
A/C	1.14mg/mmol	
P/C	340.91mg/mmol	

7.3.2 Button functions

Query

Click “Query”button, the following dialog box will pop up.



User can select sample condition information that needs screening as required, the information includes checking status, printing status, LIS status, name, sample number, sample test dates (start date to end date), and then click “Confirm" button, sample data will be screened in sample database, and the sample information meeting the setting conditions will be displayed in sample list area.

Validate

After singleor multiple unchecked sample records are selected in the list area, the sample lines will be highlighted;

Click “Check” button, “Check” column of sample record will bechecked, meanwhile, set the sample background color to white and skip to next unchecked sample.

Cancel

After single or multiple unchecked sample records are selected in the list area, the sample lines will be highlighted;

Click “Cancel Check” button, “Check” column of sample record is not checked, meanwhile, set the sample background color to yellow.

Print

Select the sample records that need printing in the list area, click “Print” button to print.

LIS

Select the sample information that needs transmission.

Click “LIS”button to upload the sample information to LIS.

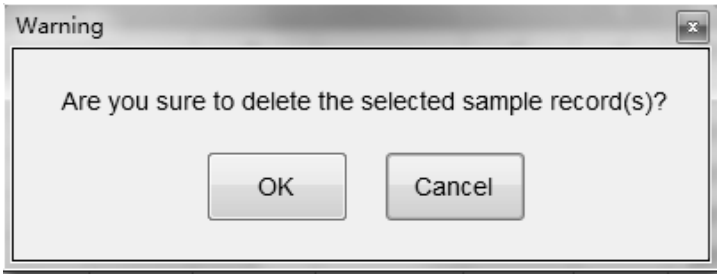
Select All

Click “Select All”, all the sample records in the list area will be selected.

Delete

Select single or multiple sample records to be deleted in the list area;

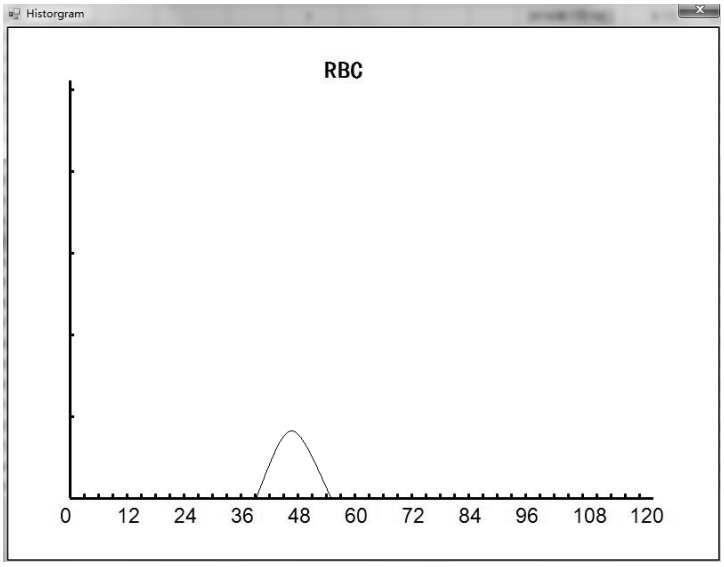
Click “Delete”button，the following dialog box will pop up.



Click “OK” and delete the selected sample and close the dialog box.

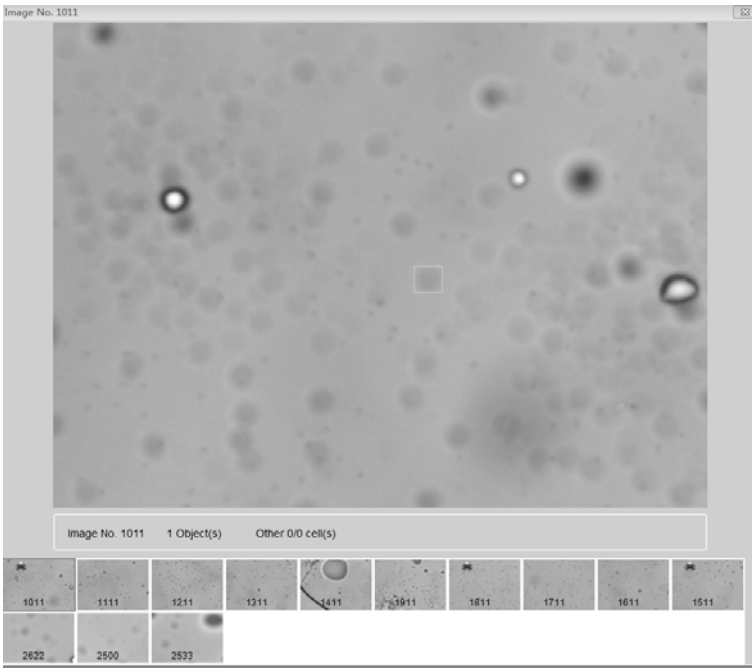
Histogram

After selecting a sample record in the list area, click “Histogram” button, the histogram information of this sample will be displayed.



Realistic picture

After one sample record is selected in list area, click “Realistic picture” button, the photo information of all the test results of this sample will be displayed, user can modify the sample photo results in this area.



Sample editing method under realistic picture:

Result value modification

Modify the result value in “Result” interface directly and there is no change for number of marks in the photo.

Photo information bar

Photo number:

Display the current photo number in result amplifying display area.

Number of objectives:

Sum of test parameters identified by instrument in sample result photo

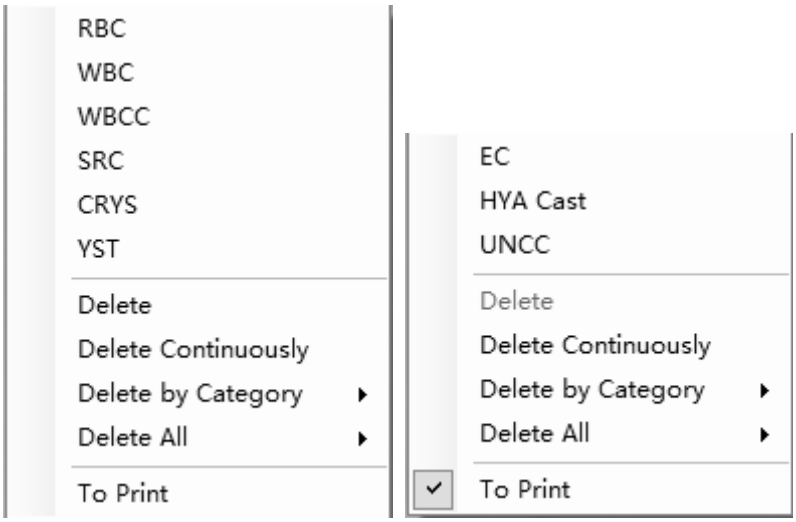
Unclassified:

Other 0/1 cell(s)

Number of samples which are not classified in some photo (high-power microscope)/sum of all unclassified in the sample

Right-key function

Right lick low-power microscope result photo amplifying area and the following menu will pop up:



EC

1. Click right key and select ‘EC’.
2. Click on the unidentified epithelial cell in the resultant picture amplification area.

HYA Cast

1. Click on right key and select ‘HYA Cast’.
2. Click on the unidentified hyaline cast in the resultant picture amplification area.

UNCC

1. Click on right key and select ‘UNCC’.
2. Click on the unidentified pathological cast in the resultant picture amplification area.

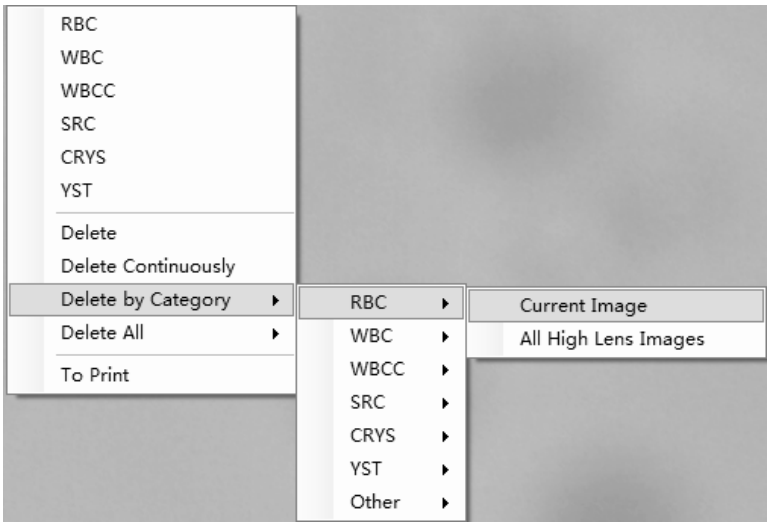
Delete

1. Select the formed element to be deleted in result photo amplifying area.
2. Right click and choose Delete.

Continuous deletion

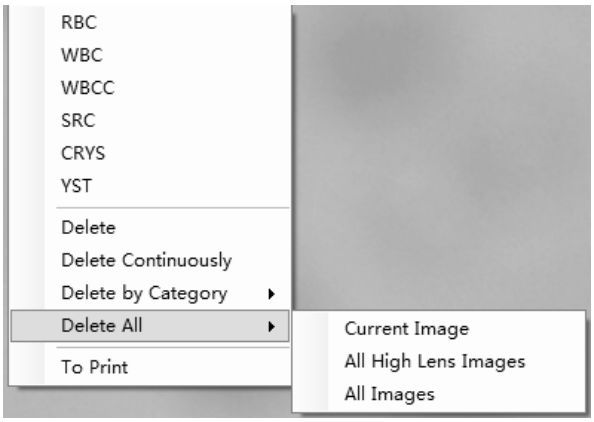
1. Right click and choose “Continuous deletion”，”√” will be displayed before continuous deletion.
2. Continuously click the formed elements to be deleted.

Delete one type



1. Right click and choose“Delete one type”
2. Select the formed elements and areas to be deleted, one type of formed elements within the specified range will all be deleted.

Delete All



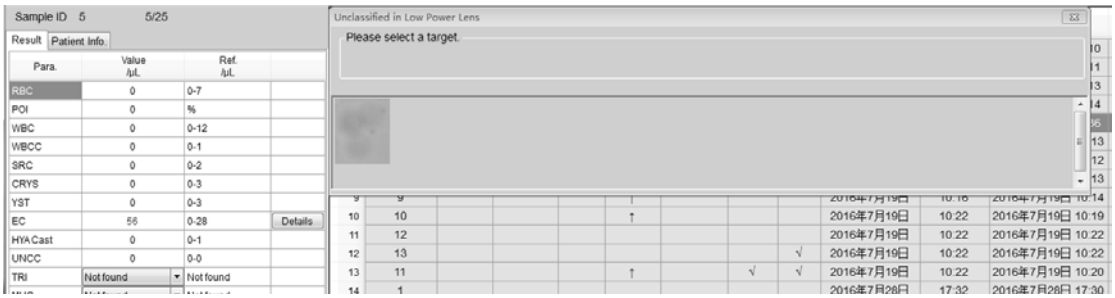
1. Right click and choose“Delete All”
2. Select the area to be deleted, the test results in this area will be all deleted.

Select to print

1. Right click and choose“Select to print”，the current photo will be set to the one to be printed and the photo will be displayed in sample result photo area.
2. Green triangular identification display will be provided in top left corner.

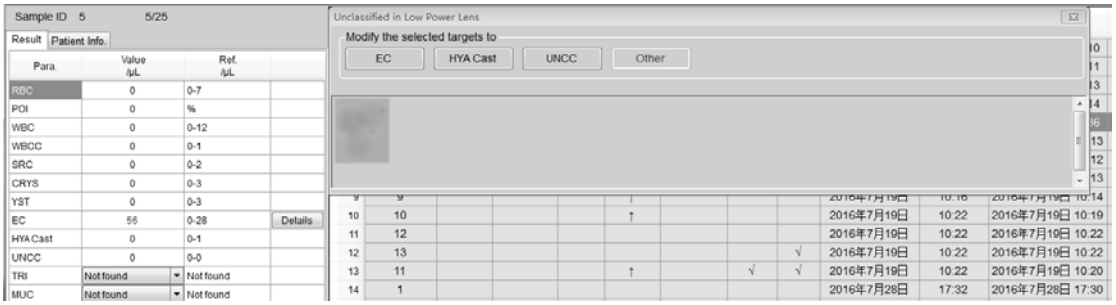
Low-power microscope unclassified

After selecting one sample record in list area, click “Low-power microscope”button, the unclassified information of this sample will be displayed.



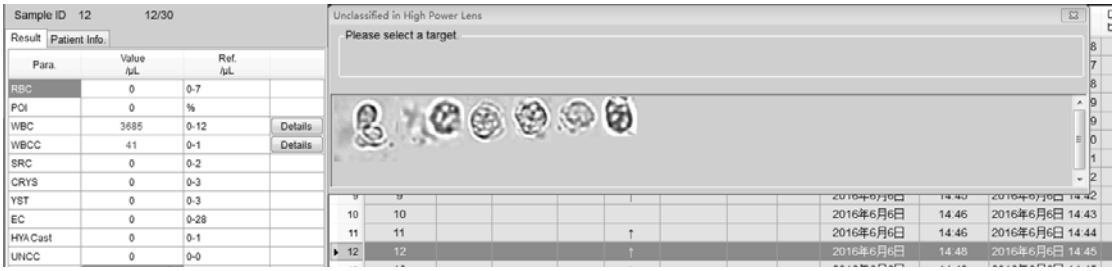
Result modification

Select the split graph to be modified, there is refresh display at the top of split graph: “Change the particle selected to”, the unclassified can be modified to epithelial cell, hyaline cast and non-hyaline cast, as shown below:



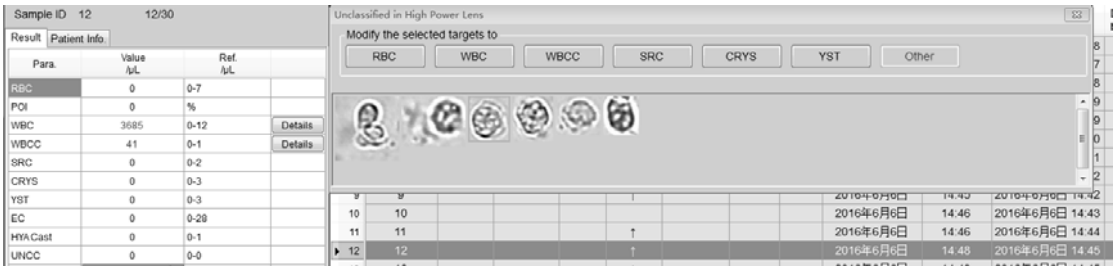
High-power microscope unclassified

After selecting one sample record in list area, click “High-power microscope”button, the unclassified information of this sample will be displayed.



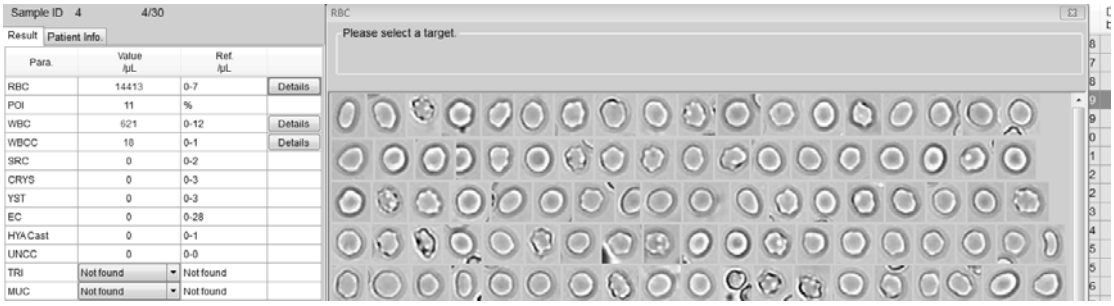
Result modification

Select the split graph to be modified, there is refresh display at the top of split graph: “Change the particle selected to”, the unclassified can be modified to erythrocyte, leukocyte, Leukocyte cluster, small round epithelium, crystallization, saccharomycopsis, as shown below:



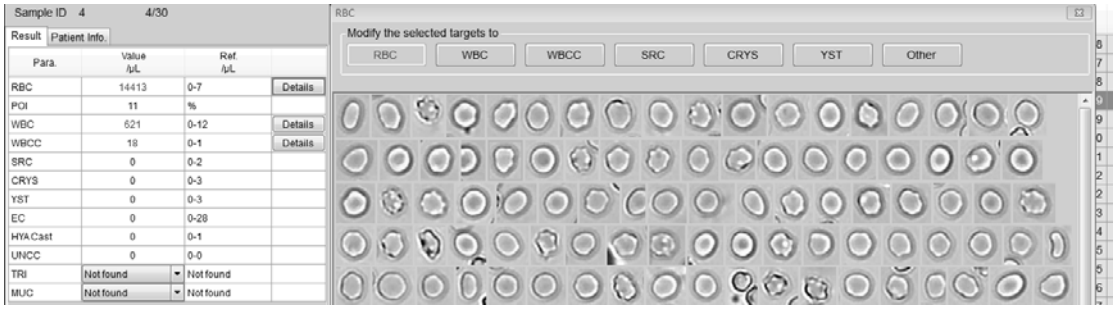
Split graph

In sample result area, click “Details” button, the formed elements identified can be displayed concentratedly.

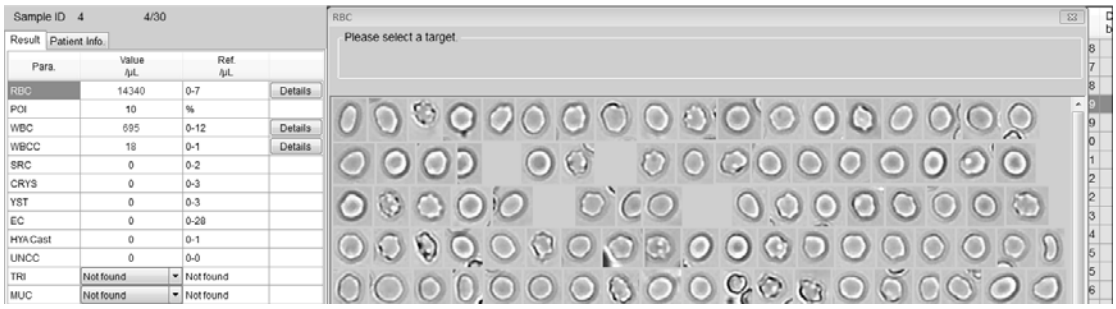


Result modification

1. Select the split graph to be modified, there is refresh display at the top of split graph: “Change the particle selected to”, as shown below:



2. Select the corresponding test parameters of split graph from the list of above list.
3. The modified split graph will be moved to the corresponding classification area automatically, e.g. part of erythrocytes are edited to leukocytes, as shown below:

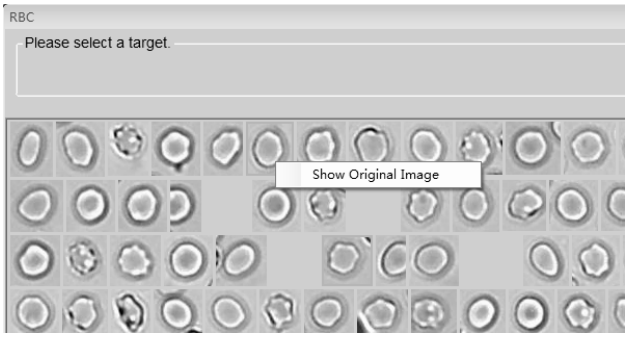


4. Closing button is provided in top right corner.

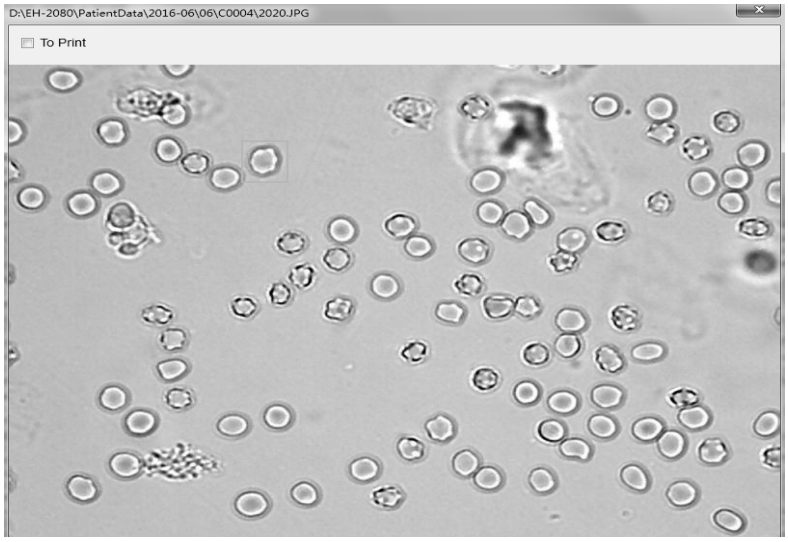
Right-key function

Display original picture

1. Select the split graph whose original picture needs to be displayed and right click, the list shown in the following picture will pop up:



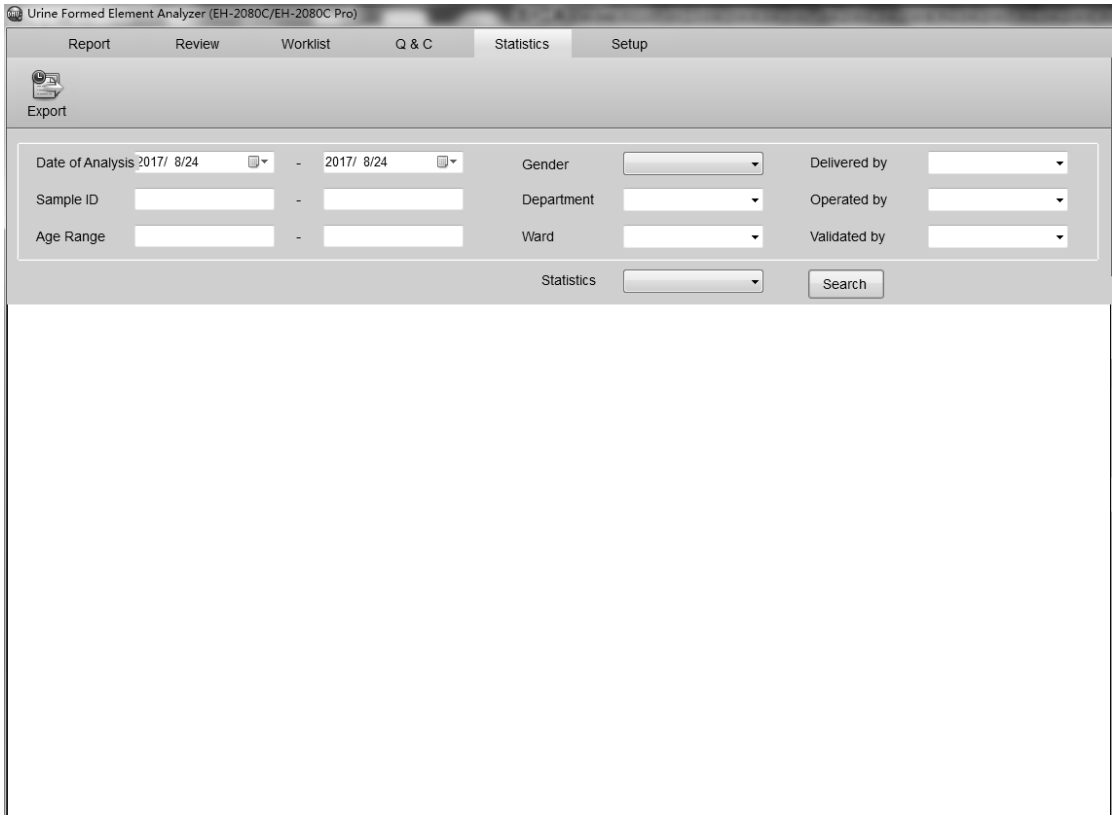
2. Select Display original picture, the windown shown in the following picture will pop up:



7.4 Statistics

User can carry out the statistics of workload on the ‘Statistics’ screen, and can export statistical result on the ‘Statistics’ screen.

Click ‘Statistics’ to go to ‘Statistics’ screen.



7.4.1 Workload Statistics

1. A or multiple statistical conditions as below are designated as required on the ‘Statistics’ screen:

- Inspection date
- Sample No.
- Age range
- Sex
- Department
- Ward
- Sample submitter
- Sample inspector
- Verification person

2. Choose a or multiple statistical items as required in the pull-down list.

The system uses statistical items to classify the result of the screening condition of step 1.

3. Click ‘Search’ to display the statistical result meeting relevant condition in the statistical result area.

7.4.2 Export

1. Click ‘Export’.
2. Choose export path in the dialog box popped up.
3. Change the file name of the exported data as required.
4. Choose the file type of exported data (default type is ‘.csv’).
5. Click ‘Save’ and start exporting.

Chapter 8 Customizing Print Template

8.1 Introduction

User can carry out modification on the default template provided by software, and can customize the format of the template.

After new template is customized and saved, user can call the customized template for print from the software, thus printing the report as per user's print template.

8.2 Entering the customization screen

Click 'Template edit' button in the system setting-up screen so as to enter the custom screen as shown in the following figure.

Information area working area templates property operation area

8.2.1Setting paper property

Paper

Click the pull-down list to choose the desired paper type from the following options:

- A3
- A4
- Half of A4
- A5
- B3
- B4
- B5
- Customized

Choose the uncustomized paper type to display corresponding numerical values in the width and height boxes at the lower part of the screen, and to display the corresponding editable area in the working area.

Choose the customized paper type, and display the numerical value of the paper type that has been chosen last time by default in the width and height boxes in the lower part of the screen; user can set the width and height of the customized paper as per his demand. The editable area in the working area can be updated in a real-time manner according to paper setting.

NOTE

- User may modify the width and height of customized paper rather than the width and height of uncustomized paper.

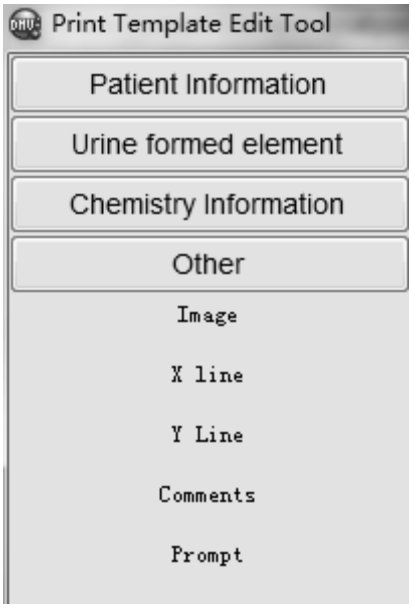
Direction

Click the pull-down list to choose the desired template direction from the following options:

- Longitudinal direction
- Horizontal direction

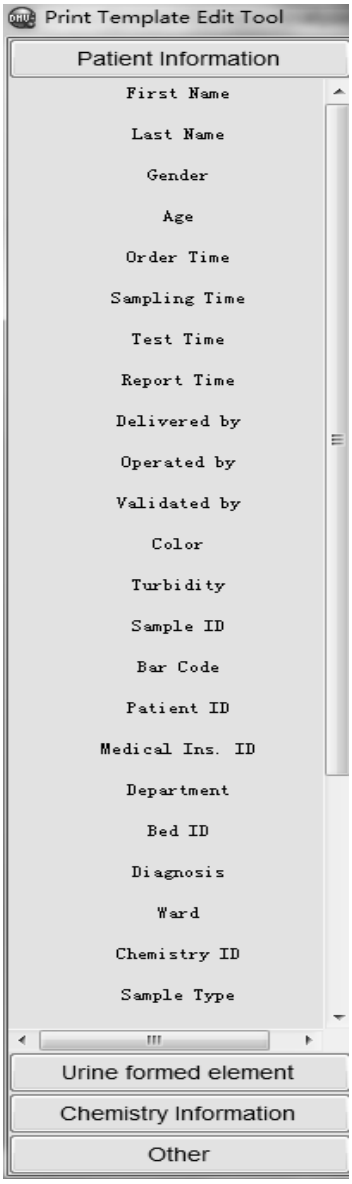
8.2.2 Editing the template

Insert control



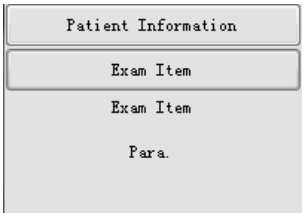
In the information area, click the control to be inserted into the working area; and then, click the mouse at certain position in the template customizing area so as to insert control, and set the position and size for the control to be inserted by moving the mouse.

- Insert basic information



Information can be added in the working area by clicking certain item of information in the basic information.

■ Insert test information



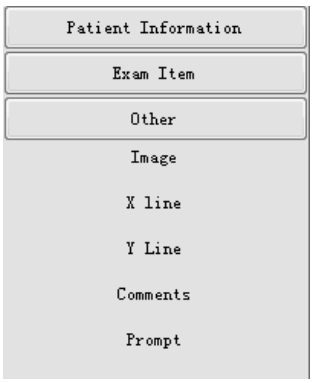
Title bar of inspection item can be added in the working area by clicking ‘Inspection item’.

Parameters can be added in the working area by clicking ‘Parameters’.

NOTE

● Inspection items are printed from the left to the right and from up do down when report is printed.

■ Insertion other items



The position and size of the pictures to be printed can be added in the working area by clicking 'Picture'.

The straight line in X direction can be added in the working area by clicking 'Line in X direction'.

The straight line in Y direction can be added in the working area by clicking on 'Line in Y direction'.

Remark textbox can be added in the working area by clicking 'Remark'.

Some fixed text information such as hospital name and report sheet name; etc can be added in the working area by clicking 'Prompt information'.

Edit the control

The control to be edited needs to be selected before control module is edited.

If you needs to choose a control, click this control and it will be selected.

If you need to edit multiple controls at the same time, multiple control modules can be chosen via any of the following methods:

■ Press and hold the [ctrl] key, and use mouse to left click the desire controls sequentially.

■ Click the template and drag the mouse, at this time, a rectangular box is displayed in which all the controls will be selected.

NOTE

● The edition of all controls is performed according to the first control or first group of selected control.

Modification of words

1. Choose the control to be modified.
2. Double click the words to be modified.
3. Enter new content; in this case, modification is completed.

Move the control

Control can be moved by the following methods:

■ Choose the control to be moved, and drag the control to the target position; and then, release the mouse.

■ Choose the control module to be moved, and depress the direction key on the keyboard to move the control to the target position.

■ Choose the control module to be moved, and click on the applicable pushbutton in the shift column of the template property operation area to move the control.

Button functions

Click ‘ ’button to move the chosen control upward;

Click ‘ ’ button to move the chosen control downward;

Click ‘ ’ button to move the chosen control leftward;

Click ‘ ’ button to move the chosen control rightward.

Alignment of controls

Controls can be aligned using any of the methods:

■ Choose the controls to be aligned; and then, click the applicable options in the alignment column of the template property operation area to execute the designated alignment.

Button functions

Click ‘ ’ button to align the chosen controls on the left;

Click ‘ ’ button to align the chosen controls to be vertical in the center;

Click ‘ ’ button to align the chosen controls on the right;

Click ‘ ’ button to align the chosen controls on the top;

Click ‘ ’ button to align the chosen controls to be horizontal in the center;

Click ‘ ’ button to align the chosen controls at the bottom.

NOTE

- The precondition for this operation is that multiple controls are chosen in the same time.

Adjustment of the size of controls

The size of control module can be adjusted using any of the following methods:

■ Choose the controls to be adjusted, and use mouse to drag the frame of the module so as to adjust its size.

■ Choose the controls to be adjusted, click the applicable options in the size column of the template property operation area to execute the designated operation.

Button functions

Click ‘ ’ button to make the width of the chosen controls identical.

Click ‘ ’ button to make the size of the chosen controls identical.

Click ‘ ’ button to make the height of the chosen controls identical.

NOTE

- The precondition for this operation is that multiple controls are chosen in the same time.

Adjustment of the spacing of controls

The spacing of control can be adjusted using any of the following methods:

■ Choose the controls to adjusted, and use the mouse to drag the object for adjusting spacing.

■ Choose the controls to adjusted, and click the applicable options in the spacing column of the template property operation area to execute the designated operation.

Button functions:

Click ‘ ’ button to make the horizontal spacing of the chosen controls identical.

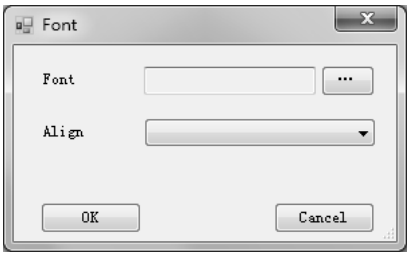
Click ‘ ’ button to make the vertical spacing of the chosen controls identical.

NOTE

- The precondition for this operation is that multiple controls are chosen in the same time.

Setting of font

1. Choose the control in which font is to be set.
2. When clicking 'Font', the following dialog box will pop up:



3. When clicking on'[...]', the dialog box will pop up; User may set font as required in the dialog box.
4. Click the pull-down list of 'Alignment', and choose the desired alignment mode of words from the options of the list.

8.3 Edit new print template

1. Open a template.

See section 8.2 for detailed operation.

2. Edit a template.

See section 8.2.2 for detailed operation.

3. After edition is completed, when clicking on 'Save as', the dialog box will pop up; Enter file name, and click 'Save'. After saving operation is completed, this template will be automatically updated to the default template list.

NOTE

- User cannot alter the save path of template without permission.

8.4 Edit of existing print template

1. Open template.

See section 8.5.1 for detailed operation.

2. Edit template.

See section 8.2.2 for detailed operation.

3. After edit is completed, click on 'save'.

8.5 Managing the Template

8.5.1 Template loading

1. When clicking the ‘load’ button in the print template customizing screen, the dialog box will pop up

2. Choose the template file to be loaded, and click ‘Open’, in this case, this template is introduced into the current template customizing area.

8.5.2Template preview

Click the ‘Template preview’ button in the system setting screen to preview the currently-chosen default template.

Chapter 9 Quality Control

9.1 Introduction

Urine analyzer may have a certain degree of error in the process of long-term use. The existence of error may lead to mistakes or unreliable analysis results.

Quality control is to use quality control material that has been assigned to monitor the performance of urinary sediment analyzer routinely. User shall test the quality control material on the urinary sediment analysis system according to the same way as the urine test and compare the result with known quality control material reference value. If the comparison result shows that the deviation is big, certain measures shall be taken.

Quality control programprovides a kind of effective method for the detection of possible error, only users are familiar with the theory of quality control and master the actual operating method can they effectively eliminate the error influence on the results of the analysis.

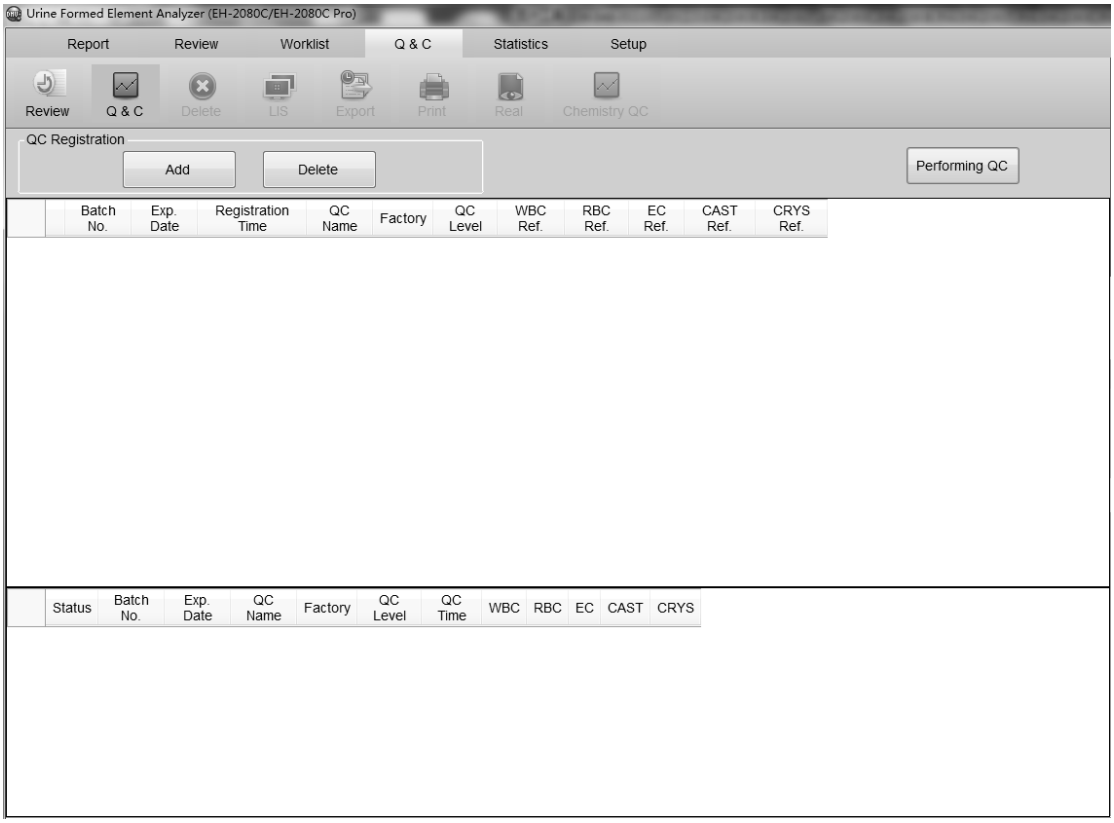
NOTE

- Operator shall use the controls and reagent designated by the manufacturer, and shall store and use them strictly according to the operator's manual of controls and reagent.
- When several items of quality control register sheet or quality control sheet is to be deleted, press Ctrl or Shift to choose multiple items.

9.2 Quality control

9.2.1 Setting

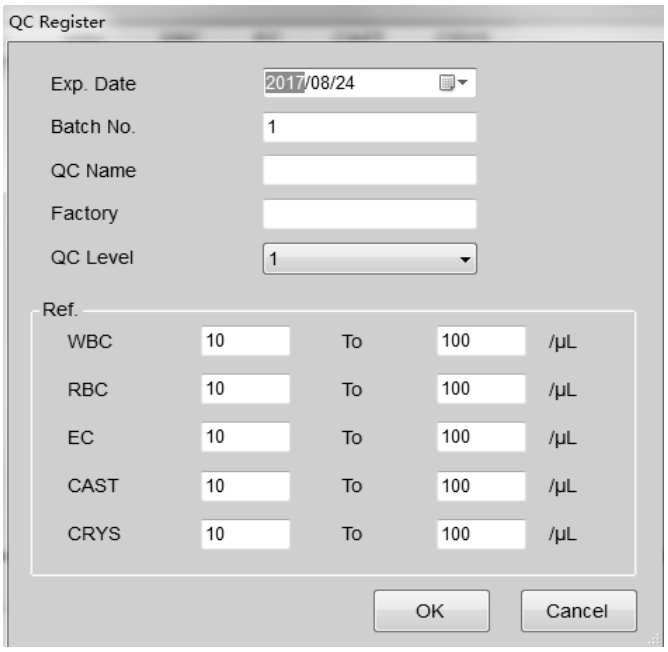
Click "Q & C" > " Q & C", as shown below:



Add the new quality control sheet

Add quality control sheet

1. Click“Add” button and enter quality control sheet setting interface, as shown below：

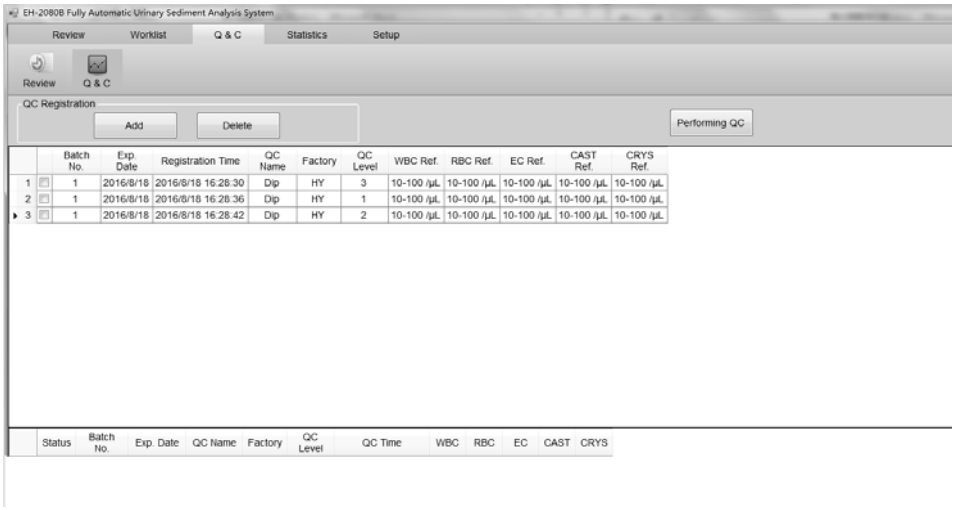


Need to complete the inputs：

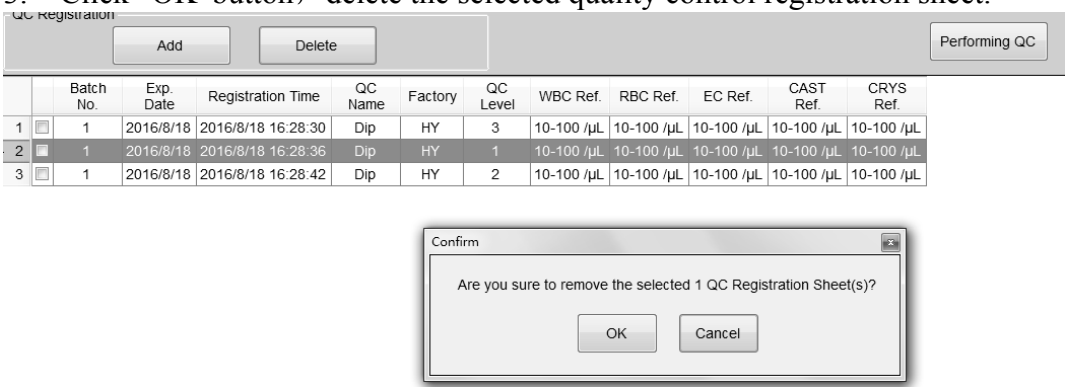
- Exp.Date (Period of validity)
- Batch No.
- QC Name (Quality control name)
- Factory (Information of manufacturer)
- QC Level (Concentration level)
- Ref (Reference range of each monitored item)

After the above information is filled in, click “Confirm” button, the adding of quality control sheet can be completed; if you want to give up the adding, click “Cancel” button;

Delete quality control sheet



1. Select quality control sheet to be deleted in list area;
2. Click “Delete”button，the following dialog box will pop up.
3. Click “OK”button，delete the selected quality control registration sheet.

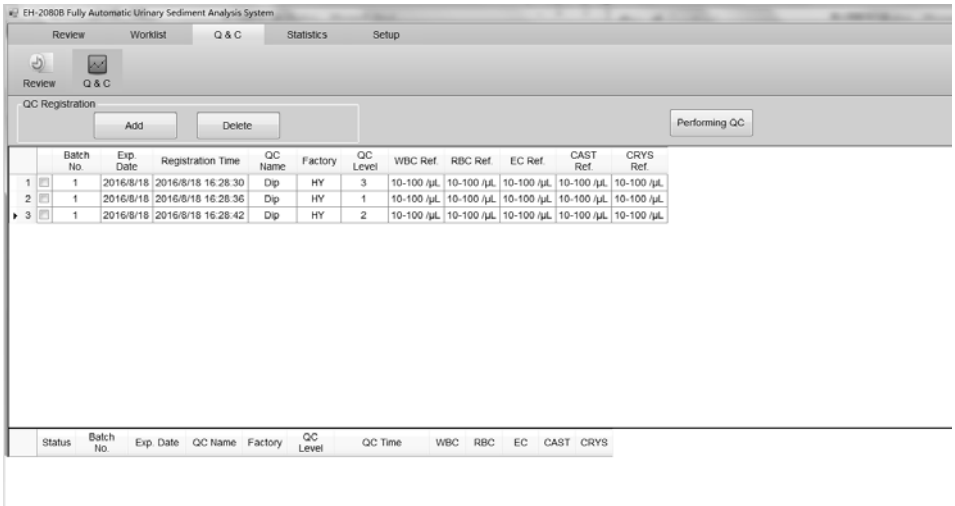


9.2.2 Quality control

NOTE

- Before executing quality control, user shall ensure valid quality control sheet;

Click “Q & C” > “Q & C”, as shown below:



- Quality control interface is mainly composed of: list area of registration sheets and quality control result browsing area on the same day;

Registration sheets area

The list area of registration sheets shows the current bit outdated quality control registration information, including: batch number, period of validity, registration time, quality control name, manufacturer, concentration level, leukocyte reference value, erythrocyte reference value, epithelial cell reference value, cast reference value and crystallization reference value.

Current result browsing area

Display the information of quality control results that have been executed on the same day, including: status (pass or fail), batch No., period of validity, quality control name, manufacturer, concentration level, quality control time, leukocyte reference and detection values, erythrocyte reference and detection values, epithelial cell reference and aging values, cast reference and aging values and crystallization reference detection values;

For the samples with quality control failure, the field in status area is red and the result field of failure item is also red.

Performing QC

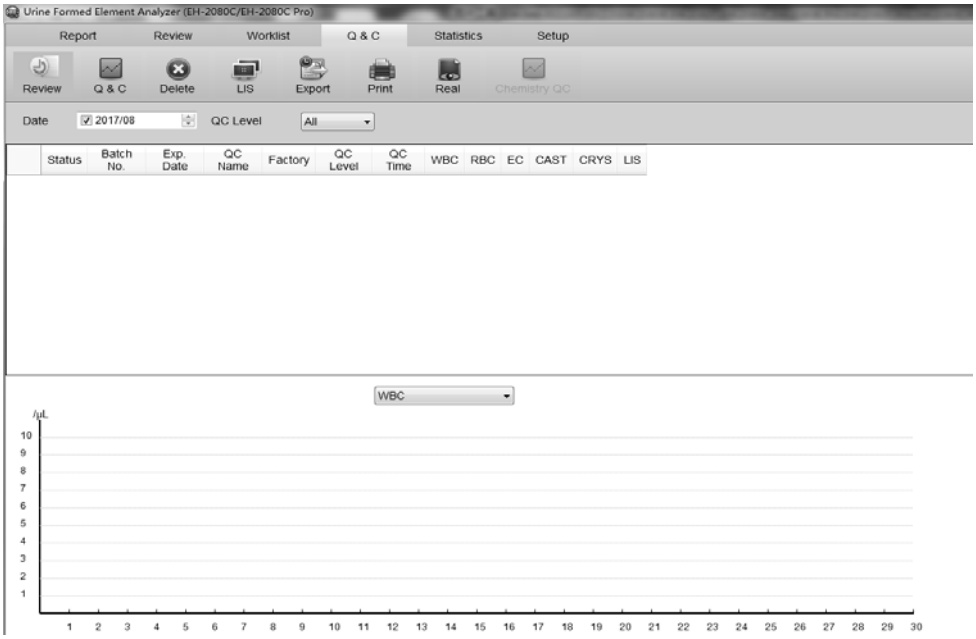
Tick the quality control sheets that need to execute quality control (multiple items can be ticked at the same time if multilevel quality control needs to be executed), then

place quality control liquid test tubes on test tube rack in the order of quality control sheets, then click “Performing QC” button.

After quality control test is completed, the corresponding test information of quality control will be updated to result area automatically.

9.2.3 Review

In quality control interface，click “Review”button to enter reviewinterface.



Review interface is mainly composed of screening area, list review area, quality control chart review area.

Screening area

Through selecting test date of quality control and concentration level, screen out the necessary information on tested quality control.

List area

In list review area, display the following screened information on tested quality control:

- Status
- Batch No.
- Period of validity
- Quality control name
- Manufacturer

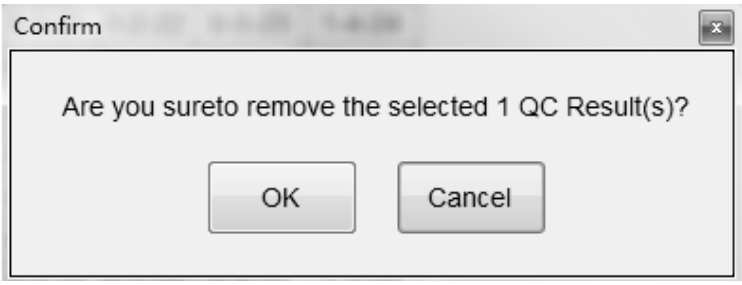
- Concentration level
- Quality control time
- Leukocyte reference and detection values
- Erythrocyte reference and detection values
- Epithelial cellreference and detection values
- Cast cell reference and detection values
- Crystallization reference and detection values
- LIS

Quality control chart review area

In quality control chart review area, display the corresponding cell type and quality control target value graph under concentration level:

Deletion function

1. Select quality control sheet to be deleted in list area;
2. Click “Delete”button, the following dialog box will pop up:



3. Click “OK”button， delete the tested QC result(s).

Export function

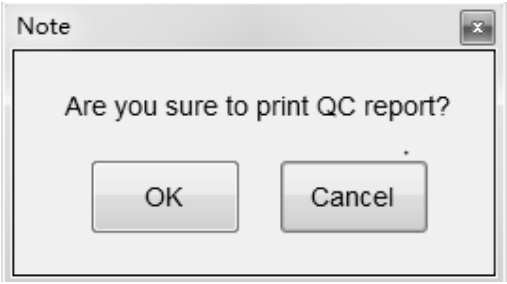
Software will export the selected quality control result data (QC results displayed in review interface) to CSV file.

Print function

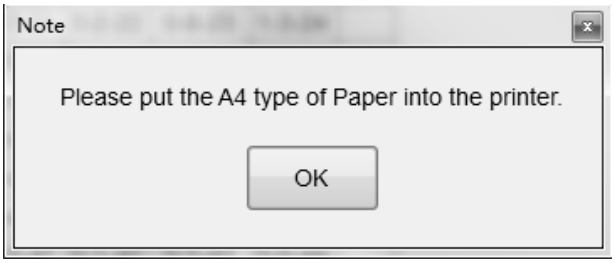
The software prints out the data of QC results displayed in review interface.

1.In the list area, select the month and level information of QC data to be printed;

2.Click “OK” button,the following prompt dialog box will pop up:



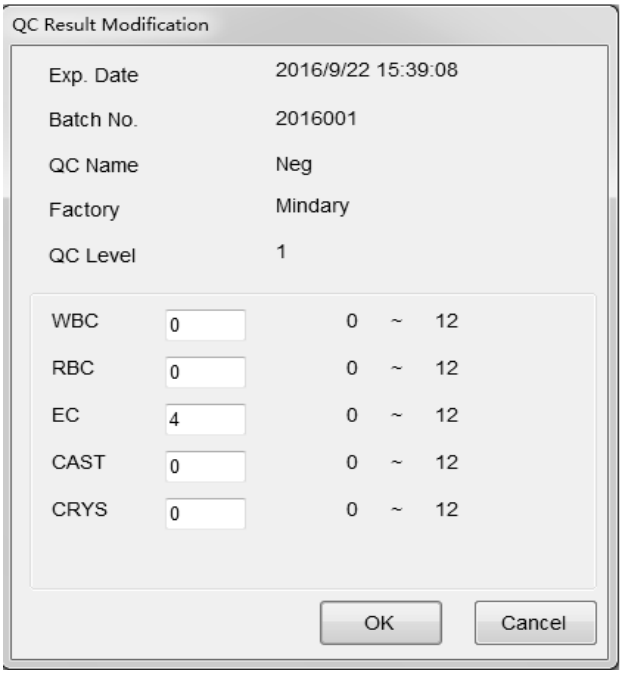
3. Click “OK” button, put A4 paper into printer.



4. Click “OK” button, print QC results

Modification function

1. In the list area, select the QC data to be modified;
2. Click "Modify" button, the following dialog box will pop up:



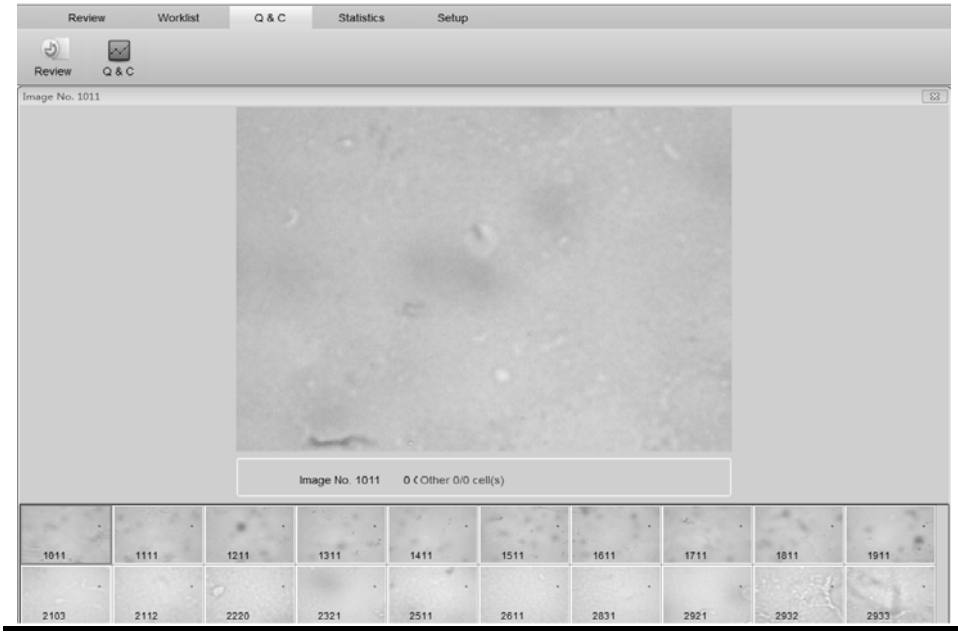
3. Modify data and click "OK" button to save the results;.

LIS Function

1. In the list area, select the QC data to LIS;
2. Click "LIS" button, the list of review area will display " √ " after the LIS;
3. In the QC LIS export folder, the QC data file is named as QC time + level.

Realview images function

1. In the list area, select the QC result to view the real view image;
2. Click "real" button, the following prompt dialog box will pop up:



NOTE

- The QC data is valid for two years.
- The QC data is valid for one month.
- When All is selected for concentration level, the QC curve is not displayed.

Chapter 10 Services

10.1 Introduction

To ensure the accurate and efficient operation of the analysis system, operator shall perform routine maintenance for the analysis system as per the requirements described in this Chapter. The analysis system provides several maintenance functions so that operator can perform the maintenance.

This Chapter describes various maintenance functions of the analysis system as well as remedy measures in case of error and alarm.

- All the analyzer components and surfaces are potentially infectious, take proper protective measures for operation or maintenance.

- Performing unauthorized maintenance procedures can damage your analyzer. Do not perform any maintenance procedures that are not described in the operator's manual.
- In case of problems not specified in this manual, contact the manufacturer's Customer Service Department or your local distributor for recommendations on maintenance.
- Only parts supplied by the manufacturer can be used for maintenance. For any questions, contact the Customer Service Department.
- Exercise caution to avoid contact with the sharp sample probe when performing maintenance.

10.2 Reagent management

- The object to be tested with this project is urine. Since the detailed pollution degree or infectivity of urine cannot be determined, it is recommended that operators wear breathing masks and gloves during operation, and be completely disinfected after operation; precautions shall be taken during operation and maintenance.

- The substance such as waste and waste sample that contain biocontamination must be treated according to local regulations.
- The sample probe is sharp, and the sample probe possibly contains biocontamination and harmful substances. For this reason, care shall be taken to avoid coming in contact with sample probe during the operation of the analysis system.

- Do not mix the new container of reagent with the residue in the replaced container to ensure accurate measurement.

NOTE

- Please keep the reagent container from severe shock or crashing against other object. Otherwise, the alarming would be unreliable.

You need to replace reagents promptly when it is low, depleted, expired, or required to remove an error. Do as follows:

- Replace with a new container of desired reagent (see section 10.6 for details).
- Finish reagent setup on the machine terminal (see section 5.2.2 for details).

You can also check the reagent information (like open-container date, expiration date, residual volume, etc.) of the reagents at the reagent setup screen of the analyzer.

10.3 Maintenance

10.3.1 When and Why

Procedure	When to Clean	Why to Clean
Booting maintenance	Maintenance after startup	Manually maintain and clean the fluidic system and counting cell of the analyzer.
Clean sample probe	There are impurities on the surface or in the middle of the sampling probe.	Avoid cross contamination among samples; remove the impurities in the sampling probe.
Normal maintenance	The counting cell is dirty or there are impurities in the counting cell.	Maintain and clean the counting cell and tube line
Forward flush	There are impurities in the tube line and counting cell	Remove impurities/clean tube
Backward flush	There are impurities in the tube line and counting cell	Remove impurities/samples

10.3.2 Startup liquid line initialization

1. Pour proper amount of curing solution into the test tube.
2. Place the test tube at the position that can add sample with the instrument in a state of rest (for detail operation, refer to 6.6.3)
3. Click on ‘maintenance’>‘startup curing’, the instrument implements startup curing operation automatically.

10.3.3 Clean the sample probe

Click on ‘maintenance’>‘cleaning the sample probe’, the instrument implements the sample probe cleaning operation automatically.

10.3.4 Normal curing

1. Pour proper amount of curing solution into the test tube.
2. Place the test tube at the position that can add sample with the instrument in a state of rest (for detail operation, refer to 6.6.3)
3. Click on ‘maintenance’>‘normal curing’, then the instrument automatically carries out the normal curing operation.

10.3.5 Forward flush

Click on ‘maintenance’>‘forward flush’, then the instrument automatically carries out forward flush operation.

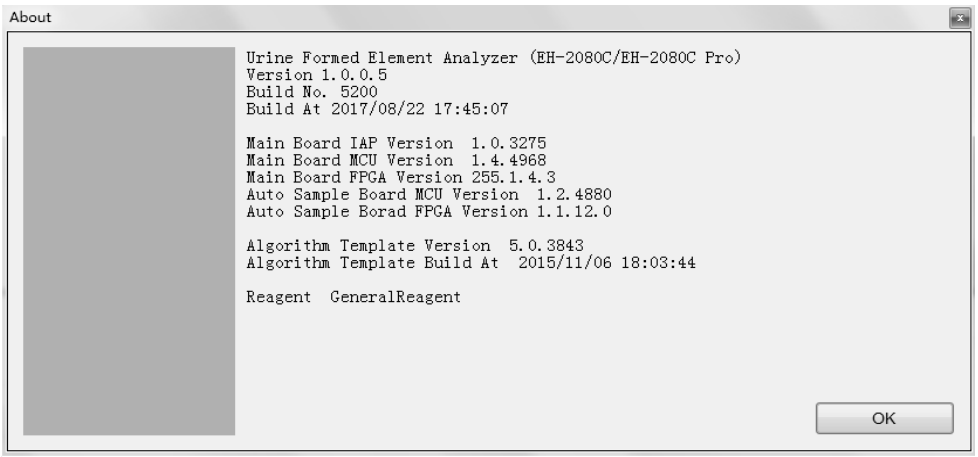
10.3.6 Backward flush

Click on ‘maintenance’>‘backward flush’, then the instrument automatically carries out backward flush operation.

10.4 Version Information

The version information of the current software and hardware of the analyzer can be viewed.

Click ‘ ’ button at the top right corner of the software to enter ‘Version information’ screen.



10.5 Manually cleaning

10.5.1 Cleaning of the surface of counting cell

User needs to maintain the surface of counting cell regularly; it is recommended to perform maintenance over one time every week.

In case there is pollutant visually, wipe the surface of counting cell gently using cotton swab.

10.5.2Cleaning the enclosure of the analyzer

Clean the enclosure of the analyzer if necessary; use cotton cloth to wipe the enclosure under shutdown condition.

10.5.3 Cleaning the lens of microscope

It is recommended to use soft brush or gauze to remove dust.

For stubborn stain, such as finger print and grease, may use soft cotton cloth and lens paper dipped with mixed liquor of anhydrous alcohol and aether to wipe off slightly.

The mixing solution ratio of anhydrous alcohol and aether is as follows: alcohol: aether =1:1.5

10.6 Replacement

10.6.1 When to Replace and Tools Needed

Procedure	When to Replace	Tools You May Need
Replace Reagent A and B	As needed	/
Replace waste container	The current waste container is full	/

10.6.2 Replacing the Reagent A and B

1. Take a new reagent container, open the cap of the reagent container, and place the container beside the reagent container to be replaced.

2. Anticlockwise open the cap of the reagent container to be replaced, and carefully remove the cap groupware of the reagent container.

3. Insert the guide tube of the cap groupware of the reagent container into the new reagent container, and clockwise tighten the cap of the reagent container.

4. Put the cap of the new reagent container on the old reagent container, and properly treat the old reagent container.

10.6.3 Replacing the waste container

1. Take an empty waste container, open container cap, and put the container beside the waste container to be replaced.

2. Anticlockwise rotate the cap of the old waste container, and carefully remove the cap groupware of the container.

3. Insert the cap groupware of the old waste container into the new waste container as vertical as possible; and clockwise tighten the cap of the waste container.

4. Put the cap of the new waste container on the old waste container, and treat the old waste container properly.

10.7 Transportation and storage

When the analyzer is being transported or storaged ,should fulfil following conditions:

- Temperature -10℃～40℃
- Relative humidity ≤93%
- Atmosphere 50 kPa～106kPa

10.8 Long-time not use dispose

If instrument will be stopped more than 15 days ,please do these jobs before stopping:

Remove the reagent joint of reagent A and B from instrument ’s back panel , and in maintain interface under the set up interface ,conduct Forward flush and backward flush 5 times in each channel .Then ,close instrument and remove waste water joint.
Please keep instrument in proper temperature ,humidity and atmosphere when stop using them.
For the instruments which stopped for more than 15 days ,before using them again ,please connect reagent A’s ,reagent B’s and waste-water’s joint ,connect them reagent accurately , and in maintain interface under the set up interface ,conduct Forward flush and backward flush 5 times in each channel .Then use the instrument again .

Chapter 11 Troubleshooting

11.1 General

This chapter contains information that is helpful in locating and correcting problems that may occur during operation of your analyzer.

NOTE

- This operator's manual is not a complete service manual and it only provides the measures to be taken by operator in the event of fault alarm on the analysis system.

11.2 Error Information and Handling

The following table lists the possible faults and corresponding help information for the analysis system:

Error Code	Error Description	Fault Cause	Troubleshooting Information	Remark
1	Startup on-line failed	1. Wires on serial port have not been connected. 2. Serial port error 3. Output interface board error 4. Communication board error.	1. Reconnect the wires on the serial port. 2. Replace the serial port of computer, and check if the drive is normal. 3. Replace the output interface board. 4. Replace the communication board or restart the software; if the error still exists, contact the manufacturer's after-sales service department.	/
2	There is no liquid or not full in the sampling pipe of forward	1. There is no solution B entering the cleaning cell; 2. Peristaltic	1. Replace solution B or check the tube line. 2. Replace the pipe of the peristaltic pump. 3. Replace the tube	/

	flush.	pump does not operate normally. 3. The pipe is aged	line; if the error still exists, contact the manufacturer's after-sales service department.	
3	Startup initialization failed	1.Maloperation occurred in the last shutdown. 2. The position of the counting cell is changed.	1. Start up/shut down the system again, and restart the software; and then, carry out initialization; if the error still exists, contact the manufacturer's after-sales service department.	/
4	The internal of the counting cell is dirty	1. Routine maintenance is insufficient.	1. Perform manual maintenance, and forward/backward flush. 2. Replace the counting cell; if the error still exists, contact the manufacturer's after-sales service department.	/
5	Chemistry on-line operation failed	1. Wires on serial port have not been connected. 2. The serial port of computer error.	1. Reconnect the wires on the serial port. 2. Replace the serial port of computer, and check if the drive is in normal condition; if the error still exists, contact the manufacturer's after-sales service department.	/
6	Chemistry on-line	1. Wires on serial port have	1. Reconnect the wires on the serial	/

	operation failed	not been connected. 2. The serial port of computer error.	port. 2. Replace the serial port of computer, and check if the drive is in normal condition; if the error still exists, contact the manufacturer's after-sales service department.	
7	There is no image in the sample test area.	1. The USB wire of microscope has not been connected. 2. USB connecting wire error.	1. Connect the USB wire, and check if the drive is in normal condition. 2. Replace the connecting wire; if the error still exists, contact the manufacturer's after-sales service department.	/
8	Sample test image is fuzzy.	1. There is dust on the surface of counting cell. 2. There is dust on the surface of object lens.	1. Use cotton swab to gently wipe the surface in the same direction. 2. Use dust-free cloth to gently wipe the lens; if the error still exists, contact the manufacturer's after-sales service department.	/
9	Sample result cannot be printed.	1. Printer has not been turned on. 2. Printer is out of paper. 3. Printer has not been connected with	1. Turn on the power supply of the printer. 2. Put paper in the paper box of the printer. 3. Connect the printer with the USB wire of the computer.	/

		computer. 4. The drive of printer is missing.	4. Update the drive of the printer; if the error still exists, contact the manufacturer's after-sales service department.	
10	The instrument does not have any response after powered on.	1. The power cable of the instrument has loose contact. 2. Fuse is blown.	1. Check the connection of power cable. 2. Replace the fuse (there is standby fuse in the document bag)	Disconnect power cable before replacing the fuse; uncover the fuse cover below the socket, replace the damaged fuse, and close the cover.

Appendix A Specification

A.1 Classification

EH-2080B/EH-2080T/EH-2080C/EH-2080B Pro/EH-2080T Pro/EH-2080C Pro is classified according to the management regulations of China's medical appliance:

This product belongs to clinic inspection analysis instrument (6840), and its management class is Class II.

A.2 Associated reagents of the product

The associated reagents of the analysis system include Reagent A 、Reagent B and Cleanser.

Reagent name
Cleanser
Reagent A
Reagent B

A.3 Parameters

Parameters	Abbreviation	Reference range
Red blood cell	RBC	(0-99999999) cell/μL
White Blood Cell	WBC	(0-99999999) cell/μL
WBC clump	WBCC	(0-99999999) cell/μL
Epithelial cell	EC	(0-99999999) cell/μL
Small round cell	SRC	(0-99999999) cell/μL
Hyaline cast	HYA Cast	(0-99999999) cell/μL
Unclassified Cast	UNCC	(0-99999999) cell/μL
Crystal	CRYS	(0-99999999) cell/μL
Yeast	YST	(0-99999999) cell/μL
Poikilocytosis rate	POI	(0-100)%

A.4 Sampling features

Sample volumes required for each analysis

Sample demand: 2ml

Sample intake: 1ml

A.5 Parameter range

parameter	reference range（cell/μL）			
	general	man	woman	child
RBC	0-7	0-5	0-7	0-6

WBC	0-12	0-6	0-12	0-4
WBCC	0-1	0-1	0-1	0-1
EC	0-28	0-4	0-28	0-4
SRC	0-2	0-2	0-2	0-2
HYA Cast	0-1	0-1	0-1	0-1
UNCC	0-0	0-0	0-0	0-0
CRYS	0-3	0-3	0-3	0-3
YST	0-3	0-3	0-3	0-3

A.6 Carryover

The carryover of all analysis parameters does not exceed 0.02%..

A.7 Input/output devices

Input device: keyboard, mouse

Output device: display, printer

A.8 Display resolution

display resolution: ⩾1440×900

A.9 Ports of the analyzer

The ports of the analyzer include:

Communication port: USB interface, serial port

Power port: power cord

Fluidic inlet: liquid detector, fluidic pipe

A.10 Power supply

Name	Voltage	Input power	Frequency
Analyzer	(100V-240V～)±10%	300VA	(50/60Hz)±1Hz

A.11 Operating environment

- Indoor use
- Altitude: do not exceed 2000m
- Temperature: 5℃- 40℃
- Relative humidity: ≤80%
- Rated contamination class: contamination class 2

Note: ‘contamination class 2’ means there is usually only non-conductivity pollution, but there might be occasionally short-time conductive due to agglomeration.

A.12 Electromagnetic compatibility

- The analyzer complies with the emission and immunity requirements described in EN 61326-1:2013 and EN 61326-2-6:2013.

- The analyzer has been designed and tested to CISPR 11 Class A. In a domestic environment it may cause radio interference, in which case you may need to take measures to mitigate the interference.

A.13 Dimension and weight

	Analyzer
Length (mm)	740
Width (mm)	600
Height (mm)	525
Weight (Kg)	65

A.14 Inventory of attachment

Serial number	The name of attachment
1	Bottle cap of reagent
2	Bottle cap of waste
3	Tubes
4	Null modern cable with 9 core
5	Power cord
6	Fuse

A.15 Performance of products

Types	performance
Speed of testing	Detection time of single product：not exceed 120 seconds Continuous testing sample count in an hour ：The testing speed for dual-channel sample exceeds 80test/h, and the testing speed for for three-channel sample exceeds 90test/h; the testing speed for four-channel sample exceeds 100test/h,.
resolution ratio	exceed 800×600
Request of popline	Liquid system works regularly
detection	The analyzer will detect the red cells and white cells with concentration of 5/μL
CV value	The analyzer's counting results for the sample of 50/μL should not exceed 25% ,for the sample of 200/μL should not exceed 15%.

Recognition rate	1.The analyzer can recognize red cell ,white cell and tube type at least .Coincidence rate of single result and microscopic examination :red cell ≥70% ,white cell ≥80% , tube type ≥ 50%
	2. The false negative rate of analyzer's test results should not exceed 3%.
Stability	Within 8 hours from starting up ,the CV value of cell counting result should not exceed 15%.
Carryover rate	It should not exceed 0.02%.

046-011548-00(1.0)